

Journalist Ideology and the Production of News: Evidence from Movers

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Abstract

What role do journalists play in determining the political slant of the news they produce? We develop and estimate a model where journalists and newspaper outlets contract over both slant and wages. The model implies a set of conditions under which we can consistently estimate the role of journalist preferences in driving the observed variation in slant across outlets by leveraging journalist transitions between outlets. To measure slant, we train a transformer-based, machine learning model using articles tweeted by politicians and apply it to a full-text database of 9+ million newspaper articles published in the US between 2013 and 2018. Under our model and identifying assumptions, our estimates (a) reject the hypothesis that journalists have zero ideological preferences over the content they produce and (b) imply that 10% of the variation in slant can be explained by journalists. When journalists move across outlets, their average slant shifts by 77% of the gap between their destination and origin outlet averages. In counterfactuals that reweight journalist party composition, moving from the observed left-leaning distribution to an even distribution of Democrats, Republicans, and non-partisans shifts average article slant in a more conservative direction, but only by 0.04 standard deviations in slant.

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1 Introduction

In the United States, 28 percent of journalists identify as a Democrat whereas only 7 percent identify as a Republican (Weaver et al. 2019).¹ In a 2019 Gallup survey, 42 percent of U.S. adults described the national news media as too liberal, compared with 13 percent who described it as too conservative. At the same time, trust in the mass media has fallen sharply: the share of Americans reporting a great deal or fair amount of trust declined from 51 percent in 2000 to 28 percent in 2025, and from 47 percent to 8 percent among Republicans (Gallup 2000; 2019; 2025).

One proposed explanation is that the political preferences of journalists themselves affect the content produced by news organizations.² Prior academic work provides theoretical mechanisms through which journalists’ preferences can generate persistent media bias (e.g., Baron 2006). Whether these mechanisms operate empirically, and to what extent individual journalists shape the political content they produce, remains an open question.

Answering this question has proven difficult because journalists sort across outlets. More liberal journalists may work for more liberal news organizations, while more conservative journalists may sort into more conservative outlets. At the same time, outlets differ in their readership, editorial policies, and incentives to produce politically slanted content.

In this paper, we develop and estimate a model where journalists and newspaper outlets contract over both slant and wages. This framework allows us to empirically examine the extent to which individual journalists shape the ideological content of the articles they write and the market provision of slant as a whole.

We start by building a simple model of the journalist labor market and the production of news. Our model highlights how observed slant is a weighted combination of three components: journal-

¹In 1971, 25.7 percent of surveyed journalists in the US identified with the Republican party and 35.5 percent identified with the Democratic party. By 2013, only 7.1 percent of surveyed journalists identified with the Republican party whereas 28.1 percent continued to identify with the Democratic party (Weaver et al. 2019). When focusing on Washington correspondents, the partisan skew is more striking—with 89 percent of surveyed correspondents indicating they voted for Clinton in the 1992 election. The distribution of self-reported partisan identification among journalists is mirrored by the overall distribution of political contributions from members of the news and media industry (Bonica 2013). See also Figure 1.

Academics and media critics have also expressed concerns about an overarching liberal media bias in the US—with these concerns being expressed in a diverse set of outlets including *The Washington Post*, *The Wall Street Journal*, and *The Quarterly Journal of Economics* (e.g., Groseclose and Milyo 2005; Wemple 2017; Sauter 2020).

²Matthew Yglesias, co-founder of Vox, stated “*I think content is getting dragged left because that’s the labor market (and to an extent the audience) rather than as deliberate strategy*” (Smith 2021). See also Groseclose (2011). High-profile exits of certain journalists are also linked to within-outlet political representation. For example, in 2017, *The New York Times* hired Bari Weiss to diversify the paper’s opinion page. Yet, in 2020, Weiss resigned—citing “*constant bullying by colleagues who disagree with my views*” (Weiss 2020).

ist preferences, outlet average preferences, and a search-specific outlet shock. Our model provides testable predictions for whether journalists have preferences over slant, and it provides guidance for estimating the magnitude of these preferences given potential selection in job transitions induced by search-specific shocks to outlet preferences.

In order to take the model-informed estimators to the data, we construct a novel measure of media bias leveraging recent advances in machine learning. Using a new dataset of articles tweeted by politicians, we fine-tune RoBERTa (a transformer-based model of human language) to predict the political party of the politician who shared a given article. Our fine-tuned model performs well on both in-sample cross-validation tests and out-of-sample comparisons to human coders, and it represents a significant improvement over bag-of-words methods used previously in the literature.

We then apply our measure of slant to predict the partisan slant of 9+ million articles published between 2013 and 2018 from almost 300 newspapers in the ProQuest US Newsstream database (accessed via TDM Studio). Importantly, the ProQuest database also includes the byline for each article—allowing us to construct an author-level panel dataset that tracks the ideological slant of the articles written by a given journalist as they move across outlets.

Our empirical strategy for the observational data builds on the rich literature in labor economics that estimates worker and firm wage fixed effects (e.g., Abowd et al. 1999, AKM) or worker preferences (e.g., Sorkin 2018) using employee transitions between firms.³

We start by estimating an event-study specification using a sample of consistent movers.⁴ Under our model and assumptions, if we are able to reject the hypothesis that journalists close the entire origin-destination gap between outlets, we can also reject the hypothesis that journalists have zero preferences over the slant they produce. In other words, if journalists simply write content that mirrors the preferred slant of an outlet and have zero ideological preferences, the gap in slant between the origin and destination outlets would close entirely when journalists switch outlets. On the other hand, if journalists have strong ideological preferences, we would expect there to be little-to-no change in observed slant when a journalist switches outlets.

Under these assumptions, our event-study estimates reject the hypothesis that journalists have

³Our results are consistent with a persistent journalist component in measured slant. Estimated journalist fixed effects also differ by recorded party affiliation in the voter-linked sample. We interpret these patterns through a model of ideological preferences, but the empirical designs do not isolate those preferences as the causal mechanism. Other persistent attributes, such as source networks, beat expertise, writing style, or reporting skills, may contribute to these patterns and may themselves be related to ideology.

⁴Among journalists, there are a wide variety of work arrangements including both freelancers and salaried employees. In our event study analysis, we constrain our analysis to likely salaried employees by focusing on journalists who wrote T or more articles at the origin outlet before switching and writing T or more articles at the destination outlet.

zero ideological preferences over the slant they produce. Using our sample of consistent movers, journalists close roughly 77 percent of the gap in average slant between the origin outlet and the destination outlet. The 95 percent confidence intervals on our event study estimates exclude both 0 percent and 100 percent.

We also use the AKM fixed effect framework to decompose the variation in article-level slant using journalist and outlet fixed effects. In our preferred specification, restricted to the set of movers and using a leave-out correction, we estimate that journalist fixed effects explain 8 percent of the observed variation in slant, outlet fixed effects explain 20 percent, and sorting between journalists and outlets explains another 7 percent.⁵

We also examine the relationship between journalist fixed effects and other observable characteristics of journalists. As we would expect, the distribution of journalist fixed effects for Republican journalists is further to the right (i.e., more conservative) than the distribution for other journalists. Women and non-white journalists tend to have more liberal fixed effects than men and white journalists.

AKM decompositions restrict complementarities between journalists and outlets. As a robustness check, we re-estimate the decomposition using the framework of Bonhomme et al. (2019), which discretizes journalists and outlets into types and allows for nonparametric interactions between the two. The estimated interaction component is negligible: about 0.06 percent of the variation in slant.

While interesting, we view these decomposition results as primarily descriptive given the search-specific shocks to outlet preferences highlighted in our model which may violate the conditional mean independence assumption of these models.

We derive a new estimator to provide causal estimates of the *between* outlet variation in slant driven by journalists. By focusing on only the *between* outlet variation in slant, we are able to provide causal estimates under a weaker set of assumptions than the conditional mean independence assumption of the AKM model.⁶ Rather than assuming no selection on the search-specific shocks (as the AKM model assumes), we allow movers to be selected on these shocks, but require that, within each origin–destination pair, the average shock deviation from the outlet mean is the same at the origin and destination.

⁵The remaining variation is explained by time fixed effects, other covariance terms, and the residual.

⁶Moreover, we may be particularly interested in the role of journalists in driving the *between* outlet variation in slant, rather than both the *between* and *within* outlet variation. For example, if consumers are choosing between outlets to subscribe to, between outlet variation in slant may be more relevant for understanding how variation in slant affects consumer welfare.

Our preferred estimates from this specification suggest journalists explain 14 percent of the observed variation in average slant between outlets. This is broadly similar in magnitude to the 20 percent of outlet-level variation in slant attributed to consumers in Gentzkow and Shapiro (2010).

Finally, we model selection directly, adapting our framework into an empirical model of journalists and outlets contracting over slant. In the estimated model, journalists explain 10 percent of the variation in article slant. We use the estimated model to conduct counterfactuals that change the party composition of the journalist pool to be representative of the population. Setting the pool to equal thirds of Democrats, Republicans, and non-partisans moves mean slant by only 0.04 standard deviations.

Overall, these results suggest that while journalists have ideological preferences that influence the production of slant, they play a relatively moderate role in driving the market provision of slant.

This paper relates to several strands of literature.

First, our paper relates to a literature on media bias. Within this literature, our paper makes both a methodological and a substantive contribution.

Methodologically, previous work has used bag-of-words or citation based measures of media bias (e.g., Gentzkow and Shapiro 2010; Groseclose and Milyo 2005). We leverage recent advances in machine learning, BERT-based transformer models, to measure media bias and show how these more advanced methodologies can show substantial improvement in predicting the human perceived slant of an article. This builds on separate work using recent advances in machine learning to measure non-verbal media bias based on the facial expression of politicians in images used on news websites (Boxell 2021).

We also make substantive contributions to the literature on media bias. Much of the prior empirical work on media slant has focused on demand-driven motives for media bias (e.g., Gentzkow and Shapiro 2010; Martin and Yurukoglu 2017; Widmer et al. 2026), conglomeration and owner effects (e.g., Martin and McCrain 2019), or professional norms (e.g, Baum and Groeling 2008; Shapiro 2016). Recent work also uses article-level slant measures to study polarization in news consumption on social media (Braghieri et al. 2025).

The three closest related papers empirically are Cagé et al. (2025), Srinivasan (2021), and Xu (2024).⁷ Cagé et al. (2025) use an AKM-style decomposition to show that much of the variation in slant on French television and radio shows can be explained by host- or journalist-specific fixed effects. In contemporaneous work, Srinivasan (2021) finds smaller average responses to outlet

⁷Baron (2006) showed theoretically how journalist discretion and preferences could drive persistent media bias.

slant in cross-outlet comparisons of journalists’ writing at eight US online news outlets. Our broader sample and event-study focus on sustained outlet transitions may contribute to the different estimates. Xu (2024) also leverages journalist turnover to examine how financial journalists’ social networks influence the favorability of coverage towards firms in the US.⁸

Our model, however, highlights the limits of the AKM-style variance decomposition estimates given the potential for selection into journalist transitions. We solve this selection issue by providing novel estimators and associated conditions with which we can recover the role of journalist preferences in driving observed variation in slant across outlets. Overall, our empirical results corroborate Cagé et al. (2025) and Xu (2024) in finding important effects of journalists, but in the broader US print newspaper market as opposed to French broadcasting or US financial news.

Second, our paper relates to a literature on employee preferences and compensating differentials. Some work examines the importance of compensating differentials in general (e.g., Sorkin 2018), whereas other work focuses on more specific dimensions such as a firm’s corporate social responsibility behavior (e.g., Hedblom et al. 2019) or the co-partisanship of co-workers (e.g., Gift and Gift 2015; Colonnelli et al. 2025). Interpreted through our model, our results provide evidence on employee preferences over the ideological content of their production in the news industry.

2 Model of News Production and Empirical Implications

We build a simple model of news production with heterogeneous journalists and outlets.

2.1 Journalists

Journalist i has utility from a job at outlet j

$$u_{ijt}(x_{ij}, \omega_{ij}) = \omega_{ij} - \beta_i(x_{ij} - \delta_i)^2 + \varepsilon_{ijt}$$

where x_{ij} is the average slant of articles written, δ_i is the bliss point for journalist i , ω_{ij} is an unobserved utility transfer from the outlet to the journalist (e.g., wages and benefits). The outside option gives utility $\omega_{i0t} + \varepsilon_{i0t}$. The shocks ε_{ijt} and ε_{i0t} are alternative-specific type-1 EV shocks that are independent across individuals, alternatives, and time. Note that for a given match, utility can be transferred by giving journalists assignments closer to their bliss point δ_i or increasing the unobserved utility transfer (e.g., wages).

⁸Barberá and Sood (2018) find a strong correlation between journalist campaign contributions and the ideological slant of the content they produce. See also Hassell et al. (2020) and Graves et al. (2016).

2.2 Outlets

An outlet j has preferences over their portfolio of journalists A_j

$$u_j(A_j) = R_j(A_j) - \sum_{i \in A_j} \omega_{ij}$$

where

$$R_j(A_j) = \sum_{i \in A_j} [v_j - \beta_j(x_{ij} - \delta_j - \xi_{ij})^2].$$

Here, v_j is fixed revenue per journalist, δ_j is the outlet's baseline slant ideal point, β_j measures its strength of slant preferences, and ξ_{ij} is a shock independent across *searches*.

Outlets have preferences over revenue $R_j(A_j)$ minus the wage transfers to journalists $\sum_{i \in A_j} \omega_{ij}$.

2.3 Matching and Contracting

There is an initial distribution of journalists across outlets. Each period, firms draw a vector of search-specific preference shocks ξ_j of length s_j . Journalists are then matched to firm-searches, where journalist i is matched to firm j 's search s with some probability $p_{ijs}(\cdot)$.⁹ We do not take a stand on the inputs to $p_{ijs}(\cdot)$ —it could include previous values of slant x , previous wages ω , previous firms j , and/or previous or current search-specific shocks ξ .

The prospective outlet j observes the journalist's current contract (x_c, ω_c) , type (δ_i, β_i) , and outside option ω_0 , and offers a contract (x, ω) .¹⁰ The journalist then draws the alternative-specific shocks described above, which are independent of variables realized before the choice, and chooses whether to remain at current outlet c , move to prospective outlet j , or take outside option 0. We assume journalist's output is perfectly observable to the firm, contract enforcement is costless, and both journalists and outlets are myopic.

Given a match with journalist i who has current contract (x_c, ω_c) , the prospective outlet j chooses a contract (x, ω) that solves

$$\max_{x, \omega} \underbrace{\Delta U_j(x, \omega)}_{\text{Change in Profit}} \times \underbrace{M_i(x, x_c, \omega, \omega_c, \omega_0)}_{\text{Prob. Journalist Accepts Offer}},$$

⁹If journalist i is matched to outlet j 's search s , we write $\xi_{ij} = \xi_{js}$.

¹⁰We suppress journalist and time subscripts on contract terms and outside options when there is no ambiguity.

where

$$M_i(x, x_c, \omega, \omega_c, \omega_0) = \frac{\exp(\omega - \beta_i(x - \delta_i)^2)}{\exp(\omega_0) + \exp(\omega - \beta_i(x - \delta_i)^2) + \exp(\omega_c - \beta_i(x_c - \delta_i)^2)}.$$

and

$$\Delta U_j(x, \omega) = v_j - \beta_j(x - \delta_j - \xi_{ij})^2 - \omega.$$

Proposition 1. *If $\beta_i, \beta_j \geq 0$ and $\beta_i + \beta_j > 0$, the outlet's unique optimal contract has slant $x_{ij}^* = \frac{\beta_i \delta_i + \beta_j(\delta_j + \xi_{ij})}{\beta_i + \beta_j}$ and wage $\omega_{ij}^* = h_{ij}(x_{ij}^*, x_c, \omega_c, \omega_0)$, where h_{ij} is determined by the journalist and outlet primitives.*

Proof: See Appendix A.

Holding the journalist and outlet primitives fixed, the optimal wage can depend on the journalist's current contract (x_c, ω_c) and outside option ω_0 , whereas the optimal slant contract cannot.

The distribution of ξ_{ij} across *accepted* matches can differ from its distribution across *searches*. For example, if journalists have strong slant preferences and if journalists are all more conservative than the most conservative outlet, then journalists will be more likely to accept a slant contract with a higher ξ_{ij} , potentially raising the mean shock among accepted matches. In this sense, *realized* variation in ξ can also be a function of journalist preferences.

Zero journalist slant preferences eliminate this acceptance-stage selection, but do not by themselves eliminate selection into the offered pool.

Lemma 1. *For any outlet j with $\beta_j > 0$, if $\beta_i = 0$, then, conditional on the journalist's current wage and outside option, accepting outlet j 's offer does not further select on ξ_{ij} : the conditional mean of ξ_{ij} among journalists who accept outlet j 's offer equals its conditional mean among journalists who receive outlet j 's offer.*

Proof: See Appendix A.

Proposition 2. *For any outlet j with $\beta_j > 0$, if $\beta_i = 0$ for all i and*

$$\mathbb{E}[\delta_j + \xi_{ij} \mid \omega_c, \omega_0, i \text{ receives } j\text{'s offer}] = \mathbb{E}[\delta_j + \xi_{ij}],$$

then

$$\mathbb{E}[x_{ij}^* \mid i \text{ accepts } j\text{'s offer}] = \mathbb{E}[\delta_j + \xi_{ij}],$$

where the unconditional expectation is across searches.

Proof: See Appendix A.

The lemma shows that zero journalist slant preferences prevent acceptance from adding further selection on outlet preferences beyond what exists at the offer stage. The proposition’s conditional mean restriction then assumes no selection on mean outlet preference at the offer stage, leading to a continued lack of selection at the acceptance stage given zero journalist slant preferences.

3 Data

Our data is derived from three primary sources: articles shared by politicians on Twitter, full-text newspaper articles from ProQuest, and voter registration data from L2. We use the politician-shared articles to build a labeled training dataset, which allows us to train a machine learning model for predicting an article’s political slant. We use the full-text data on newspaper articles to track journalists as they move across outlets and to measure how the slant of their content varies as they move. And we use the voter registration data to determine the ground-truth partisan affiliation of journalists in our data set.

3.1 Articles Tweeted by Politicians

We construct a labeled dataset of news articles shared by either Republican or Democratic politicians. The labeled nature of this dataset (either Republican or Democratic) is important for training our measure of media slant later.

We start with a list of active Twitter accounts by politicians from ProPublica’s Politwoops dataset released in July 2019. We then scraped the tweets from each username listed—giving us the tweet history from more than 1,800 usernames associated with politicians.

From the dataset of politician tweets, we locate all URLs tweeted by politicians—giving more than 1 million unique tweeted URLs.¹¹ We then randomly sample 600,000 URLs and scrape the text associated with each URL. Not all URLs are successfully scraped. We obtain 348,821 URLs with some text. We further exclude URLs with either (a) 1,000 or less characters or (b) no author detected by our scraper. We use LLMs to identify and remove non-news content, and to assign IPTC media topics to news articles.¹² This gives a final dataset of 182,538 non-unique URLs with text. 54,433 of these URLs were shared by Republican politicians, and 128,105 of these

¹¹We exclude urls including “youtu.be,” “pscp.tv,” “facebook.com,” “instagram.com,” “patreon.com,” and “google.com.”

¹²Additional detail on our data processing can be found in Appendix Section B.1. Information on our topic assignments specifically can be found in Appendix Section B.3.

URLs were shared by Democratic politicians. Appendix Figure C13 summarizes the outlet-level distribution of these article tweets by the party of the politician who shared them.

3.2 ProQuest Full-Text News

We also use a full-text database of news articles from more than 279 US newspapers, including major US daily newspapers such as *The New York Times* and *The Wall Street Journal*. The full-text nature of the data enables us to improve upon bag-of-word or keyword-based approaches to measuring media slant (e.g., Gentzkow and Shapiro 2010) by applying recent machine learning advances that leverage textual context (e.g., BERT or RoBERTa), thereby allowing us to construct more accurate measures of slant at the article level. Moreover, the large number of outlets allows us to track journalists as they move across outlets and provides insights across both national and local newspaper markets.

We obtain the full text for all articles from the ProQuest US Newsstream (accessed via TDM Studio) published between January 2013 and December 2018. This provides more than 9 million articles. For most of the analysis, we restrict to articles labeled as “News” by ProQuest.

For articles with multiple authors on the same byline, we reshape the data to be at the author-article level. So, an article that is written by two authors would appear twice in our dataset—once for each author.¹³ (From here-on, when we refer to an “article,” we mean an “article-author” pair.) We restrict attention to authors with more than 10 articles in our data and perform various cleaning steps in an attempt to remove wire services and syndicated articles, and to handle outlet mergers.¹⁴

After the restrictions, we are left with a dataset of 9,005,189 articles from 33,571 authors published in 279 outlets. Table 1 provides summary statistics for our sample. Roughly a third of the authors in our dataset write for the same outlet throughout our analysis period, while the remaining two-thirds of authors switch outlets at some point. However, there appears to be heterogeneity in the share of movers versus non-movers across outlets. For example, many authors at *The Wall Street Journal* never write for another outlet, whereas the vast majority of authors at *The Chicago Tribune* switch outlets at some point. These differences may be a combination of differences in the use of freelancers versus staff writers, job stability, syndication agreements, and the use of wire services.

¹³We assume any individual with the same name is the same person, and we clean author names by replacing common female and male nicknames using the list of nicknames from <https://github.com/uguryi/abeR/tree/master/inst/extdata>, which is derived from Abramitzky et al. (2021).

¹⁴Additional details on duplicate removal and other data-processing steps appear in Appendix Section B.1. We combine evening and day editions, city-specific editions, online editions, and outlet mergers into a single outlet where known.

3.3 Voter Registration Data

We also use voter registration data from L2 to match journalists to their party registration status. To match journalists, we first apply a similar name cleaning procedure to the L2 data. We then assign a journalist a specific party if matched on first and last name to a set of voters with a unique *party*, after restricting to voters in the same state as the outlet headquarters. This allows a journalist to be matched to multiple voters in a given state, as long as those voters all have the same party affiliation. The party assignment is done at the author-outlet level—thus allowing authors to change party affiliation as they change outlets.

Of the 10,670 journalists we matched to voter registration records (covering 32 percent of the cleaned author sample), 51 percent registered as Democrats, 27 percent as non-partisan, and 21 percent as Republican. Appendix Table C1 compares all journalists to the subset linked to L2 records, while Appendix Table C2 summarizes journalist characteristics by party affiliation.

4 Measuring Political Slant and Stylized Facts

We use the politician-shared articles as a labeled dataset to fine-tune an existing transformer-based machine learning model (RoBERTa) to predict the political party of the politician sharing each article based on the article’s text. Our fine-tuned RoBERTa model performs well in the leave-out validation set and is a significant improvement over previous methods when evaluated on a separate human-labeled dataset of political news articles.

4.1 RoBERTa Model

To measure political slant, we use the RoBERTa model framework (Liu et al. 2019). The RoBERTa model builds on the previously successful BERT transformer architecture (Devlin et al. 2019). (See Vaswani et al. 2017 for more on transformers.)

BERT stands for Bidirectional Encoder Representations from Transformers and is an unsupervised pre-trained model. To start, BERT uses word embeddings for each token from a 30,000 token vocabulary taken from Wu et al. (2016). These word embeddings are then split into sequences at the sentence level (each sequence having two sentences). The BERT model is then trained under a dual objective function. First, the BERT model will randomly “mask” or hide certain tokens (i.e., words) and the training objective is to successfully predict the masked tokens based on the surrounding context. Second, the BERT model is tasked with predicting the next sentence in a sequence. A key contribution of BERT is the ability to use the left *and* right context simultaneously in making these masked word predictions, as opposed to the unidirectional or left-to-right predic-

tions used in many other language models (e.g., OpenAI’s GPT). The BERT model was pre-trained on data from the BooksCorpus (Zhu et al. 2015) and English Wikipedia, giving a combined total of 3.3 billion words.

After pre-training the BERT model on the masked word task, the model can be subsequently fine-tuned on various downstream tasks, such as next sentence prediction, classification, or question answering. When released in 2018, BERT achieved state-of-the-art performance on many of the standard NLP tasks including both sentence-level and token-level tasks. The base BERT model has 12 layers, a hidden size of 768, and 12 self-attention heads—giving a model with 110 million parameters. The large version of the BERT model has 340 million parameters.

RoBERTa, or robustly optimized BERT approach, is one of several refinements to the BERT model. It improves on BERT’s performance by pre-training on more data and larger batches, training on longer sequences, dynamically changing the masking pattern, and removing the next sentence prediction objective. These modifications resulted in significant improvements over BERT across multiple NLP tasks.

4.2 Fine-Tuning RoBERTa and Measurement of Slant

The RoBERTa model is a pre-trained model for masked word prediction. We use the base RoBERTa model available in the HuggingFace library (<https://huggingface.co/roberta-base>).

Our objective is to accurately predict the *perceived* political slant of an article. That is, how liberal or conservative would a reader think a given article is? To achieve this objective, we use our dataset of articles tweeted by politicians to fine-tune the RoBERTa model.

We label each article according to the party of the politician who tweeted it. We then randomly select 70 percent of the articles to be the training dataset and the remaining articles to be the validation dataset.¹⁵ To avoid class imbalance in training, we reweight observations so that articles tweeted by Democratic and Republican politicians receive equal total weight within each tweet month; see Appendix Section B.1.1 for detail.

Since the RoBERTa architecture assumes a sequence length of 512 tokens (where one token is roughly one word), we restrict the initial sections of each article when they are longer than the required sequence length and we “pad” articles that are shorter than required. Our fine-tuning task is a sequence classification task with two labels (Republican or Democratic). We use the default

¹⁵Since an article can be tweeted by more than one politician and we allow duplicate articles in our dataset in general, we exclude all articles in the validation dataset that match the text of an article in the training dataset. This helps prevent cross-contamination of our training and validation datasets.

training parameters from the HuggingFace library for RoBERTa (e.g., cross-entropy loss), except that we use: a batch size of 128, 20 percent attention dropout, 20 percent hidden dropout, a learning rate of 0.5×10^{-5} , a linear learning rate scheduler with 1000 step warmup, and 25 training epochs. These hyper-parameters were chosen by a manual search over values.

We choose the model weights that achieve the lowest cross-entropy loss in the validation dataset at the end of each epoch, which was achieved on the fourth epoch. The training loss was 0.262, the validation loss was 0.449, and the balanced accuracy in the validation dataset was 82 percent, with a ROC-AUC of 0.90. Appendix Figure C7 reports validation balanced accuracy by tweet month, and Appendix Table C3 summarizes the highest-contribution tokens in the fine-tuned model.¹⁶

Using the fine-tuned RoBERTa model, we then predict the partisanship of every article in our ProQuest dataset.¹⁷

4.3 External Validation of Slant Measure

In addition to the leave-out validation dataset of articles tweeted by politicians, we validate our measure of slant by comparing to human-coded measures of slant at the article level from Budak et al. (2016).

Budak et al. (2016) dataset contains a sample of over 10,000 political news articles from 15 popular news outlets in 2013. Each article was then rated by MTurk coders on a 5-point likert scale for its favorability towards the Republican Party and, separately, for its favorability towards the Democratic Party. Each article was assigned to two coders, one who observed the outlet name and one who did not. Following Budak et al. (2016), our validation measure uses the rating from the coder who did not observe the outlet name, with human-perceived partisanship defined as the relative difference in perceived favorability towards the Republican Party versus the Democratic Party.

For a random subset of these articles, we scraped the underlying text from the source URL and

¹⁶Appendix Table C3 also compares the highest-contribution tokens, model diagnostics, and key results across our baseline and two alternative models that use an LLM to anonymize geography and other potential location proxies in the politician-tweeted articles used as training data. See Appendix Section B.2 for details.

¹⁷One potential concern is that the model could learn non-slant text features predictive of outlet identity, rather than slant itself. To assess this concern, we compare out-of-sample predictive performance for models trained on article-stratified vs. outlet-stratified splits of a common sample; for the latter, a given outlet appears only in the training data or only in the held-out data for that split, but not both. Appendix Figure C8 reports the results. Across five split assignments, outlet-stratified mean balanced accuracy is 76.9 percent vs. 83.1 percent for article-stratified splits ($p = 0.139$ for the difference); mean ROC-AUC is 0.89 and 0.93, respectively ($p = 0.054$). Average performance is somewhat (but not always statistically significantly) lower for outlet-stratified metrics. This difference could reflect learning about outlet-specific features related or unrelated to human-perceived slant, or other differences between the splits, such as the composition of the held-out samples.

predicted the slant using three different methods (a) our RoBERTa slant measure described above, (b) a FastText model (Joulin et al. 2017) that we also train on the politician-tweeted articles, and (c) a phrase-based classifier following Gentzkow et al. (2019), re-estimated on the same article training sample as RoBERTa. We use its Republican posterior probability under equal party priors. Panel A of Figure 2 reports the Spearman rank correlation for each of the three text-based, machine-coded measures of slant and the human-coded slant measure from Budak et al. (2016). While all three machine-coded methods have a significant positive correlation with the human-coded measure, our RoBERTa-based slant measure has the highest rank correlation.

Panel B of Figure 2 reports results from a partitioning exercise where we examine the ability of each method to accurately partition the articles in half based on their relative partisan leanings. Our RoBERTa slant measure accurately categorizes over 70 percent of articles that are perceived to be clearly partisan by human coders—again, performing better than these other methods.

Overall, we believe these results suggest that both (a) our RoBERTa-based slant measure is performing well in capturing text-based variation that humans would perceive as political slant, and (b) our RoBERTa-based slant measure is a significant improvement over previous methods used in the literature.

Appendix C reports additional results comparing our model’s performance against human-coded articles, as well as comparisons to previous measures of slant or partisanship in the literature.

4.4 Stylized Facts

4.4.1 Sorting of Journalists

In Figure 3, we compare the average slant at an outlet to the share of partisan journalists matched in the L2 data who are Republicans. Overall, there is a strong, positive, and significant correlation between the average slant of outlets and the share of Republican journalists.

The New York Times and *The Wall Street Journal* are interesting reference points. While both outlets are located in New York City, and thus can pull from the same potential set of employees, the share of Republicans at *The Wall Street Journal* is roughly twice that of *The New York Times*. And, correspondingly, the average slant at *The Wall Street Journal* is to the right of the average slant at *The New York Times*.

4.4.2 The New York Times and The Wall Street Journal

We examine the distribution of author-level slant at *The New York Times* and *The Wall Street Journal* across all authors and separately for the set of authors who have written for both outlets.

Panel A of Figure 4 plots the distribution of author-level slant at *The New York Times* and *The Wall Street Journal*. Consistent with prior suggestion, we see that the average slant of all authors at *The New York Times* is shifted to the left relative to the average slant of authors at *The Wall Street Journal*.

Panel B of Figure 4 is the same as Panel A, except it restricts attention to the set of authors who have written for both outlets. Among this set of authors, we see that the distributions of slant at *The New York Times* and *The Wall Street Journal* are more similar. Through the lens of our model, the set of authors who have written at both outlets either have significantly different ideal points δ_i than the general set of authors at *The Wall Street Journal* and/or have large search-specific shocks ξ .

Lastly, Appendix Figure C1 examines the average slant of select politicians when writing op-eds at *The New York Times* and *The Wall Street Journal*. As we would expect, among these politicians, the average slant of Republican politicians is to the right of the distribution at both outlets relative to the average slant of Democratic politicians.

4.4.3 Trends in Slant

We also examine trends between 2013 and 2018 in slant at key outlets, as well as the newspaper market overall. Figure 5 shows a secular shift to the left over time in slant for key outlets as well as the market as a whole. While this trend is gradual for the market as a whole, the 2016 election marks a turning point for many national news outlets. Several national US dailies exhibit a rapid shift leftwards after the 2016 election including for *The Wall Street Journal*—an outlet typically viewed as more favorable towards Republicans. Of the major national US dailies examined, it is interesting to note that while *USA Today* had similar levels of slant to *The Washington Post* in 2013, it reaches parity with *The Wall Street Journal* in the aftermath of the 2016 election—highlighting how outlets differentially responded to the election. *USA Today* has a higher share of Republican journalists than the other outlets considered (see Figure 3).

5 Evidence from Journalist Movers

We now turn to examining the role of journalists in determining the slant of the content they produce.

We start by providing preliminary evidence on journalist content preferences using standard mover event study frameworks and AKM variance decompositions. We provide guidance on the extent to which we can consider these estimates causal, as opposed to descriptive, in light of our

model.

To make progress empirically, we assume that journalists and outlets are only heterogeneous in their bliss points δ and have common strength of preferences. That is, $\beta_i = \beta$ for all i and $\beta_j = 1$ for all j , where $\beta_j = 1$ is a normalization. Therefore,

$$x_{ij}^* = \frac{\beta}{\beta + 1} \delta_i + \frac{1}{\beta + 1} (\delta_j + \xi_{ij}).$$

Note that, in this case, the average observed slant at an outlet is also additively separable

$$\bar{x}_j = \frac{\beta}{\beta + 1} \bar{\delta}_{A(j)} + \frac{1}{\beta + 1} \delta_j + \frac{1}{\beta + 1} \bar{\xi}_j$$

where each barred term is calculated over the full sample by averaging author–article rows within outlet j . Recall that $A(j)$ is the portfolio of journalists working at firm j .

5.1 Mover Event Studies

Consider a mover from j to j' . Then

$$x_{ij'} - x_{ij} = \frac{1}{\beta + 1} [(\delta_{j'} + \xi_{ij'}) - (\delta_j + \xi_{ij})].$$

Ideally, we would be able to estimate

$$\hat{x}_{it} = \alpha_q + d_{m(i,t)} + \sum_{l=-T, l \neq -1}^{T-1} \rho_l [(\delta_{d(q)} + \xi_{i,d(q)}) - (\delta_{o(q)} + \xi_{i,o(q)})] \mathbf{1}_{\{t=l\}} + \varepsilon_{it},$$

where $\hat{x}_{it}$ is the estimated slant of the article at event time t for move q involving journalist i , $t = 0$ is the first article at the destination outlet, α_q are move-pair fixed effects, and $d_{m(i,t)}$ are year-by-month fixed effects. $o(q)$ and $d(q)$ denote the origin and destination outlets. We omit $t = -1$, so each ρ_l is measured relative to the last article at the origin outlet.

The difference between ρ_l before and after the move would be an estimate of $\frac{1}{\beta+1}$. However, the outlet preference terms $(\delta_{j(i,t)} + \xi_{i,j(i,t)})$ are unobserved. We estimate the following event-study specification.¹⁸

$$x_{it} = \alpha_q + \sum_{l=-T, l \neq -1}^{T-1} \rho_l [\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}] \mathbf{1}_{\{t=l\}} + \varepsilon_{it},$$

¹⁸Before estimation in the mover event-study and outlet-level-decomposition exercises (Section 6), we residualize article slant by publication-month means estimated once on the full eligible article universe. Throughout those exercises, x and $\bar{x}$ denote article outcomes and outlet means constructed from the resulting measure.

replacing the outlet preferences with observed average slant at the destination and origin outlets. We calculate both outlet means over the full sample using the author–article panel, but remove every article on which the moving journalist appears.¹⁹ The same destination-minus-origin gap scales every pre- and post-move event-time indicator, so the coefficients are expressed throughout as fractions of the observed outlet gap.

Let $m_j = \mathbb{E}[\delta_j + \xi_{ij}]$ denote outlet j ’s mean preference across searches. In general, $\mathbb{E}[\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}] \neq m_{d(q)} - m_{o(q)}$ and so our estimates $\hat{\rho}$ do not have precise mappings to the underlying model primitives. However, if journalists have zero slant preferences, selection into and out of outlets is balanced, and other technical conditions hold,²⁰ then the population benchmark is $\rho_l = 1$ for $l \geq 0$. Under our model and identifying assumptions, by testing for whether $\hat{\rho}_{t \geq 0} = 1$, we are testing for whether journalists have preferences over the ideological content which they produce, i.e., $\beta > 0$. Thus, while this exercise does not allow us to pin down the relative magnitude of β , it is still informative about the existence of journalist preferences. An important caveat is that unobserved shocks ξ still function as measurement error in the independent variable, which could attenuate our estimates of ρ .

In estimating the event study specification, we restrict to “consistent” moves which we define as when a journalist has written T or more articles consecutively at the origin outlet before writing T or more articles consecutively at the destination outlet.²¹ Our baseline uses $T = 5$; we retain $T = 10$ as a horizon robustness check. Lastly, to improve precision of our estimates, we weight observations by the square of the relative move size between outlets $(\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)})^2$. Figure 6 plots the distribution of move sizes $\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}$ for our estimating sample of consistent movers when $T = 5$ and $T = 10$. As we would like to see, the distribution of moves is approximately symmetric around zero and has many “large” moves.

Our event study estimates allow us to reject the hypothesis that $\rho_0 = 1$ and, under our model and identifying assumptions, reject the hypothesis that $\beta = 0$. Figure 7 reports the estimates of ρ from our event study specification when $T = 5$ and when $T = 10$. All preperiod coefficients in these two specifications are individually insignificant at the 5 percent level. Upon move, both specifications show a quick transition from approximately zero, reaching 0.77 at $t = 0$ in the baseline. The magnitude of this shift suggests journalists close roughly 77 percent of the observed slant

¹⁹For a coauthored article, we remove every author-expanded row associated with that article.

²⁰Proposition 2 establishes balance for newly accepted offers; see Appendix A.5 for the additional conditions.

²¹Prior to defining a “consistent” move, we drop outlet spells with fewer than five articles to reduce the potential influence of syndications and news wires. Appendix Tables C4 and C5 report journalist- and move-level summary statistics for the resulting mover samples.

gap between outlets when they move between outlets. Appendix Figure C15 shows a qualitatively similar pattern after restricting to articles with a single author.²² We see no evidence of learning behaviors—with the estimates being relatively flat across time in the post-move window of analysis. If we were to interpret our event study coefficient estimates as estimates of $\frac{1}{\beta+1}$, they would imply $\beta \approx 0.29$.

In Figure 8, Panel A repeats our estimation separately for the set of movers who move to a more conservative outlet (e.g., $(\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}) > 0$) and the set of movers who move to a more liberal outlet (e.g., $(\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}) < 0$). Panel B examines the relationship between the change in average outlet slant $\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}$ with the change in average journalist slant using pooled pre- and post-move mean changes, with the fitted line constrained through the origin. Both panels suggest a high degree of symmetry across the direction and size of journalist moves across outlets. Appendix Figure C17 reports analogous event-study estimates by party-affiliation group.

5.2 AKM Decompositions

In this section, we continue to examine the role of journalists across a broader set of movers using a variance decomposition of journalist and outlet fixed effects following Abowd et al. (1999, AKM).

To highlight the appropriate interpretation of our AKM estimates, we can write and re-arrange the optimal slant contract as

$$x_{ij}^* = \underbrace{\frac{\beta}{\beta+1}\delta_i}_{\gamma_i} + \underbrace{\frac{1}{\beta+1}\delta_j}_{\alpha_{j(i)}} + \underbrace{\frac{1}{\beta+1}\xi_{ij}}_{\varepsilon_{ij}}$$

and so,

$$x_{ij}^* = \gamma_i + \alpha_{j(i)} + \varepsilon_{ij}.$$

As explained in Abowd et al. (1999), the outlet fixed effects α_j are identified by the set of journalists moving between outlets. In order for our estimates of the journalist and outlet fixed effects (γ_i and $\alpha_{j(i)}$) to be unbiased, we need AKM's conditional mean independence assumption to hold

$$\mathbb{E}[\varepsilon_{ij}|i \in A(j)] = 0.$$

If the conditional mean independence assumption holds, then we can use the leave-out estima-

²²Appendix Figure C16 follows the main event-study specification after replacing article slant with the mean slant of the article's topic, calculated across all eligible articles using the same publication-month adjustment. The resulting coefficients are near zero. See Section B.3 for details on topic classification.

tor proposed by Kline et al. (2020, KSS) to consistently estimate the variance of the journalist fixed effects $V(\gamma_i)$, which is the combination of the variation in journalist ideal points and their relative strength of preferences

$$V(\gamma_i) = \left(\frac{\beta}{\beta + 1} \right)^2 V(\delta_i).$$

However, given the importance of selection in our model, we do not expect the conditional mean independence assumption to hold. As such, we interpret our AKM variance decompositions as a descriptive exercise familiar to many readers—not as estimates of underlying model primitives.

In practice, we estimate

$$\hat{x}_{ijt} = \gamma_i + \alpha_{j(i,t)} + \tau_t + e_{ijt}$$

to allow for systematic temporal variation in slant due to the timeliness of different topics, where each $\hat{x}_{ijt}$ is the slant of an article published by journalist i at outlet j in month t , γ_i are journalist fixed effects, $\alpha_{j(i,t)}$ are outlet fixed effects, τ_t are time fixed effects, and e_{ijt} is the error term.²³

Relative to our event study sample, we are less restrictive in the set of articles we include. We still drop outlet spells with fewer than five articles (to reduce the influence of syndicated articles and news wires) and journalists with more than 10 moves (to reduce the influence of freelancers), but make no other restrictions after the restrictions defined in Section 3. This will likely still include some freelancers, salaried journalists, and other writers. We use these observations directly in the AKM and KSS specifications, with publication-month fixed effects. In total, there are 30,504 journalists (with 4,091 of those being movers), 266 outlets, and 7,065,499 observations in the connected sample.²⁴

Table 2 reports the variance decomposition estimates using both the plug-in estimator (Panel A) as well as the leave-out variance estimator proposed by Kline et al. (2020), which addresses the limited mobility bias that is often present in these employee-employer transition datasets (Panel B). Both panels use the leave-out connected set of outlets. The different columns report estimates on various samples of the data.

Panel B is our preferred set of estimates. Our baseline specification (Column 1 of Panel B) estimates 16 percent of the variation in article slant is explained by journalist fixed effects, 19 percent is explained by outlet fixed effects, and 8 percent is explained by the covariance between journalist and outlet fixed effects. These results are reasonably robust when the estimation sample

²³In Appendix B.4, we use the BLM framework of Bonhomme et al. (2019) to allow for interactions between journalist types and outlet classes. The interaction residuals account for 0.06% of observed calendar-adjusted article variance.

²⁴In the AKM analysis, movers are journalists observed at more than one retained outlet.

includes both movers and stayers (e.g., Columns 1, 4, and 5 of Panel B), with the exception of the specification where we replace outlet fixed effects with parent company fixed effects (Column 10 of Panel B) or restrict to “top” outlets (Column 6 of Panel B). We report complementary evidence on ownership changes in the appendix.²⁵

When our estimation sample is restricted to a set of movers only (Columns 2, 3, 7, 8, and 9 of Panel B), the estimated shares attributed to journalist and outlet fixed effects vary across the alternative mover samples. However, even in the most restrictive set of movers (Columns 8 and 9 of Panel B), journalist fixed effects still explain 15 percent and 13 percent of the variation in slant, respectively.

When restricting the sample to spells around large moves (absolute outlet slant gaps greater than 0.1), Column (3) still estimates a role for journalist fixed effects. Column (8), which uses the large-move subset of the $T = 5$ event study sample defined by the event-study gap cutoff, reports a larger role for journalist fixed effects than Column (3), and a smaller role for outlet fixed effects.

It is also interesting to note that both sorting and the variance of outlet fixed effects is larger among low-tier outlets. When restricting to the top 20 outlets by circulation, our estimated correlation between γ_i and α_j is 0.085 (Column 6, Panel B) compared to 0.245 in our full sample (Column 1, Panel B). These results could be consistent with a model where top-tier outlets differentiate themselves vertically, whereas low-tier outlets differentiate themselves horizontally.

Figure 9 plots the distribution of estimated journalist fixed effects for various subsets of journalists. Panel A reports the distribution for Republican, Democratic, and non-partisan journalists, where we see that the distribution of fixed effects for Republican journalists is significantly to the right of other journalists. Panels B and C suggest that women have more left-leaning fixed effects than men and non-white journalists have more left-leaning fixed effects than white journalists.

5.3 Slant Under Counterfactual Journalist Party Compositions

These heterogeneous AKM estimates by party affiliation also allow us to conduct a simple partial-equilibrium counterfactual exercise, analyzing how average slant would change under different distributions of journalists’ party affiliations. We can decompose our average estimated slant as:

$$\bar{x} = \left(\sum_p s_p \bar{\gamma}_p \right) + \bar{\alpha} + \bar{\tau}$$

²⁵Appendix Figures C18, C19, and C20 report ownership-change event-study analyses and the underlying collapsed DID variation. Appendix Section B.5 describes the specifications. These exercises examine whether target-outlet slant changes around ownership changes in ways related to the acquiring owner’s portfolio.

where s_p and $\bar{\gamma}_p$ denote the share and average journalist fixed effect among those with party affiliation p , while $\bar{\alpha}$ and $\bar{\tau}$ denote overall averages of outlet and time fixed effects.²⁶ This allows us to analyze changes in mean slant as journalist party-affiliation shares vary, holding fixed the average journalist fixed effect by party as well as the overall average outlet and time fixed effects.

Appendix Figure C21 reports the resulting counterfactual means together with benchmark observed means for selected outlets. Because the reweighting exercise is defined over journalists with Democratic, Republican, or non-partisan affiliations, the natural baseline is the observed mean slant in the non-missing-party subsample. Relative to that baseline, moving from the observed party distribution to an even one-third Democratic, one-third Republican, and one-third non-partisan distribution shifts mean slant in a more conservative direction by 0.003 points, or only 0.01 standard deviations of article-level slant. Even the most extreme comparison, moving from an all-Democratic to an all-Republican journalist distribution, changes mean slant by only 0.035 points, or 0.12 standard deviations. These shifts are therefore modest relative to the variance in slant across articles and to the dispersion in benchmark outlet means shown in the figure. In this reweighting exercise, changes in the composition of journalist party affiliations generate modest changes in average slant, holding within-party mean journalist effects, outlet effects, and time effects fixed.²⁷

6 Outlet-level Decompositions

Both the event study and AKM results are limited by their ability to address the selection driven by the search-specific shock process in our model. In this section, we relax the conditional mean independence assumption by focusing on estimating the role of journalists in driving the observed variation in *outlet-level* slant.

6.1 Model-Driven Specification

Recall that the average observed slant at an outlet is

$$\bar{x}_j = \frac{\beta}{\beta + 1} \bar{\delta}_{A(j)} + \frac{1}{\beta + 1} \delta_j + \frac{1}{\beta + 1} \bar{\xi}_j,$$

²⁶By construction, the residual (e_{ijt}) averages to zero across articles.

²⁷Note that this partial equilibrium exercise holds the supply of outlets and the number of articles per outlet fixed. Changing journalist ideologies could lead to the expansion or entry of ideologically aligned outlets, changes in journalist-outlet bargaining strength, or other effects which might plausibly generate changes larger than our estimates.

Let

$$\psi_j = \frac{\beta}{\beta + 1} \bar{\delta}_{A(j)}$$

denote the “journalist effect” at outlet j .

Our goal is to estimate $\text{Var}_p(\psi_j)$.²⁸

Our identification strategy uses a movers identification strategy to difference out most of the non- ψ_j components, and we provide conditions under which we can ignore the remaining selection. Let q index observed moves. For move q , let $i(q)$ denote the journalist and let $j = o(q)$ and $j' = d(q)$ denote the origin and destination outlets. Write ξ_{qj} and $\xi_{qj'}$ for the search shocks on the corresponding origin and destination contracts. Then

$$x_{i(q)j'} - x_{i(q)j} = \frac{1}{\beta + 1} [(\delta_{j'} + \xi_{qj'}) - (\delta_j + \xi_{qj})],$$

and we can write the difference in average slant between outlets j' and j as

$$\bar{x}_{j'} - \bar{x}_j = \frac{\beta}{\beta + 1} (\bar{\delta}_{A(j')} - \bar{\delta}_{A(j)}) + \frac{1}{\beta + 1} (\delta_{j'} - \delta_j) + \frac{1}{\beta + 1} (\bar{\xi}_{j'} - \bar{\xi}_j).$$

Let $y_q = (\bar{x}_{j'} - \bar{x}_j) - (x_{i(q)j'} - x_{i(q)j})$. Combining the two equations above, we get

$$y_q = \frac{\beta}{\beta + 1} (\bar{\delta}_{A(j')} - \bar{\delta}_{A(j)}) + \frac{1}{\beta + 1} (\bar{\xi}_{j'} - \bar{\xi}_j) - \frac{1}{\beta + 1} (\xi_{qj'} - \xi_{qj}),$$

which we can re-arrange and relabel as

$$y_q = \underbrace{\frac{\beta}{\beta + 1} \bar{\delta}_{A(j')}}_{\psi_{j'}^d} - \underbrace{\frac{\beta}{\beta + 1} \bar{\delta}_{A(j)}}_{\psi_j^o} - \underbrace{\frac{1}{\beta + 1} [(\xi_{qj'} - \bar{\xi}_{j'}) - (\xi_{qj} - \bar{\xi}_j)]}_{e_q}$$

to get

$$y_q = \psi_{j'}^d - \psi_j^o - e_q. \quad (1)$$

Note that $\psi_j^d = \psi_j^o$ for a given outlet j .

Intuitively, $x_{i(q)j'} - x_{i(q)j}$ differences out journalist $i(q)$ ’s fixed preferences $\delta_{i(q)}$ that remain the same across outlets, thereby leaving the difference in outlet components $(\delta_{j'} + \xi_{qj'}) - (\delta_j + \xi_{qj})$. Then, by subtracting $x_{i(q)j'} - x_{i(q)j}$ from $\bar{x}_{j'} - \bar{x}_j$, we difference out the outlet components that

²⁸Throughout this subsection, baseline outlet-level variances and covariances use fixed weights p_j proportional to each outlet’s full-sample number of articles and the same fixed connected outlet support.

remain the same for all journalists (i.e., δ_j). Thus, leaving the difference in journalist components ψ_j between origin and destination outlets, as well as the difference in the structural errors for movers and non-movers at each outlet, $[(\xi_{qj'} - \bar{\xi}_{j'}) - (\xi_{qj} - \bar{\xi}_j)]$.

We would like to estimate equation (1) using origin and destination fixed effects among the sample of movers and to construct $\text{Var}_p(\hat{\psi}_j^d)$, an estimate of $\text{Var}_p(\psi_j)$.

Two empirical issues remain.

First, it is well known that plug-in estimates of quadratic forms, such as variances, are biased (e.g., Kline et al. 2020). Motivated by this concern, we base our estimator on the covariance between origin and destination fixed effects rather than either estimated variance.

Second, our estimates of ψ_j^d and ψ_j^o are potentially biased themselves if they are correlated with e_q —which is likely under any form of journalist preferences over slant or selection in job transitions. We postulate a set of conditions for which we can derive consistent estimates for $\text{Var}_p(\psi_j)$ below.

Let T_{qj}^o be an indicator for $o(q) = j$, and let T_{qj}^d be an indicator for $d(q) = j$. The supported move pairs are

$$\mathcal{P} = \{(j, k) : \Pr(o(q) = j, d(q) = k) > 0\}.$$

Assumption 1. (*Pair-Conditional Difference Balance*)²⁹ For every $(j, k) \in \mathcal{P}$, conditional on the fixed full-sample article-production weights w ,

$$\mathbb{E}[(\xi_{qk} - \bar{\xi}_k) - (\xi_{qj} - \bar{\xi}_j) \mid o(q) = j, d(q) = k, w] = 0.$$

Let N_m denote the number of observed moves, let $\mathbf{Y}$ stack y_q , and let T_j^o and T_j^d stack the corresponding indicators across moves. Consider the linear model

$$\mathbb{E}[\mathbf{Y}|\mathbf{T}] = \mathbf{T}\boldsymbol{\theta}$$

where $\mathbf{T} = [-T_1^o, \dots, -T_J^o, T_1^d, \dots, T_J^d]$, a $N_m \times (2J - 1)$ matrix and $\boldsymbol{\theta}$ is a $(2J - 1)$ parameter vector which we can partition as $\boldsymbol{\theta} = [\boldsymbol{\theta}^o, \boldsymbol{\theta}^d]'$. Then

$$\hat{\boldsymbol{\theta}} = (\mathbf{T}'\mathbf{T})^{-1}\mathbf{T}'\mathbf{Y}$$

²⁹This condition permits nonzero, pair-specific selection in both shock deviations as long as their conditional means balance. When the weights are fixed independently of the shocks, the standard AKM conditions of conditional mean independence and exogenous mobility imply this condition.

is the OLS estimator. When the same moves estimate both role effects, shared sampling errors can enter their covariance and bias the plug-in estimate upward. We therefore cross-fit the covariance (see also Kline et al. 2020). For each repeated split s , we partition journalists into two groups, A and B , and estimate equation (1) separately in each group. Let $\hat{\theta}^{r,h,s}$ denote the estimate for role $r \in \{o, d\}$ from group $h \in \{A, B\}$. Our estimator is

$$\hat{C}_{CF} = \frac{1}{S} \sum_{s=1}^S \frac{1}{2} \left[\text{Cov}_p(\hat{\theta}^{d,A,s}, \hat{\theta}^{o,B,s}) + \text{Cov}_p(\hat{\theta}^{d,B,s}, \hat{\theta}^{o,A,s}) \right].$$

Each covariance therefore combines role-effect fits estimated on disjoint sets of movers.

Proposition 3. *If pair-conditional difference balance holds, $\hat{C}_{CF}$ is a consistent estimator for $\text{Var}_p(\psi_j)$.*³⁰

Proof: See Appendix A.

Remark 1. $\text{Var}_p(\hat{\theta}^d)$ or $\text{Var}_p(\hat{\theta}^o)$ would also be consistent estimators for $\text{Var}_p(\psi_j)$, as would the full-sample plug-in covariance $\text{Cov}_p(\hat{\theta}^d, \hat{\theta}^o)$. We prefer the cross-fit covariance in finite samples because the disjoint halves avoid reusing the same mover observations in both role-effect fits while preserving the same probability limit.

The proposition conditions on the realized full-sample weights. It also applies when estimated weights converge to fixed population production shares and the stated moment and regularity conditions hold jointly.

The pair-conditional difference balance assumption is admittedly strong. However, it is weaker than the standard conditional mean independence and exogenous mobility assumptions commonly used in AKM and related models. Intuitively, while the standard AKM assumptions imply no selection on the search-specific shocks, our assumption allows movers on a given route to be selected on both their origin and destination shocks, but requires those two forms of selection to balance on average.

Agreement between the origin and destination estimates is a necessary implication of the model and can provide a partial diagnostic.³¹

Under certain conditions, $\text{Var}_p(\psi_j)$ is a lower bound for the variance in observed outlet-level slant driven by journalist effects *and* selection.

³⁰See Appendix A for normalization details and additional regularity conditions.

³¹Disagreement may reflect a failure of pair-conditional difference balance or sampling error, while agreement alone does not establish the assumption.

Assumption 2. (Non-negative Covariance) The covariance between the average journalist ideal point and the average structural error at outlet j is non-negative. That is, $\text{Cov}_p(\bar{\delta}_{A(j)}, \bar{\xi}_j) \geq 0$.

Corollary 1. If the pair-conditional difference balance and non-negative covariance assumptions hold, then $\text{Var}_p(\psi_j + \frac{1}{\beta+1} \bar{\xi}_j) \geq \text{Var}_p(\psi_j)$ and $\hat{C}_{CF}$ consistently estimates the lower-bound value $\text{Var}_p(\psi_j)$.

Proof: Note that

$$\text{Var}_p(\psi_j + \frac{1}{\beta+1} \bar{\xi}_j) = \text{Var}_p(\psi_j) + \text{Var}_p(\frac{1}{\beta+1} \bar{\xi}_j) + 2\text{Cov}_p(\frac{\beta}{\beta+1} \bar{\delta}_{A(j)}, \frac{1}{\beta+1} \bar{\xi}_j) \geq \text{Var}_p(\psi_j),$$

with the last inequality holding due to the non-negative covariance assumption. Proposition 3 then implies $\hat{C}_{CF}$ consistently estimates the lower-bound value $\text{Var}_p(\psi_j)$.

6.2 Results

To visualize each of these differences in y_q , we partition outlets into above- and below-median groups based on average slant using our full dataset and compute $\bar{x}_{d(q)}$ and $\bar{x}_{o(q)}$. We then compute $x_{i(q),d(q)}$ and $x_{i(q),o(q)}$ for the set of moves from above-median outlets to below-median outlets, and vice-versa, from our baseline $T = 5$ event study dataset as used in Section 5.1.³² (Note that this implicitly restricts to larger moves.)

Figure 10 plots the results. The “Outlet Gap” in each panel of Figure 10 reports $(\bar{x}_{j'} - \bar{x}_j)$, the difference in average slant between destination and origin outlet. The “Change on Move” in each panel reports $(x_{ij'} - x_{ij})$, the average change in slant for a journalist that moves from the origin to the destination outlet.

If outlets in the set of above-median outlets have homogeneous preferences (and likewise for below-median outlets), then the difference between the “Outlet Gap” and the “Change on Move” for each panel would give us estimates of ψ^d and $-\psi^o$ for one set of outlets normalized relative to the other set. Using the below-median outlets as the reference group, our example implies $\hat{\psi}^d = (0.25 - 0.17) = 0.08$ and $\hat{\psi}^o = -(-0.26 + 0.17) = 0.09$ for the above-median set of outlets. Together, the two role effects account for about 33 percent³³ of the difference in average slant between the two outlet groups.

³²This descriptive construction is distinct from the event-study regression estimand, which uses leave-focal-author-outlet gaps and squared-gap weights.

³³ $(0.08 + 0.09) / [0.25 - (-0.26)] \approx 0.33$.

The results in Figure 10 are from a restrictive set of authors and estimate first-moment differences across sets of outlets. Table 3 provides our cross-fit variance decomposition, $\widehat{C}_{CF}/\text{Var}_p(\bar{x}_j)$, using outlets identified in every split half.

Our baseline estimate in Column 1 of Table 3 suggests 14 percent of the outlet-level variation in slant can be explained by journalist effects, with a 95 percent confidence interval of 11.3–18.9 percent from a bootstrap that reweights journalist clusters.³⁴ A wild cluster bootstrap test of the model-implied equality between the origin and destination effects yields $p = 0.209$ (see Appendix B.6).

We can compare our baseline to the estimated 20 percent of observed variation in outlet-level average slant that is explained by consumer demand in Gentzkow and Shapiro (2010). Although the two estimates are not perfectly comparable, taken at face value they suggest that journalist effects are nearly as important as consumer demand in explaining the variation in observed slant at the outlet level.

7 Modeling Selection

In this section, we model selection directly, adapting the framework of Section 2 into an estimable model by restricting preferences, grouping outlets into classes, and using nested stay/move choices.

Journalist i has a slant ideal point $\delta_i \in (0, 1)$ and aversion to mismatch β ; outlets are grouped into $K = 5$ classes,³⁵ class k having a median preference $\delta_k \in (0, 1)$ before selection, and outlet aversion is normalized to $\beta_j = 1$. The optimal contract offered at article choice t by class k is

$$x_{ikt}^* = \frac{\beta \delta_i + \delta_k + \xi_{ikt}}{\beta + 1}, \quad (2)$$

where $\delta_k + \xi_{ikt} = \Lambda(b_k + \zeta_{ikt})$, $b_k = \log(\delta_k/(1 - \delta_k))$, and $\Lambda(u) = e^u/(1 + e^u)$. This is the optimal contract of Proposition 1 with $\beta_i = \beta$ and $\beta_j = 1$. The log-odds shock ζ_{ikt} keeps offers and contracts between 0 and 1; ξ_{ikt} is the corresponding slant-scale deviation from δ_k .

We also write:

$$\delta_i = \Lambda(a_{p(i)} + \tau_{\delta, p(i)} v_i), \quad v_i \sim N(0, 1); \quad \zeta_{ikt} \stackrel{\text{iid}}{\sim} N(0, \tau_\zeta^2), \quad (3)$$

³⁴Without cross-fitting, the plug-in estimate is 15.5 percent on its natural 238-outlet connected support and 18.0 percent on the same 154-outlet support as the cross-fit estimate. The equal-outlet-weight cross-fit estimate is 15.7 percent, with a 95 percent confidence interval of 12.0–20.8 percent.

³⁵We group outlets with similar distributions of journalists' average slant during their spells at the outlet. This captures both the level of slant and the variation within an outlet. The classes are ordered from most liberal to most conservative by their average slant.

where $p(i) \in \{\text{Dem, Rep, NP, NA}\}$ ³⁶ is journalist i 's party, a_p and $\tau_{\delta,p}$ capture party-specific heterogeneity in slant. The type δ_i is fixed over time. Before each article is written, one shock ζ_{ikt} is drawn for every outlet class, and the journalist observes these draws before deciding whether to stay or to move to another outlet. If they move to k , their contract carries the accepted ξ_{ikt} throughout their time at that outlet. Their own match is not redrawn while they stay.

For a spell entered at t in class k , observed article slant contains a month-level unobservable and measurement error: $x_{ia} = x_{ikt}^* + \mu_{m(i,a)} + e_{ia}$, $e_{ia} \sim N(0, \sigma_e^2)$. Write $\hat{x}_{ia} = x_{ia} - \mu_{m(i,a)}$ for calendar-adjusted article slant.

For each article, journalist i chooses among $\{\text{stay}\} \cup \{\text{move to class } 0, \dots, 4\}$, with systematic utilities:

$$\begin{aligned} u_{it}^S &= s + \kappa_n \log(1 + n_{it}) - \beta (\bar{x}_{it} - \delta_i)^2 & (\text{stay}) \\ u_{ikt}^M &= \alpha_k + \gamma \mathbf{1}\{k = c_{it}\} - \kappa_d \log(1 + d_{ikt}) \\ &\quad - \frac{\beta}{(\beta+1)^2} [\delta_i - (\delta_k + \xi_{ikt})]^2 & (\text{move to } k) \end{aligned} \quad (4)$$

with c_{it} the current outlet class and γ a same-class premium. The stay utility rises with prior article count in the current spell (n_{it}).³⁷ d_{ikt} is the distance in km from the current outlet's city to the nearest outlet of class k (own outlet excluded), so κ_d captures the cost of moving to a distant outlet.

The coefficient $\beta/(\beta+1)^2$ is the cost for the journalist of writing slant at an outlet of class k evaluated at the anticipated contract: moving to k means writing x^* from (2) and $\beta(x^* - \delta_i)^2 = \frac{\beta}{(\beta+1)^2} [\delta_i - (\delta_k + \xi_{ikt})]^2$. In estimation, we proxy the current contract by $\bar{x}_{it}$, the observed calendar-adjusted mean over the full current spell. This proxy stays fixed within the spell.

We use a nested logit with one stay option and a common nest of five destinations, with nesting parameter $\lambda \in (0, 1]$. Writing $I_{it} = \lambda \log \sum_{h=0}^4 \exp(u_{iht}^M / \lambda)$, the choice probabilities are

$$P(\text{stay}) = \frac{\exp(u_{it}^S)}{\exp(u_{it}^S) + \exp(I_{it})}, \quad P(k) = \frac{\exp(I_{it})}{\exp(u_{it}^S) + \exp(I_{it})} \frac{\exp(u_{ikt}^M / \lambda)}{\sum_{h=0}^4 \exp(u_{iht}^M / \lambda)}.$$

Ideology plays three roles: (1) in writing through $\beta/(\beta+1)$; (2) in whether to move through

³⁶NP=Non partisan, NA corresponds to journalists that we could not match to the voter file, either because of name discrepancy or because they are not registered to vote. We treat them as their own party.

³⁷Tenure is unknown for spells already in progress at the start of the panel; for these articles the tenure term is replaced by a separately estimated stay level s_c . These spells account for most articles, since every journalist present at the panel's start begins in one.

$\beta/(\beta+1)^2$; and (3) in where to move through $\beta/((\beta+1)^2\lambda)$.

Estimation

Estimation proceeds in two steps. The first step recovers the amount of measurement noise σ_e through stayers; the remaining 24 parameters $\theta = (\beta, \{a_p, \tau_{\delta,p}\}_p, b_1, \dots, b_4, \lambda, \gamma, \alpha_1, \dots, \alpha_4, s, \kappa_n, s_c, \kappa_d, \tau_\zeta)$ are estimated by maximum simulated likelihood with σ_e held fixed. We normalize the class-0 location b_0 to recover the observed mean slant at class-0 outlets,

$$\bar{x}_0 = \frac{\beta}{\beta+1} (\sum_p s_{0p} E[\delta_i | p]) + \frac{1}{\beta+1} E[\Lambda(b_0 + \zeta)], \quad (5)$$

where s_{0p} are the class-0 party shares. The year-month fixed effects μ_m are absorbed by residualizing article slant on the calendar month before estimation.

First Step Articles written by stayers within a spell belong to the same contract, so all the dispersion around the spell mean is noise; the within-spell variance of article slant therefore estimates the noise variance,

$$\hat{\sigma}_e^2 = \frac{\sum_\ell \sum_{a \in \ell} (\hat{x}_{ia} - \bar{x}_\ell)^2}{\sum_\ell (n_\ell - 1)}, \quad (6)$$

where ℓ runs over all journalists' spells, n_ℓ is the number of articles in spell ℓ , and $\bar{x}_\ell$ is its calendar-adjusted mean.

Second Step Conditional on the accepted draw, all articles in an entered spell are drawn around the same contract. Let m_{it} indicate a move at article t to destination k_{it} , $\mathcal{A}_{it}$ its entered spell's articles, and ζ_{it} the five offer shocks. The likelihood can be written as:

$$L_i(\theta) = \int \prod_t \left[\int P(\text{stay} | \delta, \zeta)^{1-m_{it}} \right. \\ \left. \times \left[P(k_{it} | \delta, \zeta) \prod_{a \in \mathcal{A}_{it}} f(\hat{x}_{ia} | k_{it}, \delta, \zeta) \right]^{m_{it}} dG(\zeta) \right] dF_{p(i)}(\delta), \quad (7)$$

where f is the normal density of calendar-adjusted article slant around the accepted contract, F_p is the ideal-point distribution implied by (3), and G the distribution of the offer draws.³⁸

The component of a journalist's slant that persists across their own spells recovers the journalists' component $(\frac{\beta}{\beta+1})^2 \text{Var}(\delta)$; since δ_i enters writing with weight $\beta/(\beta+1)$ but choices with weight $\beta/(\beta+1)^2$, comparing how strongly that persistent component predicts writing versus

³⁸The mobility likelihood conditions on the observed initial outlet and treats spell means as fixed. Spells with unobserved entry contribute to mobility but have no writing-density term.

where journalists move separates β from the dispersion of δ . Finally, the variation coming from movers entering the same outlet class with the same previous contract but who write at different levels there identifies τ_ζ . We report analytical standard errors, clustered at the journalist level.

Estimates

We estimate that journalists carry a weight of $\beta/(\beta + 1) = 0.40$ in the slant contract. The gap in mean ideal points between journalists registered with the Democrats and with the Republicans is 0.10 in slant units, which is relatively small compared to the 0.71 distance between the most liberal and the most conservative outlet classes; non-partisan journalists and those we could not match to the voter file are closer to Democrats than to Republicans. The small nesting parameter λ makes the destination choice, conditional on moving, highly sensitive to journalists' ideology, while the decision to move at all is dominated by the mobility shock, with ideology playing a more moderate role.

At the estimated parameters, we can obtain the average accepted log-odds draw among movers, $E[\zeta \mid \text{accepted offer}] = +0.04$; it varies from -0.17 at the most conservative class to $+0.30$ at the most liberal class. The mean absolute value of $E[\zeta \mid \text{accepted offer}]$ across movers is 0.24: each journalist accepts draws closer to their own ideal, liberal journalists lower ones and conservative journalists higher ones.

Appendix Table C11 decomposes calendar-adjusted article slant in the model's stationary distribution at observed party composition and fixed outlet-class shares. The journalist, outlet-class baseline, and accepted offer deviation components have variances equal to 6.5%, 24.3%, and 16.8% of total model variance; independent article noise accounts for 50.5%. All three signed covariance terms are reported separately.³⁹

To combine the direct journalist component with associated offer selection, we calculate

$$\text{Var}\left(\frac{\beta}{1+\beta}\delta_i + \frac{1}{1+\beta}E[\xi_{ikt} \mid \delta_i]\right).$$

When calculated over the model's stationary distribution of published articles, this journalist component accounts for 10% of the total model-implied variance of article-level slant.

³⁹The AKM shares use observed raw article variance, while the BLM shares use observed calendar-adjusted article variance in its first-move sample.

Counterfactuals

We conduct a few counterfactual scenarios, changing the party composition of the journalist pool and assessing the extent to which that impacts the distribution of slant. We implement extreme counterfactual scenarios where we replace the population of journalists to be 100% Democratic, 100% Republican, or 100% non-partisan, as well as more realistic ones with equal thirds of Democrats, Republicans, and non-partisans. We let journalists of each type move according to the estimated process until the allocation across outlets reaches an equilibrium, holding outlet positions and all estimated parameters fixed. We report two types of counterfactuals: (A) *outlet sizes fixed* and (B) *sizes free* (outlets hire whoever comes, so the allocation of jobs across classes responds to the pool). Figure 11 reports the change in mean slant under each scenario and its decomposition into the direct journalist effect, selected offers, and (under (B)) outlet-class composition. Note that under (B), even without changing the pool of journalists, we would get a small change in mean slant, since the observed distribution of journalists across outlets is not an equilibrium, so we compare the counterfactual results changing the pool to the equilibrium we would get without changing the pool.

Three patterns stand out. First, with outlet sizes fixed, changing the composition of journalists only marginally impacts equilibrium slant: even the full 100% Dem to 100% Rep swing is 0.057 in slant (0.21 article-level standard deviations).⁴⁰ Second, when letting outlet size vary freely, the same swing is 0.081 (0.29 standard deviations), with the outlet-class composition channel accounting for 32% of the 100% Rep effect as the share of journalists at the two most conservative classes rises from 26% to 35% of employment. Third, outlet polarization, measured by comparing classes, changes very little — the gap between the most conservative and the most liberal class stays between 0.39 and 0.43 across every scenario and closure — so journalists’ partisanship contributes little to outlet polarization.

8 Conclusion

This paper makes three primary contributions.

First, we provide an important advance in measuring media slant—with our fine-tuned RoBERTa model achieving a significant improvement in measuring human-perceived political slant relative to prior methods—and document new stylized facts regarding media slant.

⁴⁰Changing to a balanced 1/3 each of Democrat-, Republican-, and non-partisan-written articles shifts average slant by just 0.04 article-level standard deviations.

Second, we provide novel evidence on the role journalists play in determining the political slant of the news they produce. Under our model and identifying assumptions, our estimates reject the hypothesis that journalists have zero ideological preferences over the content they produce and imply that 10 percent of variation in slant can be explained by journalists. Our estimates also allow us to calculate counterfactual mean slant under alternative journalist party compositions. We estimate that shifting from the current distribution of journalist party affiliations to a more even or more representative distribution would have relatively modest effects on average slant, relative to the overall observed variation in slant and holding the supply of outlets fixed.

Third, to the extent our results speak to worker preferences over the slant they produce, they provide evidence on the importance of non-pecuniary worker preferences in driving sorting across firms.

There are several caveats to our analysis. First, our analysis requires strong assumptions about the selection of journalists across outlets. Violations of these assumptions will bias our estimates. Second, our notion of the role of journalists focuses on journalists as individuals. Group dynamics or the spillover effect of *other* journalists, would be attributed to the outlet. If these dynamics are important, our counterfactual exercises would be understating the role of journalists. Our estimates do not fully separate ideological preferences from other persistent journalist attributes. Third, we treat outlet preferences as a single composite, which bundles the influences of media owners, advertisers, editors, and readers, among others. Future research should continue to examine the effects of these potential limitations on our results.

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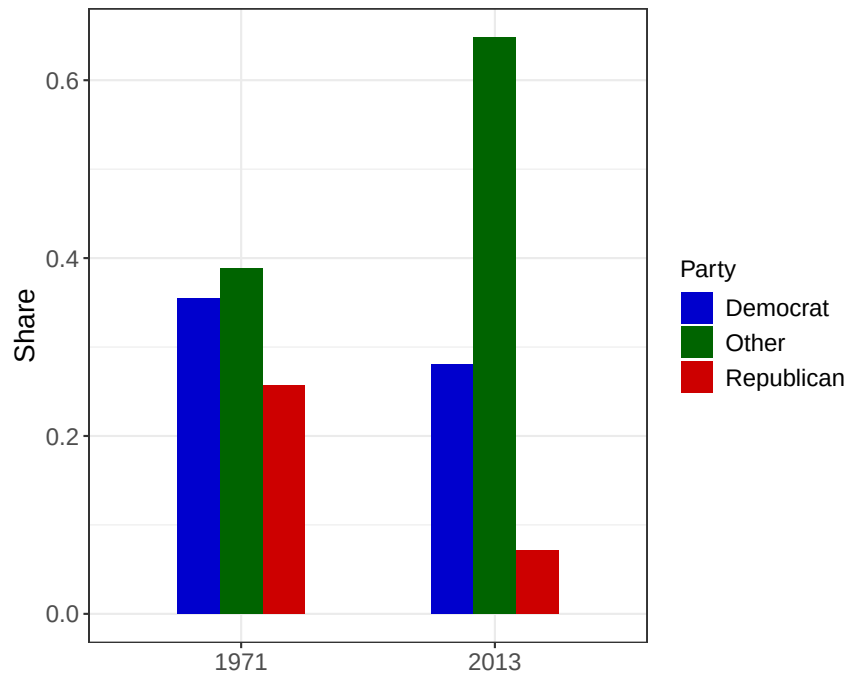
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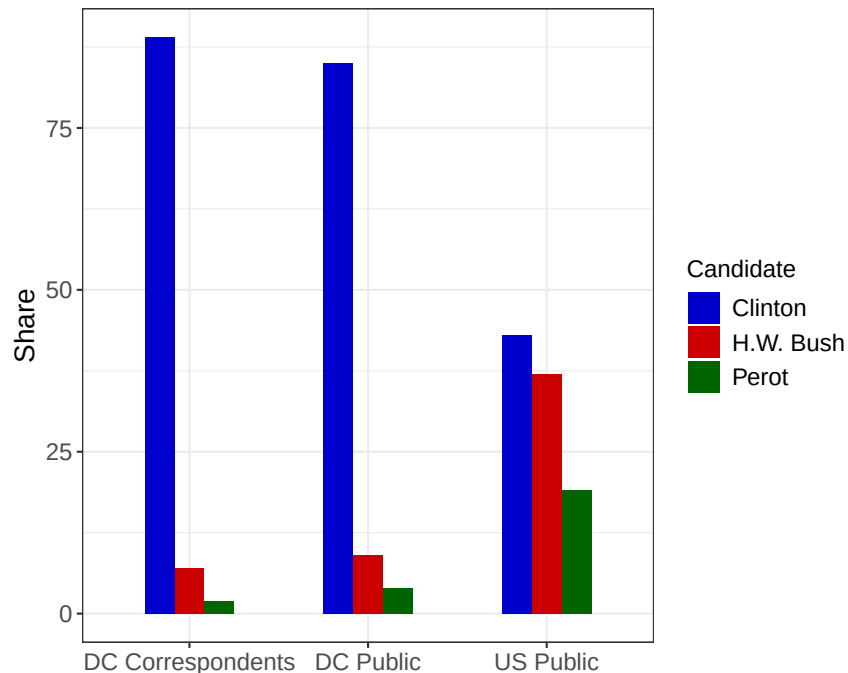
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Figure 1: Partisan Distribution of Journalists

Panel A: Surveyed Journalists' Self-Reported Partisan Identification (Weaver et al. 2019)



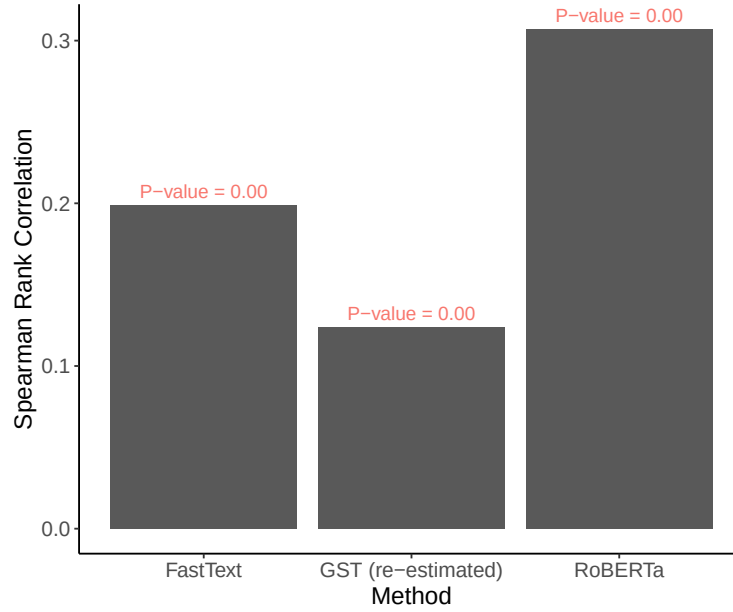
Panel B: Surveyed Washington Correspondents' Self-Reported Vote Choice in 1992 (Povich 1996)



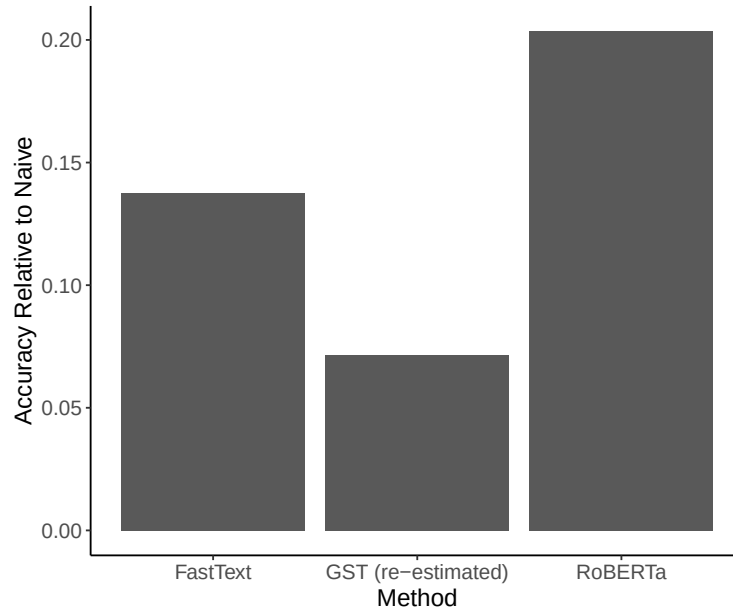
Notes: Panel A reports the share of surveyed journalists in 1971 and 2013 that self-reported as a Democrat (blue), a Republican (red), or as non-partisan or a third party (green). Data is taken from Weaver et al. (2019). Panel B reports the self-reported vote choices of Washington Correspondents from Povich (1996), as well as information on vote choices for the District of Columbia and the US population as a whole.

Figure 2: Comparing to Human Perceived Slant

Panel A: Non-parametric Rank Correlation

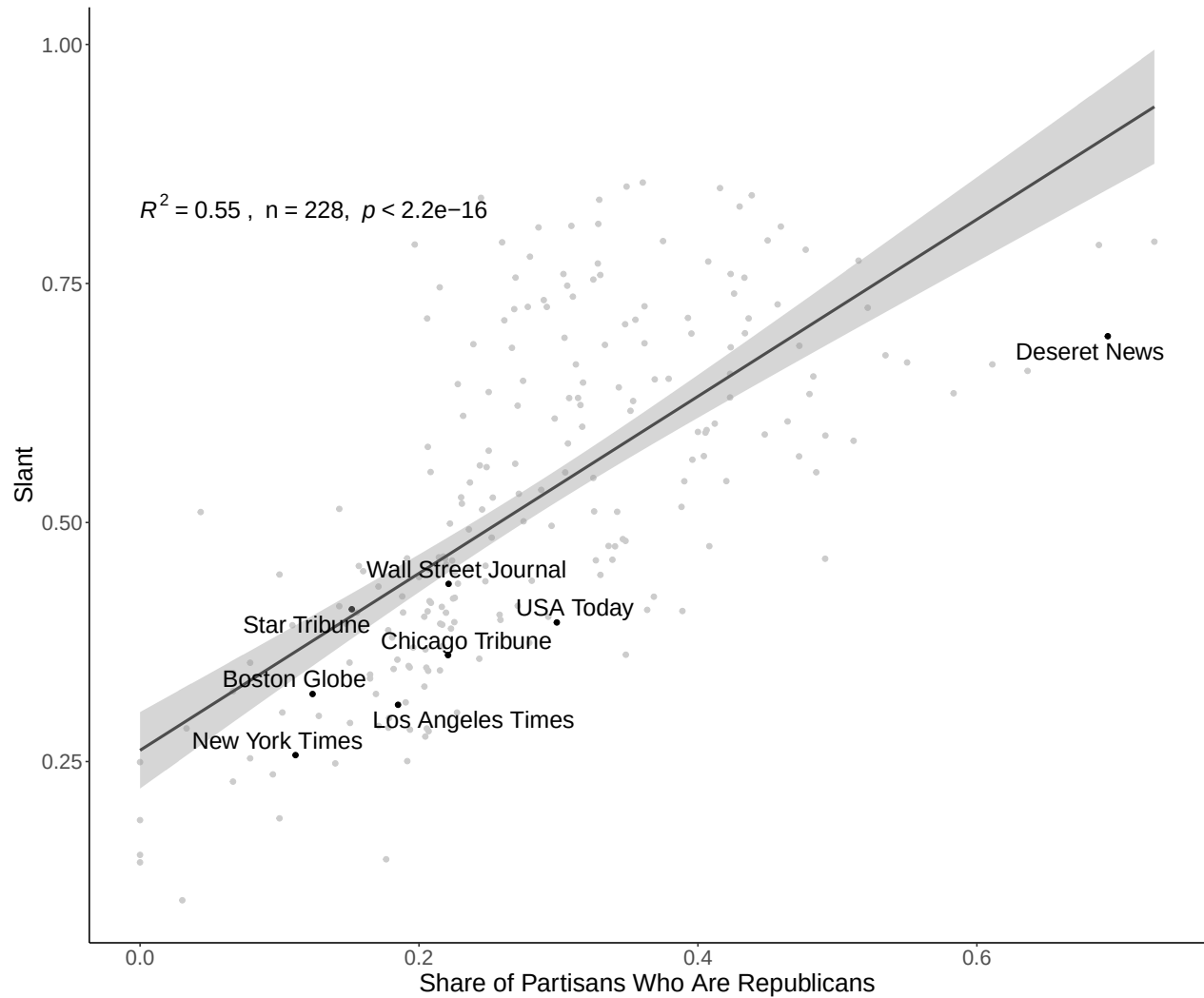


Panel B: Accuracy among Clearly Partisan Articles



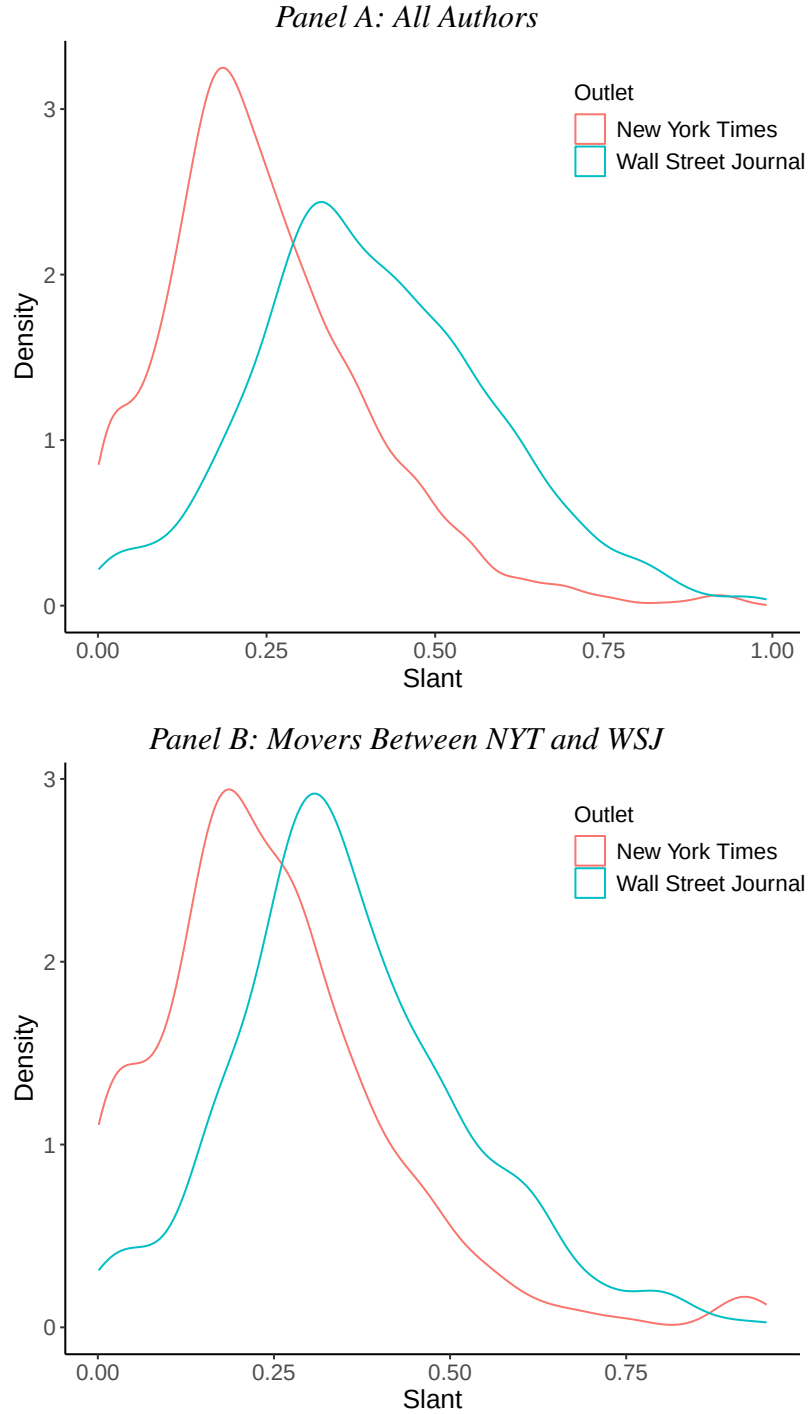
Notes: Panel A plots the rank correlation of the Budak et al. (2016) perceived slant measure for each article as defined in Appendix Table C7 versus the model-based slant measures using all articles. GST uses the re-estimated Republican posterior; its saved log-odds preserve ranks when probabilities round to one. Panel B plots the accuracy of each model-based method in partitioning the dataset in half between more liberal and more conservative articles relative to a naive algorithm. In Panel B, the sample is restricted to articles with a non-zero blinded-coder rating. Each measure and the human rating are partitioned at their respective sample medians; the figure reports accuracy minus 0.5.

Figure 3: Average Outlet Slant and Journalist Partisanship



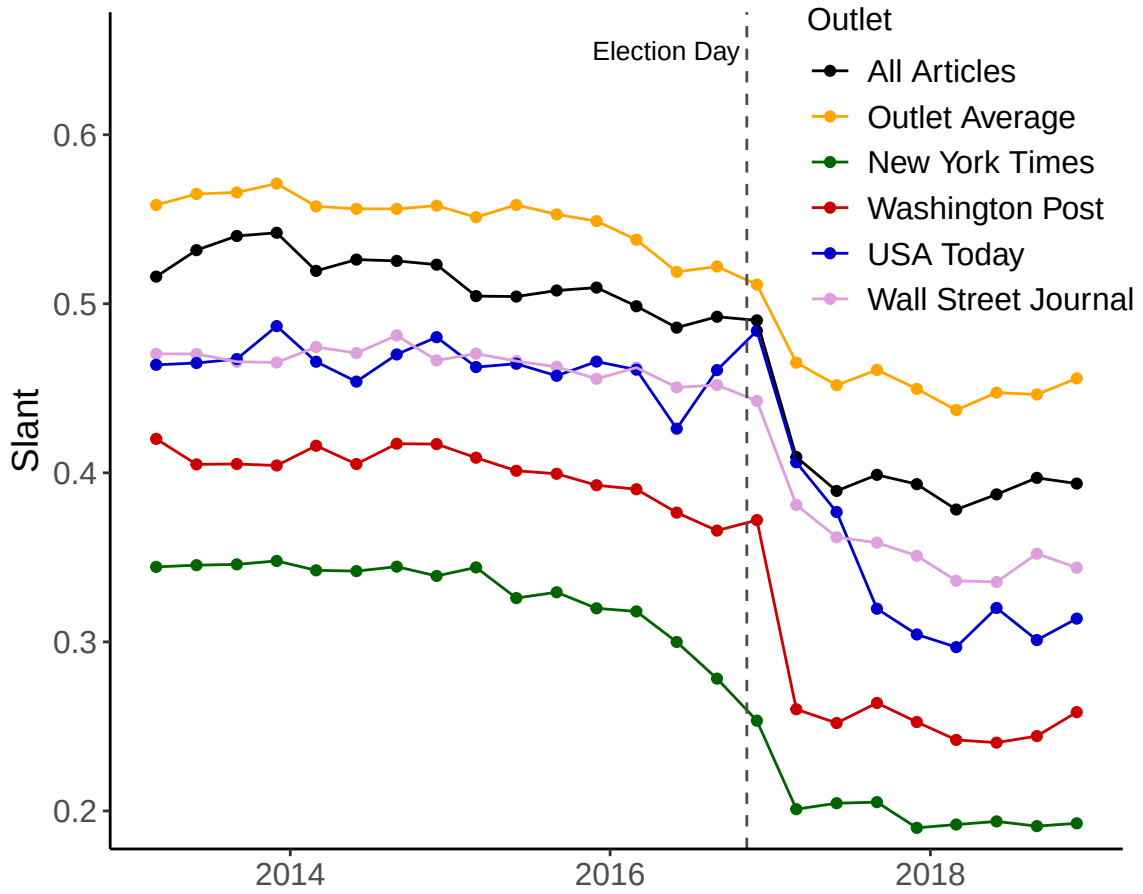
Notes: Figure plots the relationship between the average estimated slant at an outlet (y-axis) with the share of matched partisan journalists who are Republicans (x-axis). Slant ranges from 0 to 1, with larger values indicating content estimated to be more conservative.

Figure 4: Authors at *The New York Times* versus *The Wall Street Journal*



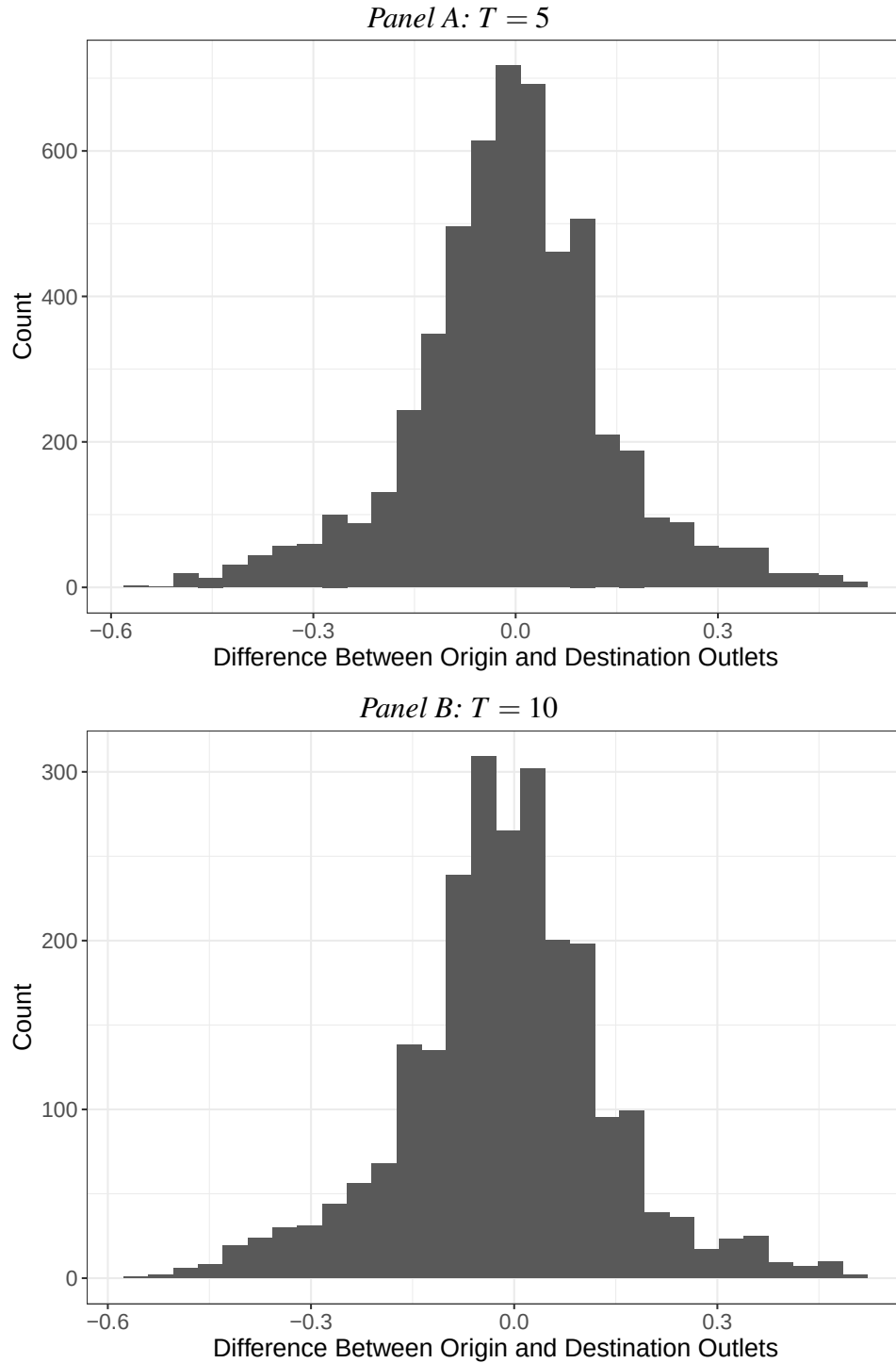
Notes: Figure plots the distribution, across authors, of the average slant of each author's article as measured by our RoBERTa model of slant for each outlet. Panel A includes all authors who appear at a given outlet. Panel B restricts attention to authors who have written at both *The New York Times* and *The Wall Street Journal*. Slant ranges from 0 to 1, with larger values indicating content estimated to be more conservative.

Figure 5: Trends in Slant



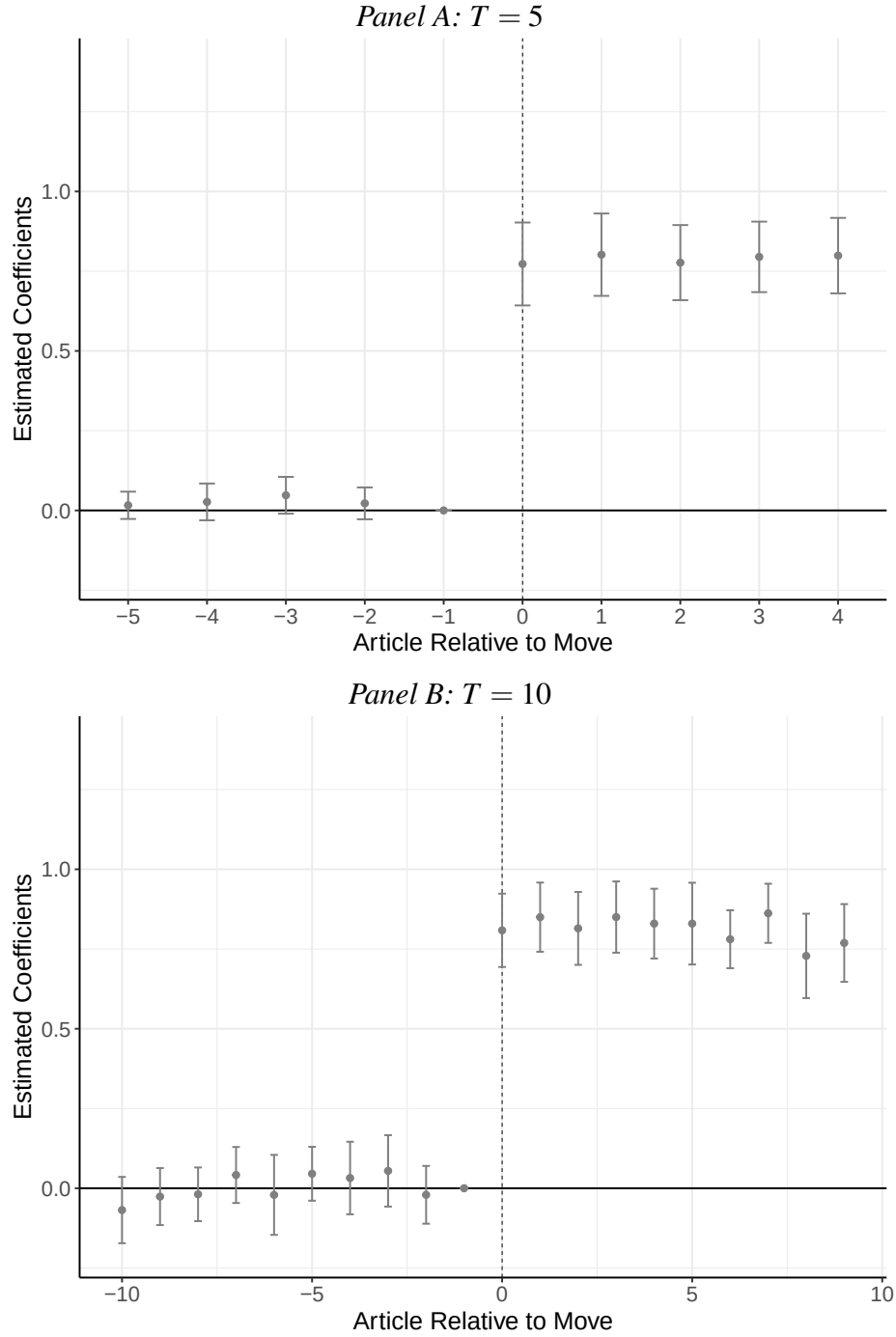
Notes: Figure plots the trends in average estimated slant over time, aggregated by quarter. This average is calculated across all articles (black) as well as for selected outlets (various colors) in our main analysis sample. We also plot an outlet-weighted (rather than article-weighted) average of slant across all outlets (yellow). The dashed vertical line is Election Day 2016. Slant ranges from 0 to 1, with larger values indicating content estimated to be more conservative.

Figure 6: Distribution of Origin-Destination Outlet Slant Scores in Event Study Samples



Notes: Figure plots the distribution of moves $\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}$ for consistent movers in the baseline $T = 5$ estimation sample (Panel A) and the $T = 10$ horizon robustness sample (Panel B). Outlet means use the full-period author–article panel after publication-month residualization, excluding every article involving the moving journalist and all associated coauthor-expanded rows.

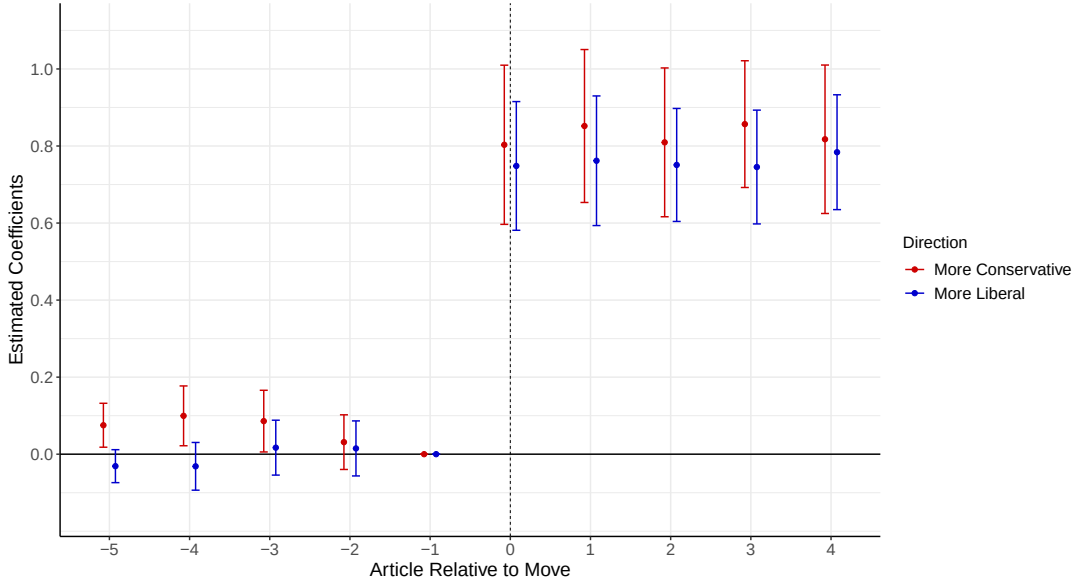
Figure 7: Event Study Changes in Slant Around Author Moves



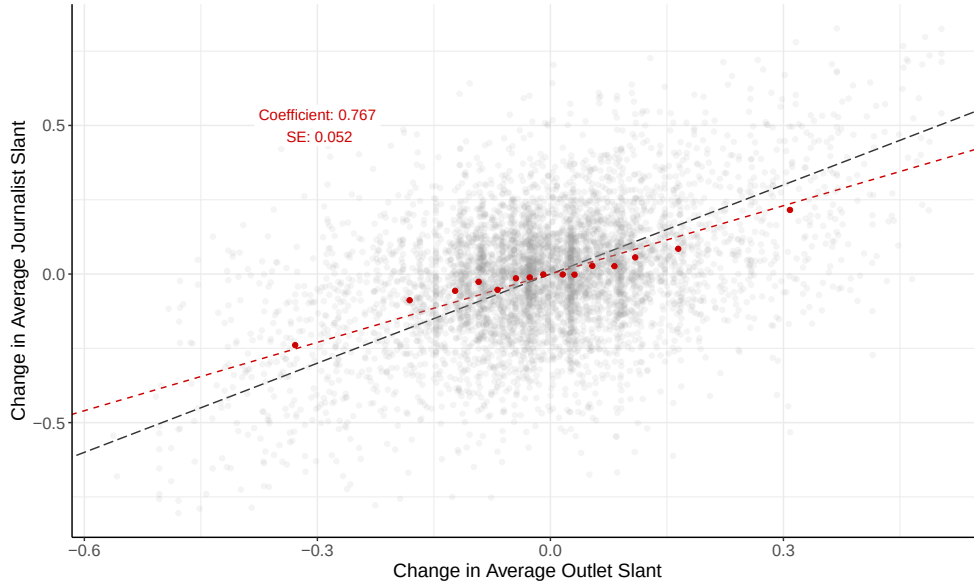
Notes: Figure plots the estimated event study coefficients $\hat{\rho}_l$ for the baseline $T = 5$ estimation sample (Panel A) and the $T = 10$ horizon robustness sample (Panel B). Event study coefficients estimate the moving author's change in slant relative to their slant at $t = -1$, expressed as a fraction of the full-period author–article-weighted leave-focal-author-out origin–destination gap after article slant is residualized by publication-month means estimated once on the full eligible article universe. Regressions weight observations by the squared leave-out gap and include move-pair fixed effects. Standard errors are two-way clustered at the origin- and destination-outlet levels.

Figure 8: Exploring Potential Asymmetry

Panel A: Event Study by Direction of Move

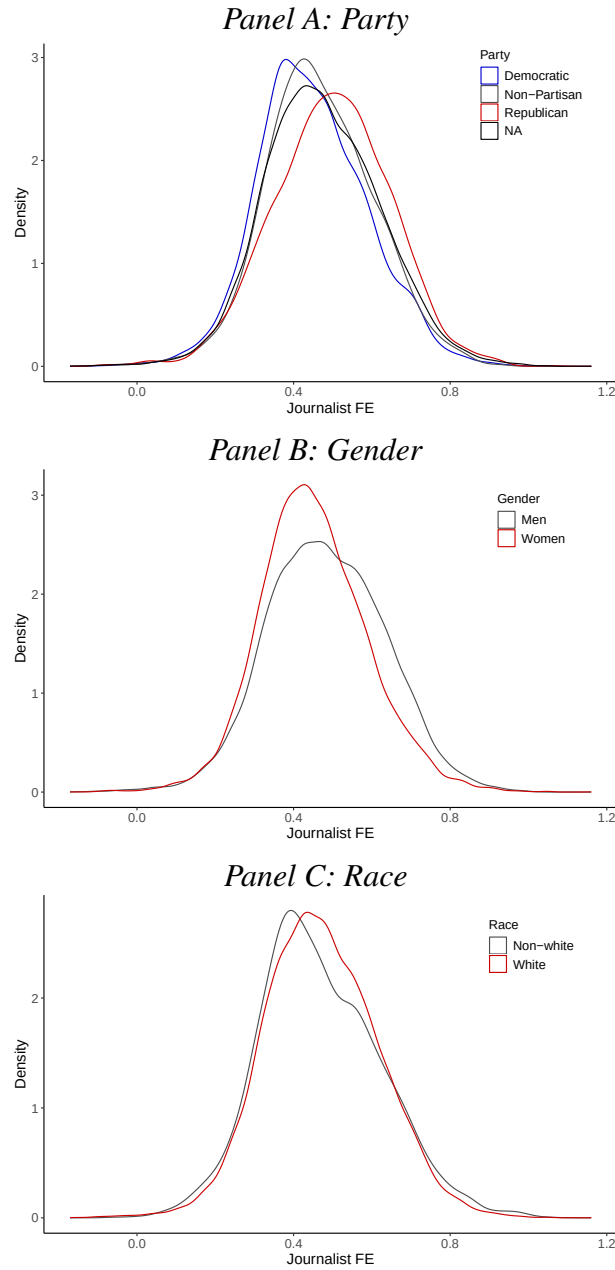


Panel B: Predicting the Change in Journalist Slant with the Change in Outlet Slant



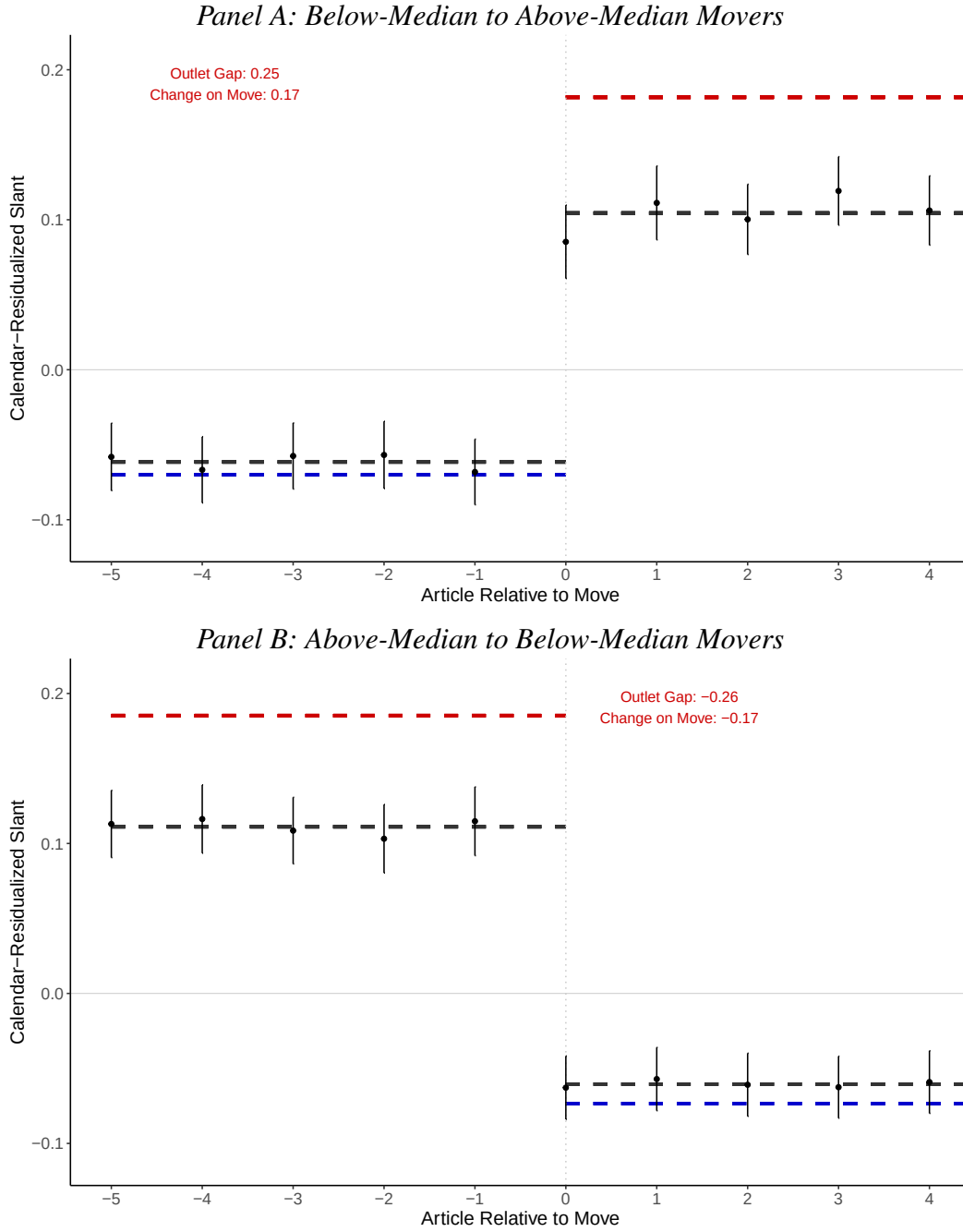
Notes: Panel A plots the event study coefficients for the baseline $T = 5$ estimation sample when the model is estimated separately among the set of movers with $\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)} > 0$ in red and the set with $\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)} < 0$ in blue. Panel B plots the mover-level scatter plot for the same $T = 5$ estimation sample with $\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)}$ on the x-axis and $\frac{1}{T} \sum_{l \geq 0} x_{il} - \frac{1}{T} \sum_{l < 0} x_{il}$ on the y-axis. In Panel B, the red points show unweighted means within 15 equal-count bins. The fitted line is constrained through the origin and weights observations by $(\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)})^2$. Both panels use the common publication-month adjustment; Panel A includes move-pair fixed effects. Panel A standard errors and the standard error reported for the Panel B slope are two-way clustered at the origin- and destination-outlet levels.

Figure 9: Heterogeneity in Journalist AKM Fixed Effects



Notes: Each panel plots the distribution in estimated AKM journalist fixed effects across various sub-groups of journalists. Panel A examines heterogeneity across journalists by their partisan affiliation. The NA group includes journalists whose party affiliation is not mapped to the Democratic, Republican, or non-partisan categories by the L2 matching and recoding procedure. Panel B examines heterogeneity by gender, as predicted by the author's name. Panel C examines heterogeneity by race, as predicted by the author's name.

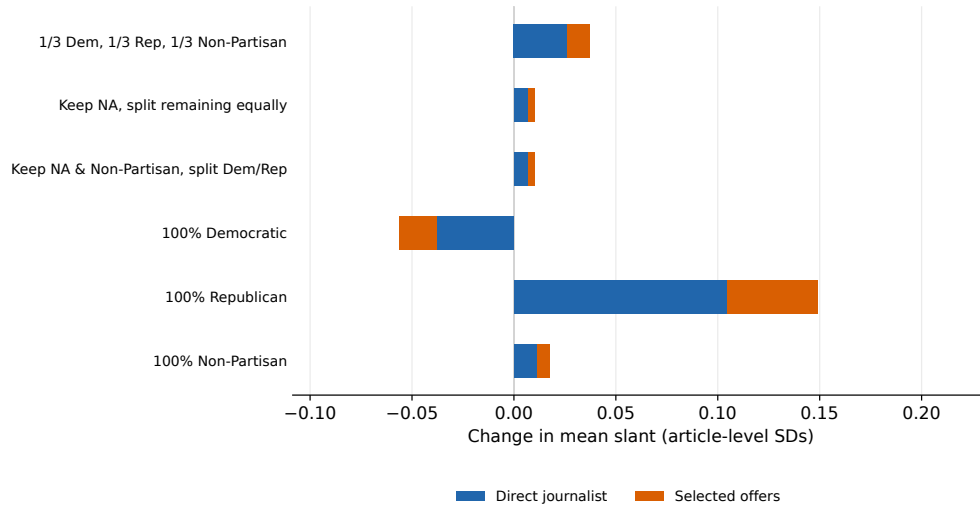
Figure 10: Decomposition Event Study



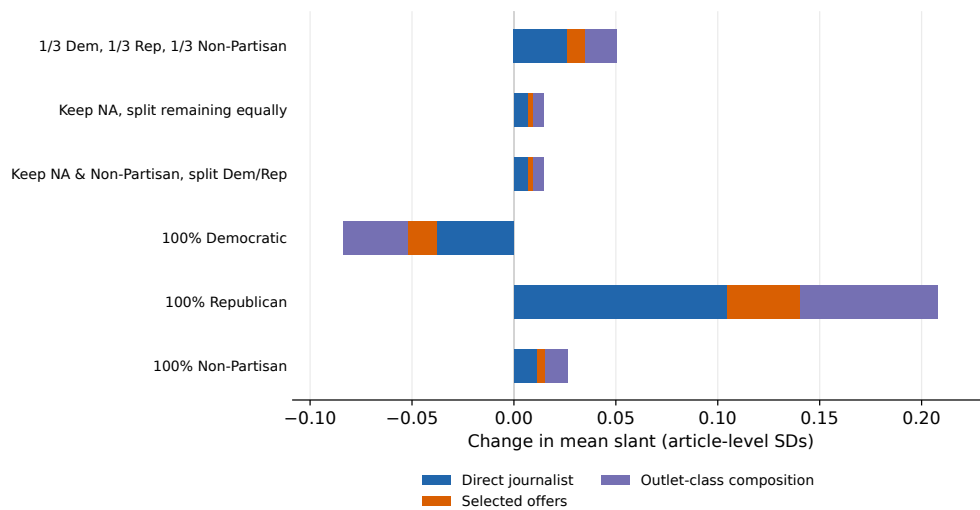
Notes: Panel A plots the average values of $\bar{x}_{j'}$, $\bar{x}_j$, $x_{ij'}$, and x_{ij} for the set of movers who move from below-median to above-median outlets using the baseline $T = 5$ event window. Outlet classifications and all plotted slant measures are constructed after removing publication-month means. The dashed blue (red) line is the average $\bar{x}_j$ for below-median (above-median) outlets. The grey points and associated confidence intervals are the average article slant for the t th article relative to the move for each journalist in the corresponding set. The dashed grey lines are the average across the grey points before and after the move respectively. Panel B is the same as Panel A, except using the set of movers from above-median to below-median outlets.

Figure 11: Counterfactual Slant Changes

Panel A: Outlet sizes fixed



Panel B: Outlet sizes free



Notes: Change in mean slant relative to the observed composition, in article-level standard deviations. Bars decompose each scenario's total change into the direct journalist effect and selected offers; in Panel B, where class employment shares adjust freely, a third component captures the change in outlet-class composition.

Table 1: Summary Statistics

	Variable					
	Articles	Authors	Slant	Slant (Single outlet)	Multiple outlets	Single outlet
<i>Boston Globe</i>	217,479	2,025	0.32	0.28	1,585	440
<i>Chicago Tribune</i>	236,470	5,607	0.36	0.38	5,300	307
<i>Los Angeles Times</i>	194,002	2,475	0.31	0.23	2,373	102
<i>New York Times</i>	288,844	2,990	0.26	0.25	2,440	550
<i>USA Today</i>	107,852	2,663	0.40	0.59	2,646	17
<i>Wall Street Journal</i>	470,103	2,927	0.44	0.46	1,751	1,176
<i>Washington Post</i>	357,805	3,033	0.35	0.35	2,790	243
All Outlets	9,005,189	33,571	0.47	0.49	21,323	12,248

Notes: Table shows summary statistics for our cleaned sample of ProQuest data. Single- and multiple-outlet counts classify publication histories in this broad sample; they do not impose the valid-move definition used in the mobility analyses.

Table 2: AKM Decompositions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Sample:	All	Dropping Stayers (Full)	Dropping Stayers (Large)	Max 2 Outlets	Drop 1 Mo. Moves	Top 20 Outlets	$T = 5$ Full	$T = 5$ Large	$T = 10$ Full	Parent Company
<i>Panel A: Plug-in Estimator</i>										
	Share of Variance, x_{ijt}									
Journalist, $\text{var}(\gamma_i)$	0.194	0.167	0.185	0.202	0.194	0.266	0.236	0.273	0.216	0.352
Outlet, $\text{var}(\alpha_j)$	0.217	0.224	0.263	0.215	0.222	0.094	0.186	0.241	0.209	0.082
Sorting, $2 \times \text{cov}(\gamma_i, \alpha_j)$	0.038	0.040	-0.001	0.030	0.031	-0.001	0.057	-0.020	0.028	0.033
	Correlation									
Sorting, $\text{cor}(\gamma_i, \alpha_j)$	0.092	0.103	-0.002	0.073	0.074	-0.004	0.136	-0.039	0.067	0.096
<i>Panel B: Leave-Out Estimator (Kline et al. 2020)</i>										
	Share of Variance, x_{ijt}									
Journalist, $\text{var}(\gamma_i)$	0.155	0.085	0.080	0.155	0.152	0.245	0.143	0.148	0.129	0.325
Outlet, $\text{var}(\alpha_j)$	0.192	0.201	0.230	0.182	0.194	0.081	0.164	0.188	0.183	0.069
Sorting, $2 \times \text{cov}(\gamma_i, \alpha_j)$	0.085	0.075	0.072	0.095	0.086	0.024	0.084	0.040	0.063	0.059
	Correlation									
Sorting, $\text{cor}(\gamma_i, \alpha_j)$	0.245	0.286	0.266	0.284	0.249	0.085	0.273	0.120	0.205	0.197

Notes: Table reports AKM variance decomposition exercises using the plug-in estimator (Panel A) and the Kline et al. (2020) leave-out estimator (Panel B) with various samples. Both panels use the leave-out connected set of outlets. All columns use raw article slant, the raw outcome-variance denominator, and publication-month fixed effects. Column (1) uses the full sample. Column (2) drops journalists observed at only one retained outlet. Column (3) retains only origin and destination spells around moves whose absolute calendar-residualized leave-focal-author-outlet gap exceeds 0.1, counting each author–article observation once. Column (4) drops journalists that appear in more than two outlets. Column (5) restricts to tenures of longer than one month in duration. Column (6) restricts to the top 20 outlets according to weekend circulation. Column (7) uses the $T = 5$ event study sample. Column (8) is the same as Column (7) except that it restricts to moves whose absolute calendar-residualized leave-focal-author-outlet gap exceeds 0.1. Column (9) uses the $T = 10$ event study sample. Column (10) treats all outlets owned by a single “Parent Company” as the same outlet and restricts attention to outlets for which we observe the parent company.

Table 3: Outlet-level Journalist Effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sample:	All	12+ Months	Max 1 Move	Drop 1 Mo. Moves	Top 20 Outlets	$T = 5$ Sample	$T = 10$ Sample	Parent Company
Share of Variance, $\bar{x}_j$								
Cross-fit, article weights	0.143	0.126	0.201	0.144	0.008	0.165	0.103	0.247
95% CI	[0.113, 0.189]	[0.089, 0.178]	[0.125, 0.293]	[0.084, 0.208]	[-0.146, 0.175]	[0.122, 0.224]	[0.022, 0.201]	[0.160, 0.345]
Cross-fit, equal outlet weights	0.157	0.127	0.176	0.155	-0.077	0.177	0.056	0.350
95% CI	[0.120, 0.208]	[0.084, 0.188]	[0.104, 0.255]	[0.100, 0.219]	[-0.293, 0.156]	[0.124, 0.244]	[-0.031, 0.150]	[0.256, 0.461]
Plug-in, cross-fit support	0.180	0.161	0.193	0.174	0.089	0.191	0.171	0.264
Plug-in, natural support	0.155	0.151	0.117	0.117	0.044	0.168	0.094	0.334
Role-Equality Diagnostics								
P-value, equal role effects	0.209	0.055	0.209	0.026	0.286	0.298	0.069	0.305
Agreement ratio, $2C/(V^o + V^d)$	0.956	0.897	0.891	0.798	0.209	0.961	0.643	0.977
Sample Size								
Moves	4,700	3,681	821	1,798	573	4,700	1,256	1,091
Journalists	2,633	1,977	821	1,426	433	2,633	956	832
Outlets, cross-fit support	154	134	48	89	15	154	65	20
Outlets, natural plug-in support	238	233	194	220	17	238	205	32

Notes: Table reports the share of variance in average outlet slant attributed to outlet-level journalist effects. Articles are residualized relative to calendar-month fixed effects before calculating both journalist-spell means and outlet-level means. The baseline averages symmetric cross-half covariances from five repeated journalist-disjoint splits and uses outlets identified in every split half. Except in the equal-outlet-weight row, outlets are weighted by their full-sample number of articles. The plug-in rows use, respectively, the cross-fit support and the larger connected support available without split completeness. Confidence intervals use a journalist-cluster weighted bootstrap. The role-equality rows compare the model's origin and destination effects; p-values use a restricted-null journalist-cluster wild bootstrap, and Appendix B.6 defines the test and agreement ratio. Move and journalist counts and the cross-fit outlet count use the cross-fit support; the final row reports the natural plug-in support. In Columns (1)–(5) and (8), moves pair strictly forward adjacent article spells with at least five articles, after the common preliminary five-article screen and reconstruction; gaps longer than 12 months are not bridged. All columns exclude journalists above 10 valid moves in the complete cleaned outlet history. The preferred baseline uses full-spell means. Columns (1)–(8) report, respectively, the full sample; journalists observed in at least 12 months; journalists with at most one retained move; spells longer than one month; the top 20 outlets by weekend circulation; the $T = 5$ and $T = 10$ event-study samples; and parent-company groups.

Table 4: Simulated Maximum Likelihood Estimates

	Estimate	SE
Journalist contract weight, $\beta/(\beta + 1)$	0.403	0.019
Slant preference, β	0.674	0.053
Mean ideal point: Democratic	0.537	0.027
Mean ideal point: Republican	0.636	0.028
Mean ideal point: Non-Partisan	0.571	0.028
Mean ideal point: Unknown party	0.562	0.017

Notes: The table reports coefficients estimated by simulated maximum likelihood. Standard errors are clustered by journalist and are conditional on the first-step $\hat{\sigma}_e$. Ideal points are in slant units; the full parameter vector is reported in Table C10.

A Theoretical Appendix

A.1 Proof of Proposition 1

Write $M_i = M_i(x, x_c, \omega, \omega_c, \omega_0)$ and suppress its arguments below. The first-order conditions for the outlet's problem with respect to the wage and slant are

$$\begin{aligned} M_i &= \Delta U_j(x, \omega) \frac{\partial M_i}{\partial \omega}, \\ 2\beta_j(x - \delta_j - \xi_{ij})M_i &= \Delta U_j(x, \omega) \frac{\partial M_i}{\partial x}. \end{aligned}$$

For the logit acceptance probability, $\partial M_i / \partial \omega = M_i(1 - M_i)$ and $\partial M_i / \partial x = -2\beta_i(x - \delta_i)M_i(1 - M_i)$. Because $M_i \in (0, 1)$, we can therefore divide the slant first-order condition by the wage first-order condition to obtain

$$\begin{aligned} 2\beta_j(x - \delta_j - \xi_{ij}) &= \frac{\partial M_i / \partial x}{\partial M_i / \partial \omega} = -2\beta_i(x - \delta_i), \\ x_{ij}^* &= \frac{\beta_i \delta_i + \beta_j(\delta_j + \xi_{ij})}{\beta_i + \beta_j}. \end{aligned}$$

For any fixed x , the derivative of the outlet's objective with respect to ω has the sign of

$$v_j - \beta_j(x - \delta_j - \xi_{ij})^2 - \omega - 1 - \frac{\exp(\omega - \beta_i(x - \delta_i)^2)}{\exp(\omega_0) + \exp(\omega_c - \beta_i(x_c - \delta_i)^2)}.$$

This continuous expression is strictly decreasing from positive infinity as $\omega \rightarrow -\infty$ to negative infinity as $\omega \rightarrow \infty$. It therefore has a unique root $\omega^*(x)$. Because the objective's derivative is positive below this root and negative above it, $\omega^*(x)$ is the unique globally optimal wage conditional on x .

After maximizing over ω , the envelope theorem and the wage first-order condition imply that

the derivative of the outlet's maximized objective with respect to x is

$$-2M_i(x, \omega^*(x)) [\beta_j(x - \delta_j - \xi_{ij}) + \beta_i(x - \delta_i)].$$

Because $M_i > 0$ and the expression in brackets has positive slope $\beta_i + \beta_j$, the outlet's maximized objective increases below x_{ij}^* and decreases above it. Therefore the outlet's problem has the unique global solution $(x_{ij}^*, \omega_{ij}^*)$, where $\omega_{ij}^* = \omega^*(x_{ij}^*)$. This defines $h_{ij}(x_{ij}^*, x_c, \omega_c, \omega_0)$. **QED.**

A.2 Proof of Lemma 1

When $\beta_i = 0$, Proposition 1 gives

$$x_{ij}^* = \delta_j + \xi_{ij}.$$

At this slant, the outlet's slant loss is zero, and the wage first-order condition is

$$v_j - \omega_{ij}^* = 1 + \frac{\exp(\omega_{ij}^*)}{\exp(\omega_0) + \exp(\omega_c)}.$$

As shown in the proof of Proposition 1, this equation has a unique solution, which is the globally optimal wage at x_{ij}^* . Thus, conditional on (ω_c, ω_0, j) , the optimal wage does not depend on ξ_{ij} . Because the alternative-specific choice shocks are drawn independently after the offer,

$$\begin{aligned} & \Pr(i \text{ accepts } j\text{'s offer} \mid i \text{ receives } j\text{'s offer}, \omega_c, \omega_0, \xi_{ij}) \\ &= \frac{\exp(\omega_{ij}^*)}{\exp(\omega_0) + \exp(\omega_c) + \exp(\omega_{ij}^*)}, \end{aligned}$$

which does not depend on ξ_{ij} . Acceptance therefore does not further select on ξ_{ij} conditional on the journalist's current wage, outside option, and prospective outlet. **QED.**

A.3 Proof of Proposition 2

When $\beta_i = 0$, Proposition 1 gives $x_{ij}^* = \delta_j + \xi_{ij}$. Lemma 1 and the stated conditional mean restriction therefore imply

$$\begin{aligned} & \mathbb{E}[x_{ij}^* \mid \omega_c, \omega_0, i \text{ accepts } j\text{'s offer}] \\ &= \mathbb{E}[\delta_j + \xi_{ij} \mid \omega_c, \omega_0, i \text{ receives } j\text{'s offer}] = \mathbb{E}[\delta_j + \xi_{ij}]. \end{aligned}$$

Taking expectations over current wages and outside options among accepted matches gives

$$\mathbb{E}[x_{ij}^* \mid i \text{ accepts } j\text{'s offer}] = \mathbb{E}[\delta_j + \xi_{ij}].$$

QED.

A.4 Proof of Proposition 3

Let $\mathcal{J}$ be the fixed set of outlets that appear as both origins and destinations in the regression, and let $J = |\mathcal{J}|$. The corresponding bipartite support graph has one origin and one destination node for each outlet, with an edge for each supported move pair among these outlets. Assume this graph is connected.

For each $j \in \mathcal{J}$, let p_j be its full-sample number of articles normalized to sum to one across $\mathcal{J}$.

Without loss of generality, let $r \in \mathcal{J}$ be the reference outlet and omit its destination indicator. For each move q , let $\mathbf{t}'_q$ collect the origin and destination indicators under this normalization; $\mathbf{T}$ stacks these move-level rows. Conditional on the fixed weights w , define

$$\boldsymbol{\theta}^o = (\psi_j - \psi_r)_{j \in \mathcal{J}}, \quad \boldsymbol{\theta}^d = (\psi_j - \psi_r)_{j \in \mathcal{J} \setminus \{r\}},$$

and let $\boldsymbol{\theta} = [(\boldsymbol{\theta}^o)', (\boldsymbol{\theta}^d)']'$. Equation (1) gives $\mathbf{Y} = \mathbf{T}\boldsymbol{\theta} - \mathbf{e}$. The outlet means entering $\mathbf{Y}$ are estimated on the full article sample. Assume their estimation error is $o_p(1)$ uniformly over the fixed set $\mathcal{J}$; by Slutsky's theorem, it suffices to establish the result at their population limits. Let N_m denote the number of observed moves. Assume that the common support $\mathcal{J}$ is eventually fixed and that, in the full sample and separately within every half of each fixed split (using the corresponding half-sample move count), every supported move pair has positive limiting frequency, e_q has a finite second moment, and the sample averages of $\mathbf{t}_q \mathbf{t}'_q$ and $\mathbf{t}_q e_q$ converge in probability to their expectations.

For every supported pair (j, k) , pair-conditional difference balance implies

$$\begin{aligned} \mathbb{E}[e_q \mid o(q) = j, d(q) = k, w] &= \frac{1}{\beta + 1} \left\{ \mathbb{E}[\xi_{qk} - \bar{\xi}_k \mid o(q) = j, d(q) = k, w] \right. \\ &\quad \left. - \mathbb{E}[\xi_{qj} - \bar{\xi}_j \mid o(q) = j, d(q) = k, w] \right\} = 0. \end{aligned}$$

Because $\mathbf{t}_q$ is determined by the origin–destination pair, $\mathbb{E}[\mathbf{t}_q e_q \mid w] = 0$. The assumed law of large numbers therefore gives

$$\frac{\mathbf{T}'\mathbf{e}}{N_m} \xrightarrow{p} 0.$$

Connected support and positive limiting pair frequencies imply that $G = \mathbb{E}[\mathbf{t}_q \mathbf{t}'_q \mid w] \succ 0$. The assumed law of large numbers gives $N_m^{-1} \mathbf{T}'\mathbf{T} \xrightarrow{p} G$. Therefore,

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta} = - \left(\frac{\mathbf{T}'\mathbf{T}}{N_m} \right)^{-1} \left(\frac{\mathbf{T}'\mathbf{e}}{N_m} \right) \xrightarrow{p} 0.$$

Under these half-specific conditions, the same argument applies separately to every half of each

fixed split, and the two complete role-effect vectors converge to $(\psi_j - \psi_r)_{j \in \mathcal{J}}$. Because J and p are fixed, the continuous mapping theorem therefore gives, for every split s ,

$$\begin{aligned}\text{Cov}_p(\hat{\theta}^{d,A,s}, \hat{\theta}^{o,B,s}) &\xrightarrow{p} \text{Var}_p(\psi_j), \\ \text{Cov}_p(\hat{\theta}^{d,B,s}, \hat{\theta}^{o,A,s}) &\xrightarrow{p} \text{Var}_p(\psi_j).\end{aligned}$$

Their symmetric average, and then the average over the fixed number of splits, has the same probability limit. The reference outlet cancels from every covariance, so its choice does not affect the result. **QED.**

A.5 Event-Study Interpretation under Zero Preferences

In this subsection, we discuss the relationship between Proposition 2 and our event-study specification. The proposition analyzes newly accepted offers. It does not by itself establish the stock, mover-selection, or measurement conditions needed to interpret the event-study coefficients. Let $\hat{d}_m$ be estimated once on the full eligible article universe and subtracted from all event-study outcomes and outlet means. Throughout this subsection, x_{it} and $\bar{x}_j$ denote the resulting residualized quantities. Under $\beta = 0$, write the residualized article outcome for move q as

$$x_{it} = \alpha_q + (m_{d(q)} - m_{o(q)})\mathbf{1}_{\{t \geq 0\}} + u_{it},$$

where u_{it} collects deviations of the mover's structural shocks from their search means and the article-level error ε_{it} . Suppose that, on the selected-mover support, the implemented full-period, author–article-weighted residualized leave-focal-author-out gap satisfies

$$\bar{x}_{d(q)}^{(-i)} - \bar{x}_{o(q)}^{(-i)} \xrightarrow{p} m_{d(q)} - m_{o(q)};$$

u_{it} is mean zero conditional on the full selected regression design at each event time; and the implemented squared-gap-weighted fixed-effects regression satisfies the usual law of large numbers and full-rank conditions. A sufficient route to this gap condition is balance in initial outlet stocks and among subsequent entrants, with retention, separation, article-production weights, and the leave-focal-author-out construction preserving that balance, while the pooled publication-month residualizer consistently removes the common calendar component and mean-zero leave-out errors satisfy a law of large numbers as outlet article samples grow. These are maintained restrictions beyond Proposition 2.

Under these conditions and $\beta = 0$, standard weighted least-squares consistency implies that the event-study coefficients converge in probability to zero before the move and one after the move relative to the omitted $t = -1$ period; in particular, $\hat{\rho}_0 \xrightarrow{p} 1$.

B Empirical Appendix

B.1 Data Processing Details

In this section, we describe key data processing steps used in the main analysis, model training, and validation exercises.

B.1.1 Tweet-Linked Article Data

Our first cleaned article dataset is built from tweets posted by politicians that link to news articles. We begin from a tweet collection containing 4.4 million tweets and approximately 1,800 politician accounts. From these tweets, we recover more than 1 million unique tweeted URLs. We exclude likely non-news urls including “youtu.be,” “pscp.tv,” “facebook.com,” “instagram.com,” “patreon.com,” and “google.com.” Next, we draw a scraping sample of 600,000 URLs from this expanded URL pool. Among these sampled URLs, 348,821 produce non-empty article text, corresponding to 328,966 unique URLs with non-empty text.

We then combine the scraped article text with politician party labels and tweet-level sentiment information to construct the tweet-linked training data used in the paper. We then use an LLM (OpenAI GPT-5-mini) to classify cases in which the politician’s tweet was explicitly criticizing (rather than endorsing or having unclear support for) the article’s content. The LLM was provided with both the text of the politician’s tweet and the scraped article text in order to make this determination. This filter drops 3% of tweets as criticizing (2% of Democrat tweets, 4% of Republican tweets). 92% of tweets are classified as endorsing the article’s content, while 5% of tweets are labeled as not clearly endorsing or criticizing, and tweets with both such LLM labels are kept in our sample.

We also apply an LLM-based cleaning pipeline to the scraped article text. In this pipeline, two models identify and remove non-text spans, such as page-navigation text, captions, advertisements, and other interface elements, while a separate binary classifier flags pages that are not valid news articles at all, such as homepages, paywalls, or other non-article content.⁴¹ Articles flagged as invalid are dropped, and detected non-text spans are removed before downstream cleaning.

After this text-cleaning stage, we impose a minimum post-cleaning text-length threshold of 1,000 characters to exclude very short, malformed, or otherwise uninformative text. We also perform language detection on unique article texts and retain only articles classified as English.

The resulting cleaned tweet-linked sample contains 182,538 article–tweet observations, of which 128,105 are linked to Democratic politicians and 54,433 are linked to Republican politicians. For the tweet-based RoBERTa exercises, we split this cleaned sample into 70 percent training and 30 percent validation data using a fixed random seed. To avoid leakage across partitions,

⁴¹We use OpenAI GPT-5-mini and Claude Haiku 4.5 to detect non-text spans and OpenAI GPT-4.1-mini to identify invalid articles.

we ensure that the same article text does not appear in both the training and validation sets by removing tweeted articles from the validation sample if that article’s text appears in the training sample (due to also being referenced in a different tweet).

For the RoBERTa training sample, we retain tweets posted on or after June 1, 2011, removing the sparse early tail in which some training months contain observations from only one party. Within each retained tweet month m , let $N_{m,D}$ and $N_{m,R}$ denote the number of training observations linked to Democratic and Republican politicians, respectively, and let $N_m = N_{m,D} + N_{m,R}$. We assign each observation from party $p \in \{D, R\}$ weight

$$w_{m,p} = \frac{N_m}{2N_{m,p}}.$$

Thus each party receives total weight $N_m/2$ within month m , while the month’s total weight remains N_m . We compute these weights from the training sample and apply the same month–party weights to validation observations.

Appendix Figure C12 summarizes the monthly counts and party shares of article-tweet observations in this sample.

B.1.2 Budak et al. (2016) Article Data

Our second cleaned article source is based on the labeled article dataset from Budak et al. (2016). We merge article text files with the associated hand-coded labels, retain observations with non-missing text and URL, and apply the same invalid-article and non-text-cleaning pipeline described above. That is, we drop pages classified as non-articles and replace raw scraped text with text cleaned of interface and other non-news fragments before constructing the final article text measure. We then apply the same minimum article-length restriction of 1,000 characters. From these cleaned records, we construct the article-level slant measures used in our validation exercises.

B.1.3 ProQuest/TDM Analysis Data

For the author-, outlet-, and event-study analyses, we construct a separate cleaned dataset from the large ProQuest/TDM article archive. Following the ProQuest workflow documented separately, we first pull and preprocess the raw TDM exports and then organize the article-level archive used in the downstream cleaning steps. We next restrict this archive to observations classified as news articles, thereby excluding non-news content or opinion/editorial content from the analysis sample. We then standardize outlet names, remove observations with missing author information, and expand multi-author bylines into author-level observations. After that, we clean and harmonize author names, including nickname-based first-name standardization, and retain only observations with parsable author identities. The ProQuest query selects the Newspapers source type plus enumerated newspaper publication IDs, excluding the separate Wire Feeds source type; downstream,

we remove duplicate author–article pairs and exact-text duplicate articles. We then merge in publication and outlet-group metadata and recover missing identifiers when possible. To focus on substantively meaningful journalistic activity, we then drop authors with 10 or fewer articles and retain only authors with more than 10 articles. After these restrictions, the resulting cleaned all-outlets sample contains 9,005,189 articles from 279 outlets. For the event-study, outlet-decomposition, AKM, KSS, BLM, and structural analyses, we first drop outlet spells with fewer than five articles, reconstruct spells, and retain spells with at least five articles. We then exclude authors with more than 10 valid moves over their complete cleaned outlet history. Parent-company robustness uses the same eligible outlet-level articles before applying its additional restrictions. We estimate AKM and KSS directly on articles from the selected samples. Event-study windows impose additional consecutive-article requirements. When defining these moves, we do not bridge gaps longer than 12 months and exclude adjacent spell pairs whose boundary articles occur on the same day. This cleaned ProQuest/TDM dataset underlies the author-slant, outlet-slant, journalist-mobility, and event-study analyses in the paper. For the mover event-study, outlet-level-decomposition, BLM, and structural analyses, we remove publication-month means estimated once on the full eligible article universe from article slant before constructing article outcomes and outlet means; the AKM and KSS specifications instead use article slant directly with publication-month fixed effects. Here, the eligible article universe is the cleaned sample after the spell screen and move cap.

We infer journalists’ gender from first names and race from surnames.

Appendix Table C6 tracks observation counts for our main datasets throughout the data cleaning procedures.

B.2 Anonymized Slant-Measure Comparison

One potential concern is that our slant measure might load on geography or other proxies that are correlated with the party of a sharing politician, but may or may not represent political slant itself. We address this in two ways. First, Section 4.3 validates our slant measure against external measures of slant, showing, for example, that our measure is more strongly correlated with political slant as perceived by human readers than are other measures in the literature. Second, Appendix Table C3 reports model diagnostics, key results, and top-contribution tokens for our baseline measure and for two alternative slant models trained on geography-anonymized or more aggressively anonymized politician-tweeted article text and then applied to the same non-anonymized ProQuest articles. The geography version makes generic replacements for U.S. state and local identifiers, including place names, local facilities, and related demonyms, while preserving national, foreign, outlet, and other non-geographic text. For example, it replaces “Texas Department of Transportation” with “the state Department of Transportation.” The aggressive version additionally replaces personal names, organizations and publications, broader geographic identifiers, and dates or time

periods, while preserving generic political content and numeric references outside dates. For both versions, GPT-5-mini is used to identify these terms and make replacements. The table holds each empirical sample and estimator contract fixed while replacing the slant outcome and the outcome-dependent quantities required by that estimator. Qualitative takeaways from these alternatives are largely similar, though the anonymized measures have lower predictive accuracy, larger event-study coefficients, and weakly larger journalist shares in some specifications. We use the non-anonymized slant measure as our baseline because it is most strongly correlated with human-perceived slant as measured in Section 4.3, while still showing robustness to these alternatives given ambiguity around whether the anonymized identifiers represent or are merely correlated with human-perceived slant.

B.3 Media Topics

Our measure of slant intentionally includes both which topics or stories are covered and how that coverage is framed. To help distinguish these dimensions and to better understand the drivers of slant across articles, outlets, and journalists, including around moves, we also use an LLM (OpenAI GPT-4o-mini) to assign each article to exactly one media topic from an existing media topic taxonomy. We source this media topic taxonomy from The International Press Telecommunications Council (IPTC). To assign articles to a topic from this taxonomy, we provide the LLM with the article text and the 17 media topics in IPTC’s Level One (“Top-Level”) list, and ask it to assign this article to exactly one topic from this list based on the article’s text. We use this same approach to apply topics to articles in our politician-tweeted and ProQuest article samples.

These assigned topics allow us to analyze the frequency of topics across datasets, and the ability of topics to explain observed slant. In Appendix Figure C10, we plot for each media topic the share across reweighted politician article tweets (Panel A) and the share across ProQuest articles (Panel B). We see that while all 17 topics are represented in both datasets and their frequency is correlated, the datasets differ in the relative frequency of different topics. The “politics and government” media topic is the modal topic among politician-tweeted articles, but is only the fourth most common topic among ProQuest articles. In Appendix Figure C11, we show by topic the share of articles tweeted by Republican (vs. Democratic) politicians in our politician-tweeted articles sample. After reweighting to balance the overall share of Democrat vs. Republican tweets, we see that most topics are roughly equally likely to be tweeted by each party. Notable exceptions include the “environment” and “society” topics, which are more frequently shared by Democratic politicians than by Republican politicians. In Appendix Figure C14, we plot the distribution of article-level slant by topic in our ProQuest sample. We see mild variation in mean slant across topics, accounting for 9.2% of the variation in article-level slant. The remaining 90.8% variation in slant occurs within topics. Appendix Figure C16 follows the main event-study specification after replacing article slant with the mean slant of the article’s topic, constructed using the same

publication-month adjustment. The resulting coefficients are close to zero.

B.4 Allowing for Worker-Firm Interactions

In this section, we extend the framework developed in Section 5.2 by adopting the approach of Bonhomme et al. 2019, hereafter BLM. Relative to the AKM specification, this framework relaxes additive separability by allowing for latent heterogeneity across journalists and for nonparametric interactions between journalist types and outlet classes.

In our application, we use all articles in the two spells around each journalist’s first move and distinguish between two journalist types and four outlet classes. We use the upstream calendar-adjusted article slant, with its common level restored, and impose one shared journalist-type by outlet-class mean matrix before and after the move. Conditional noise variances remain role-specific. Journalist types are latent and estimated from the data.⁴² Outlet classes are constructed in a preliminary step by clustering outlets based on their empirical CDF of the same adjusted article slant.

Figure C22 shows the mean level of slant produced by journalists of each type working in outlets of each type. The figure reveals variation in slant along both dimensions: type 1 journalists write articles that are 17 points more conservative than their type 0 counterparts, averaging within-class differences across outlet classes, while the most conservative outlet type (type 3) produces articles about 43 points more conservative than the most liberal (type 0), on average across journalist types. Figure C23 shows the distribution of journalist types across outlet types.

We explore what observable worker characteristics correlate with their type distribution. In particular, Figure C24 shows that registered Republicans are 13 percentage points more likely to be type 1 workers than registered Democrats. Male journalists are also 19 percentage points more likely to be type 1 than female journalists.

To quantify departures from additivity, we project the shared slant mean matrix onto additive journalist and outlet type indicators, and examine how much the actual cell means deviate from the additive fit. The interaction residuals (the difference between actual cell means and their best additive approximation) account for 0.06% of observed calendar-adjusted article variance.

Figure C25 illustrates the BLM decomposition. Its denominator is observed calendar-adjusted article variance in this first-move sample; the structural decomposition instead uses model-implied stationary adjusted-slant variance, and AKM/KSS use raw slant with conditional month effects. The upstream adjustment subtracts pooled month means on the eligible article-credit universe, which can remove composition shifts as well as common shocks. It is not equivalent to estimating month effects conditional on BLM types and classes. The decomposition attributes 9.0% of observed calendar-adjusted article variance to journalist types, 25.7% to outlet classes, and 2.2% to

⁴²Here, “conservative” and “liberal” refer to the measured slant of the articles produced.

sorting of journalists across outlets.

B.5 Ownership Event Studies

We also analyze changes in newspaper ownership to better understand the potential role of owners in shaping their newspapers' slant. We use the slant of articles published in other news outlets in an acquiring owner's portfolio as a proxy for that owner's preferred slant. We find no evidence that an acquired outlet's slant shifts towards the average article slant of that new owner's portfolio, which may be consistent with a limited role for owners in determining news slant.

More precisely, we define ownership change events as adjacent-year changes in the newspaper owner. The ownership panel links ProQuest TDM publication identifiers to year-specific owners. Owner values are based on *Editor & Publisher* group-company ownership fields, matched to ProQuest publications and supplemented with manual review for publications with incomplete automatic coverage and to remove false ownership changes.

For each event, we construct the acquiring owner's pre-acquisition portfolio from other outlets owned by the acquiring owner in the year before the acquisition, excluding the acquired outlet and same-year acquisitions. Let e index ownership-change events and l index years relative to the ownership change. The baseline sample is balanced over $l \in \{-2, -1, 0, 1\}$ and omits $l = -1$. Let $\tilde{x}_{el}$ be the acquired outlet's average slant residualized by calendar-year means estimated among articles from all other outlets that are not acquired during our sample, and let g_e be the raw pre-period slant gap between other outlets in the acquiring owner's portfolio and the acquired outlet.

In one rough analog of our journalist movers' specification, we estimate:

$$\tilde{x}_{el} = \gamma_e + \sum_{k \neq -1} \rho_k^A g_e \mathbf{1}_{\{l=k\}} + \varepsilon_{el},$$

weighting by g_e^2 . The plotted coefficients are ρ_k^A ; positive post-acquisition estimates indicate the share of the initial slant gap closed by the acquired outlet's post-period slant change.

Two alternative specifications measure the extent to which an outlet's average slant and the average slant in the owner's portfolio change around the acquisition. First, we fix the same acquiring-owner portfolio membership and define time-varying alignment as $a_{el}^{TV} = -|\bar{x}_{el} - \bar{x}_{P(e)l}|$, where $P(e)$ denotes this fixed portfolio and $\bar{x}_{P(e)l}$ is its contemporaneous average slant. Second, we define $a_{el}^{FP} = -|\tilde{x}_{el} - \tilde{x}_{P(e),pre}|$, where $\tilde{x}_{P(e),pre}$ is the acquiring owner's fixed pre-period portfolio slant after applying the same calendar-year residualization. For each alignment outcome, we estimate

$$a_{el} = \gamma_e + \sum_{k \neq -1} \rho_k^B \mathbf{1}_{\{l=k\}} + \varepsilon_{el}.$$

The plotted coefficients are relative to the year before acquisition. All standard errors are clustered

at the event level. The data contain 39 ownership changes. 31 enter the gap-slope and fixed-pre alignment baselines after restricting to a balanced panel; 16 enter the time-varying alignment baseline because it requires contemporaneous owner-portfolio slant coverage, and this value is missing in some periods.

Appendix Figures C18, C19, and C20 visualize these event-study coefficients and their underlying variation. The estimates reported in the variation-plot annotations are using $l = -1$ as the pre period and the average of $l = 0$ and $l = 1$ as the pooled post period. The pooled estimates are -0.009 (s.e. 0.039) for the movers-specification slope coefficient, -0.028 (s.e. 0.015) for time-varying alignment, and -0.020 (s.e. 0.008) for fixed-pre residualized alignment. Positive estimates would indicate movement toward the acquiring owner’s portfolio; the estimates therefore do not point toward post-acquisition convergence in slant towards average slant in the owner’s other outlets.

B.6 Cross-Fit Inference and Role-Equality Diagnostics

We construct the confidence intervals in Table 3 using a cluster-level fractional-random-weight bootstrap (Præstgaard and Wellner, 1993; Xu et al., 2020). For each draw, we assign an independent mean-one exponential weight to each journalist cluster, hold that weight fixed across the journalist’s observations and all repeated splits, and re-estimate the cross-fit statistic. Unlike re-sampling journalist clusters, the strictly positive weights retain every mover in the estimation sample and therefore preserve the realized connected support in every draw. Split assignments, outlet means, and outlet aggregation weights also remain fixed. The reported 95 percent interval uses the 2.5th and 97.5th percentiles of the resulting estimates.

The model implies equality between the origin and destination role effects, up to their common normalization. For each split, V^o is the cross-half covariance between the origin-effect estimates, V^d is the analogous covariance between the destination-effect estimates, and C is the average of the two cross-half origin–destination covariances. We average these moments across splits and define

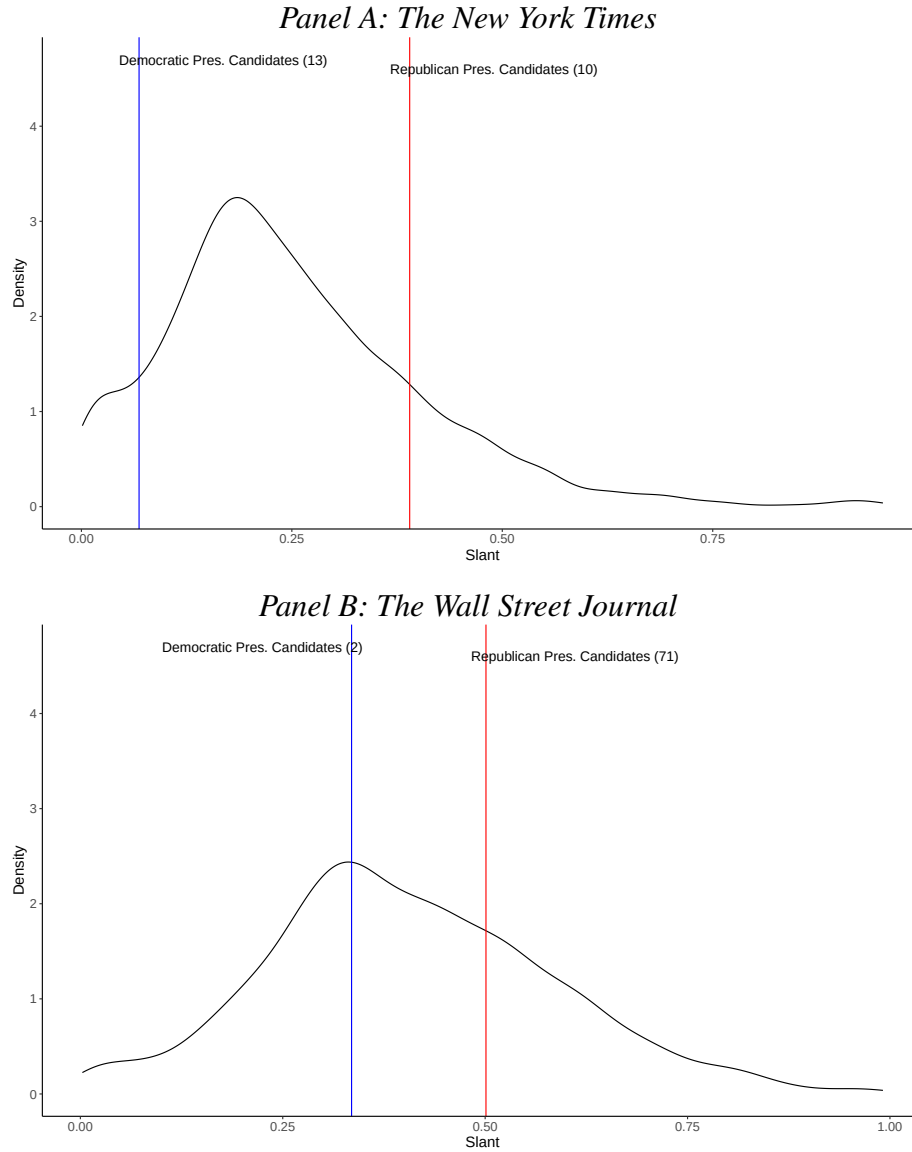
$$Q = V^o + V^d - 2C.$$

Equivalently, Q is the cross-half covariance of the centered origin–destination role gaps. Under role equality, these gaps do not vary across outlets and $Q = 0$. To test this restriction, we fit a model with common origin and destination effects, apply wild-bootstrap Rademacher multipliers at the journalist level to its residuals, and re-estimate Q . The reported p-value compares the observed statistic with this restricted-null distribution.

We also report the descriptive agreement ratio $2C/(V^o + V^d)$. The ratio equals one under role equality, but need not be bounded by zero and one in finite samples.

C Appendix Exhibits

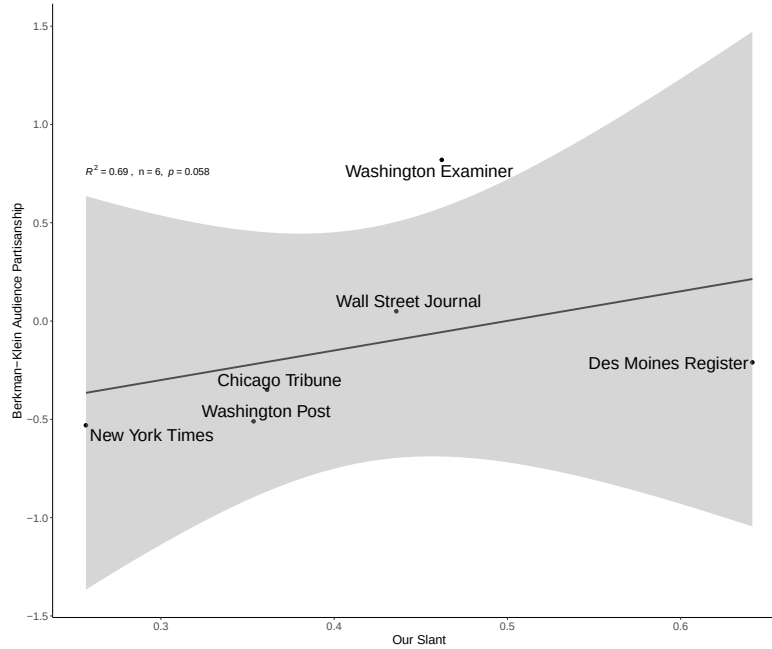
Appendix Figure C1: Politician Op-Eds



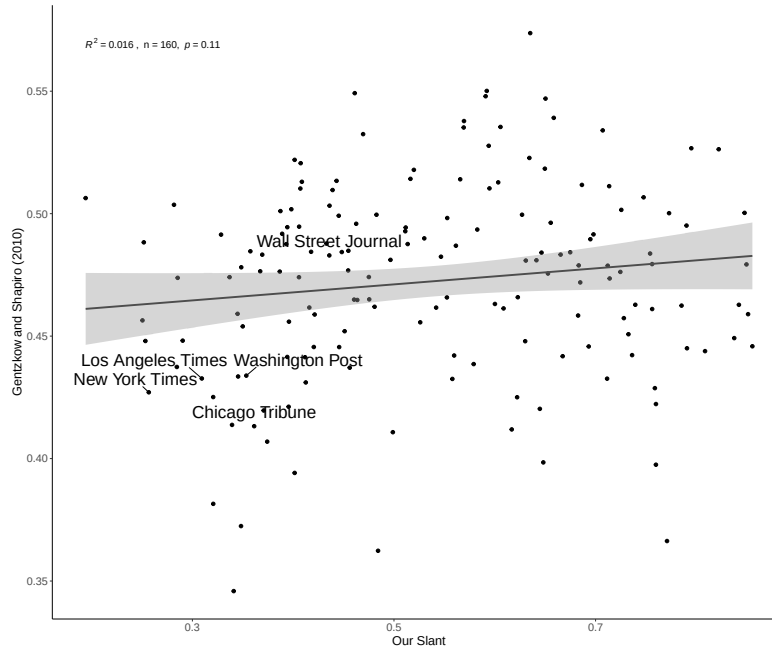
Notes: Panel A plots the distribution, across authors, of the average slant of each author's article as measured by our RoBERTa model of slant for *The New York Times* for our main analysis dataset. The vertical lines display the average slant for major Democratic presidential candidates (blue) and major Republican presidential candidates (red) in 2016 across articles written by those authors in *The New York Times*. Panel B is the same as Panel A, except for *The Wall Street Journal*. The number of articles in each set is indicated in parentheses. The politician articles include articles outside the main analysis dataset.

Appendix Figure C2: Comparison to Other Measures of Slant or Consumer Partisanship

Panel A: Comparison to Berkman-Klein

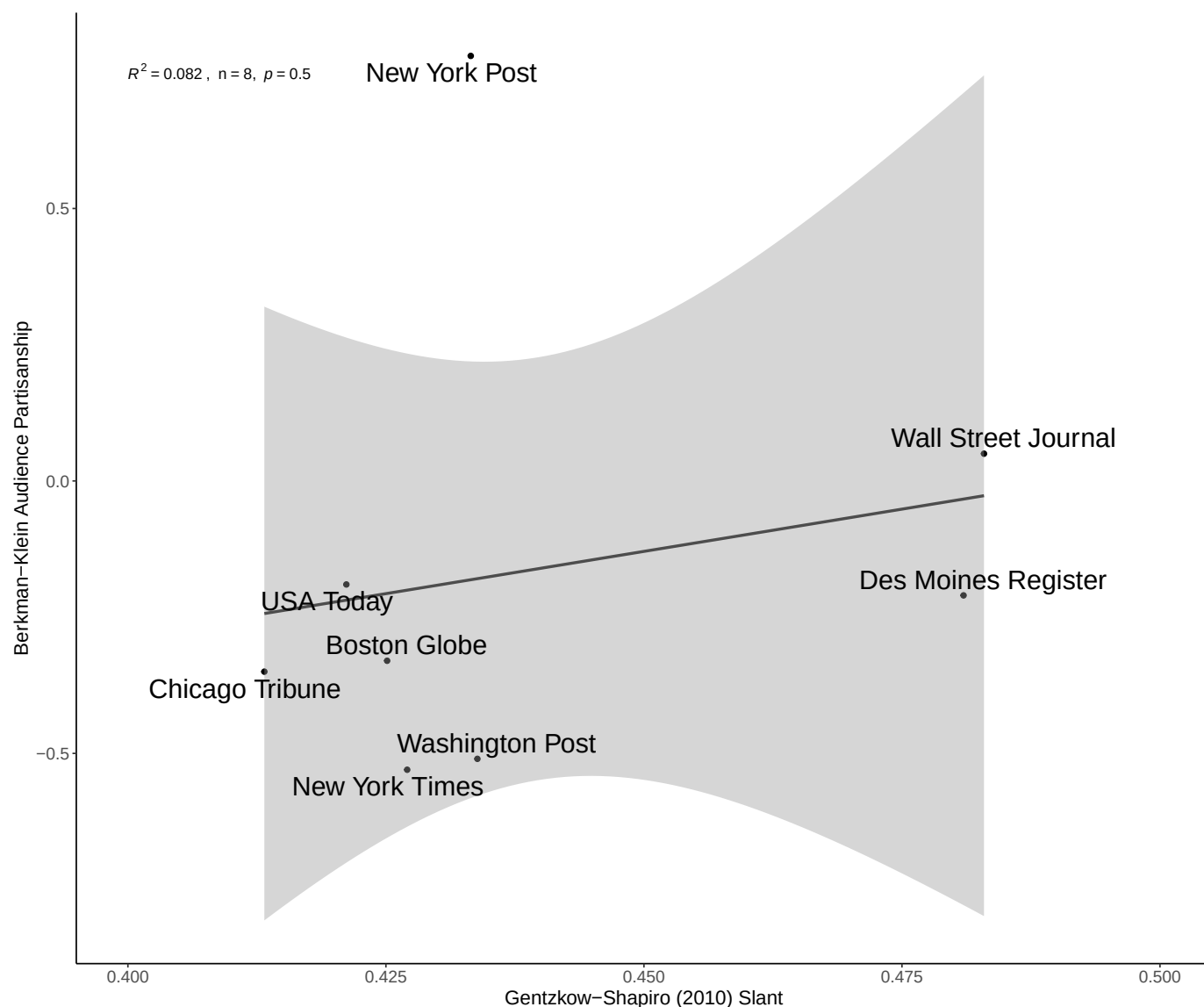


Panel B: Comparison to Gentzkow and Shapiro (2010)



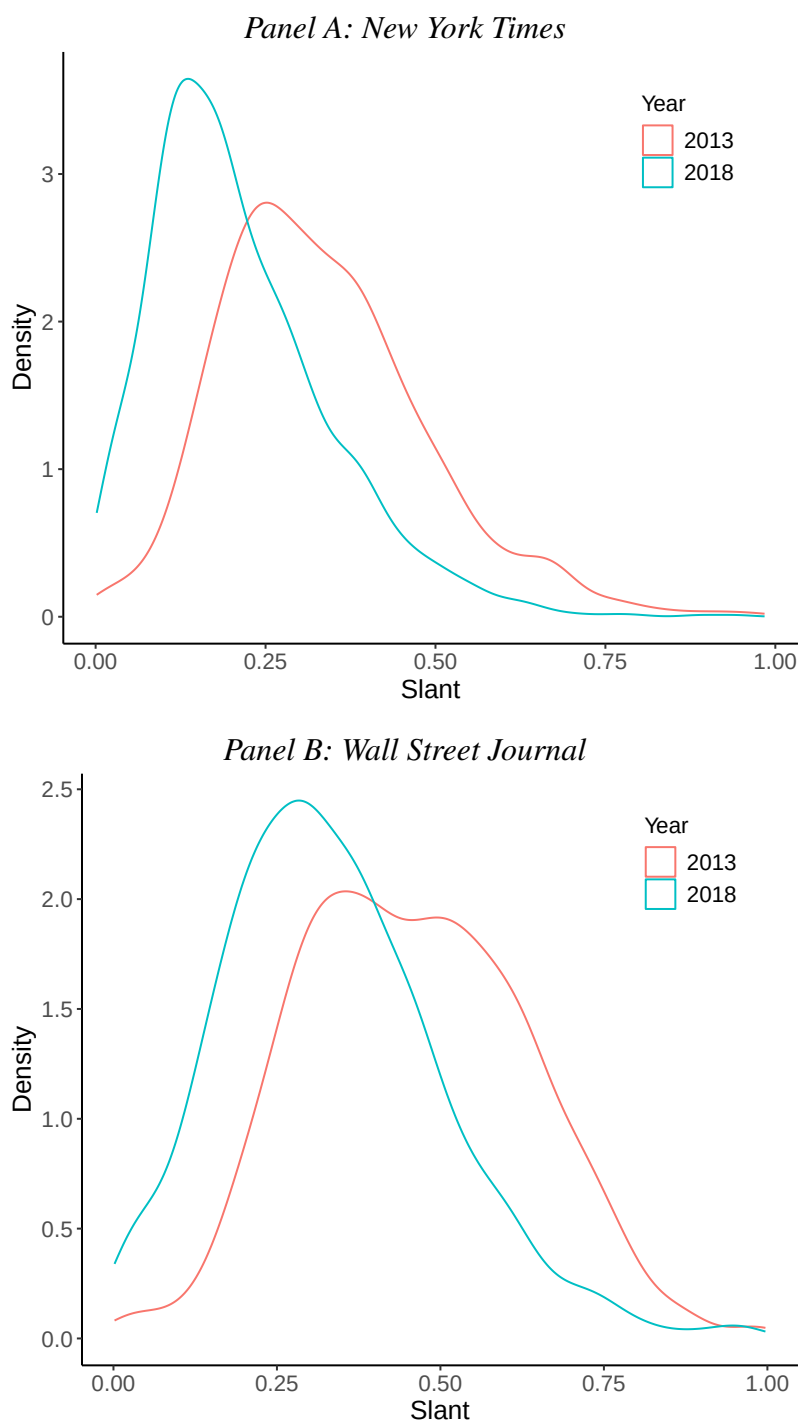
Notes: Panel A plots the comparison of our measure of slant (x-axis) to a measure of audience partisanship from the Berkman-Klein study on the 2016 election (y-axis). Panel B plots the outlet-level slant as measured in Gentzkow and Shapiro (2010) on the x-axis, and the average slant across articles as measured by our RoBERTa model of slant. Each point represents a single outlet. We match outlets between the ProQuest data and the Gentzkow and Shapiro (2010) data using a fuzzy string matching algorithm. We exclude one outlier with a negative slant score in the Gentzkow and Shapiro (2010) data.

Appendix Figure C3: Comparing Gentzkow and Shapiro (2010) to Berkman-Klein Partisanship



Notes: Figure plots the outlet-level slant as measured in Gentzkow and Shapiro (2010) on the x-axis and a measure of audience partisanship from the Berkman-Klein 2016 election study on the y-axis. Each point represents a single outlet. We exclude one outlier with a negative slant score in the Gentzkow and Shapiro (2010) data.

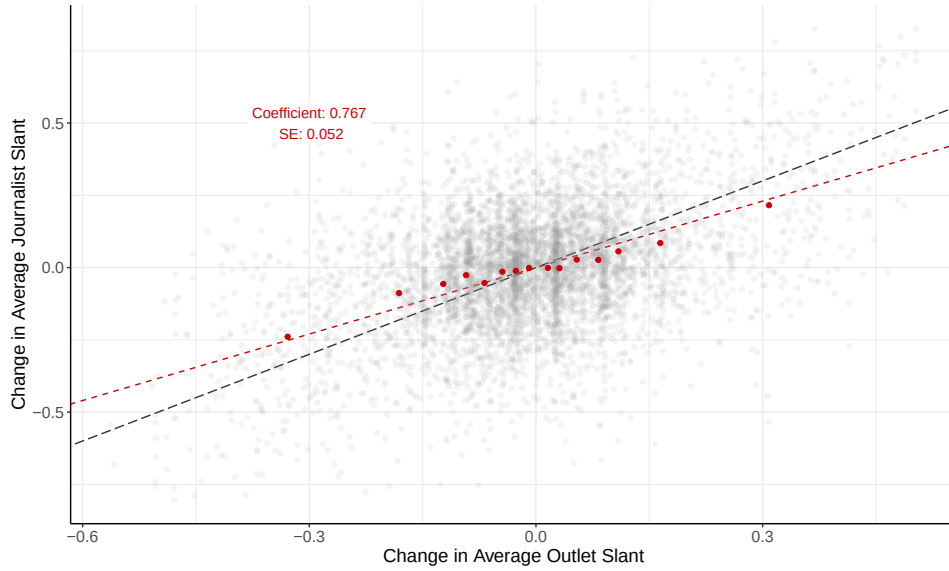
Appendix Figure C4: Authors at *The New York Times* and *The Wall Street Journal* — 2013 vs. 2018



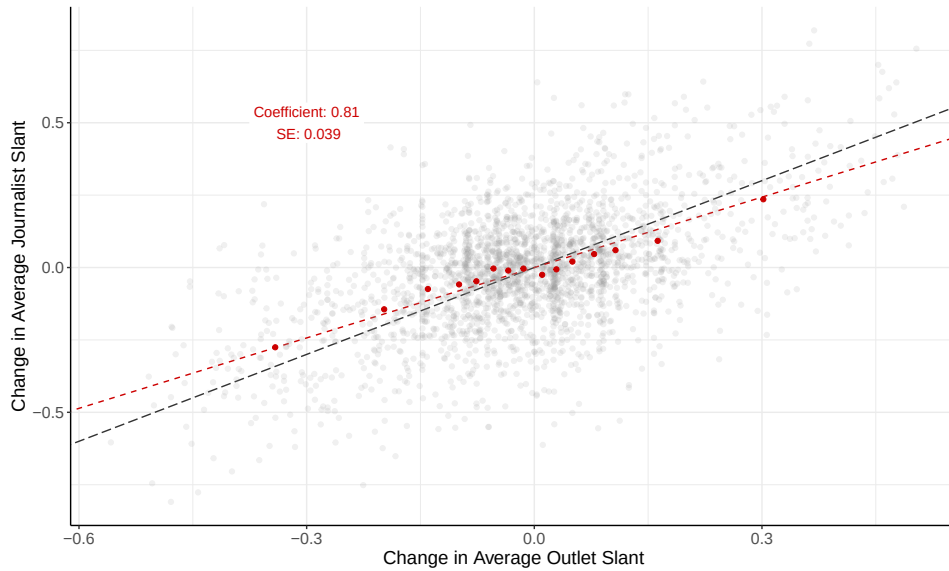
Notes: Figure plots the distribution, across authors, of the average slant of each author's article as measured by our RoBERTa model of slant for each outlet. Panel A includes all authors who appear in *The New York Times* for 2013 and 2018 separately. Panel B is the same as Panel A, except for *The Wall Street Journal*.

Appendix Figure C5: Horizon Robustness for Pooled Pre/Post Changes in Journalist Slant

Panel A: $T = 5$ Event-Study Sample

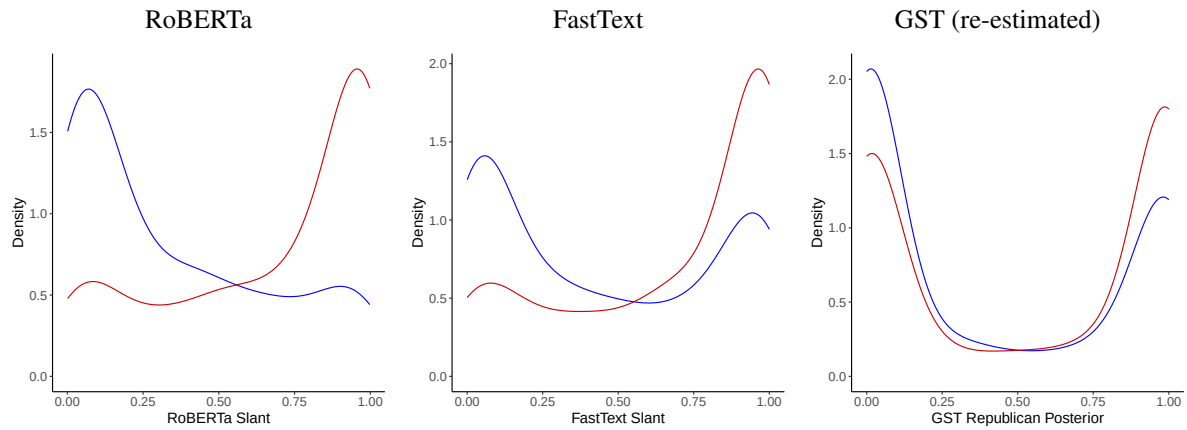


Panel B: $T = 10$ Event-Study Sample



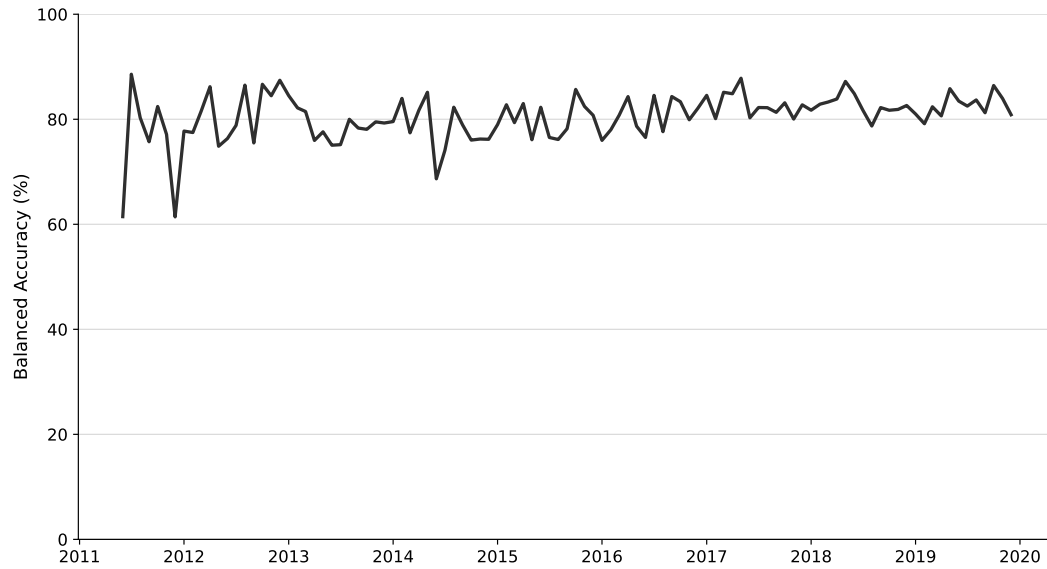
Notes: Each panel plots the mover-level change in mean journalist slant between the post- and pre-move windows against the leave-focal-author-out outlet gap. Both variables are constructed after removing publication-month means. The samples match the corresponding event studies. Panel A uses the baseline $T = 5$ window and Panel B uses $T = 10$ as a horizon robustness check. The red points show unweighted means within 15 equal-count bins. The fitted line is constrained through the origin and weights observations by the squared outlet gap; standard errors are two-way clustered at the origin- and destination-outlet levels.

Appendix Figure C6: Comparing to Human Perceived Slant — Distributions



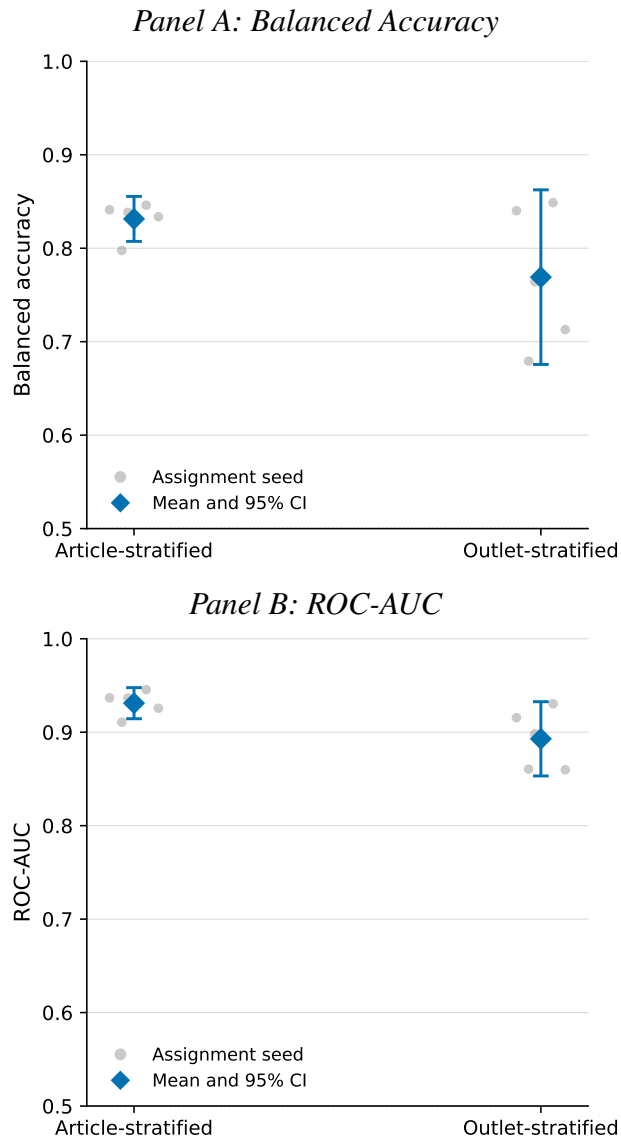
Notes: Figure plots the distribution of the text-based measure separately for articles coded as leaning Republican (red) and articles coded as leaning Democratic (blue) by the MTurk coders in Budak et al. (2016). The sample is restricted to articles with a non-zero blinded-coder rating. GST is the re-estimated Republican posterior probability. The density bandwidth is 0.1 for each measure; curves may extend beyond the score support.

Appendix Figure C7: Validation Balanced Accuracy by Tweet Month



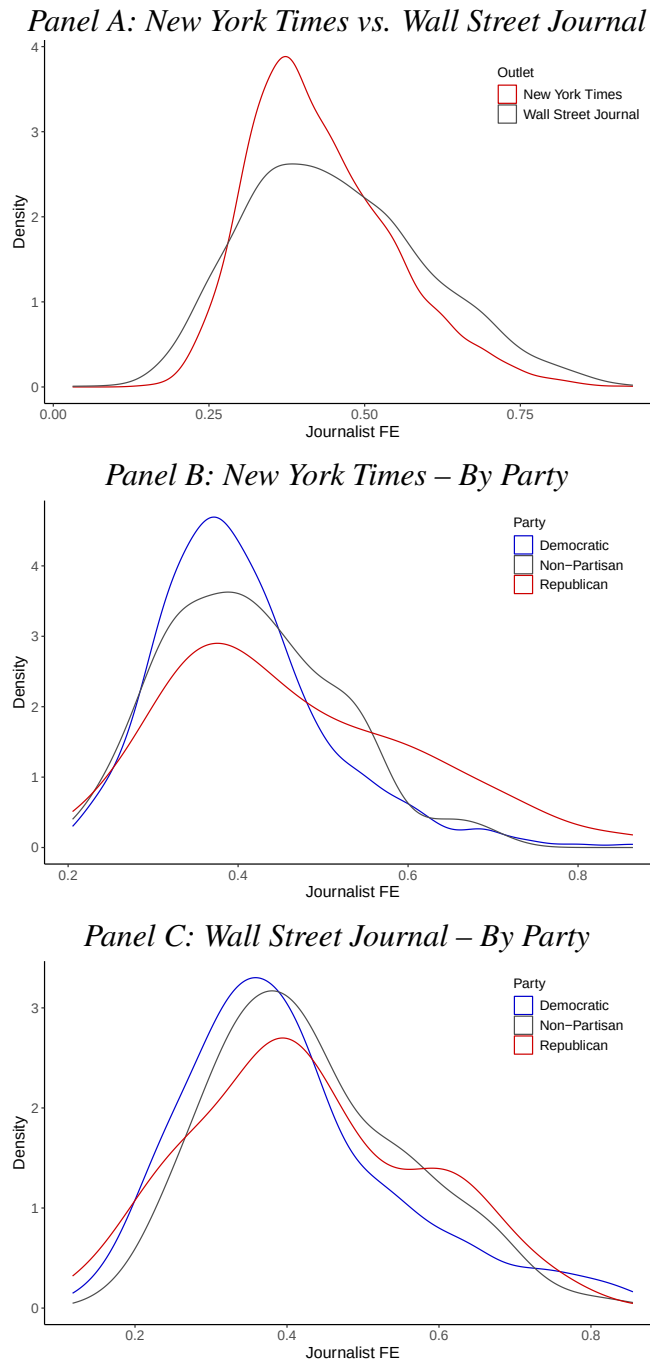
Notes: Figure plots balanced accuracy in the validation sample, calculated separately by the month in which the article was tweeted.

Appendix Figure C8: Out-of-Sample Validation for Article-Stratified vs. Outlet-Stratified Splits



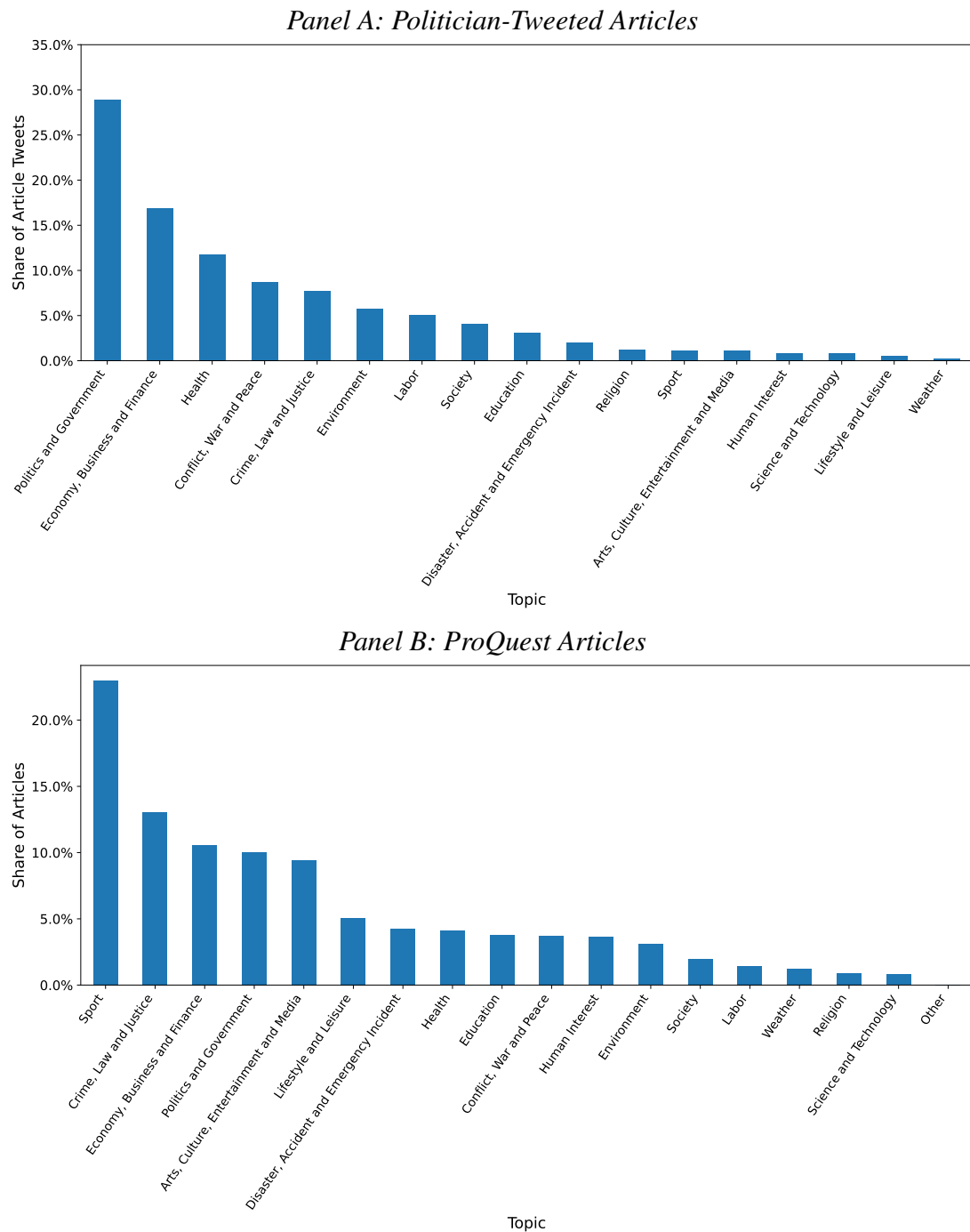
Notes: Both designs use the same sample of politician-tweeted articles from June 1, 2011 onward, excluding missing or unknown outlets and identical texts attributed to multiple outlets. Article-stratified splits assign whole groups of identical article texts; outlet-stratified splits assign whole outlets. Neither design allows identical texts in both training and validation. Dots show five random split assignments per design, diamonds show means, and whiskers show two-sided 95 percent confidence intervals across assignments. The training specification and model initialization seed are held fixed. Two-sided unpaired t-tests (allowing unequal variances) of the differences between the outlet- and article-stratified means give $p = 0.139$ for balanced accuracy and $p = 0.054$ for ROC-AUC.

Appendix Figure C9: Heterogeneity in Journalist AKM Fixed Effects — WSJ and NYT



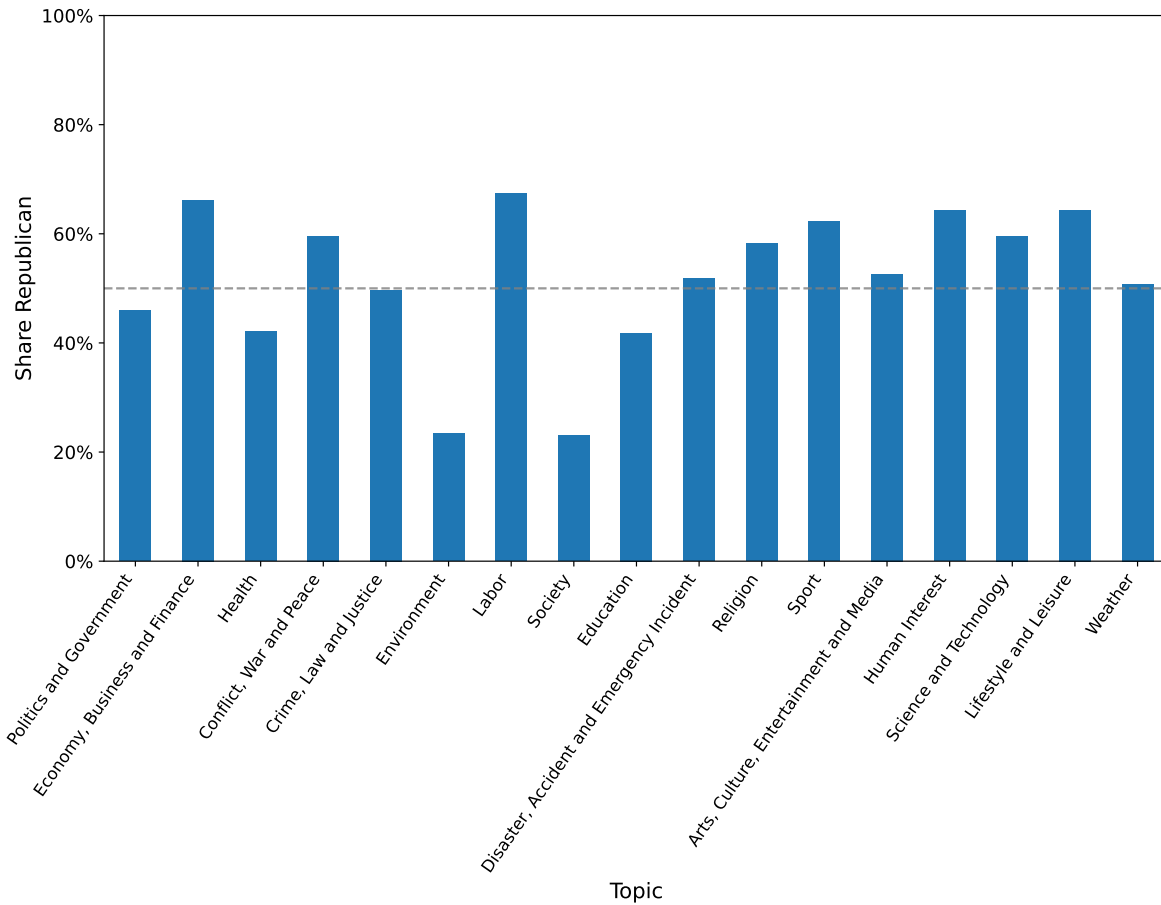
Notes: Each panel plots the distribution in estimated AKM journalist fixed effects across various sub-groups of journalists. Panel A examines heterogeneity across journalists at *The New York Times* and *The Wall Street Journal*. Panel B examines heterogeneity at *The New York Times* by partisan affiliation. Panel C examines heterogeneity at *The Wall Street Journal* by party affiliation.

Appendix Figure C10: Topic Distributions in Politician-Tweeted and ProQuest Article Samples



Notes: Panel A reports the share of reweighted politician article tweets in each IPTC topic, after reweighting article tweets to balance the overall Democratic and Republican shares. Panel B reports the share of ProQuest articles in each IPTC topic in the article sample used for the topic-slant analysis.

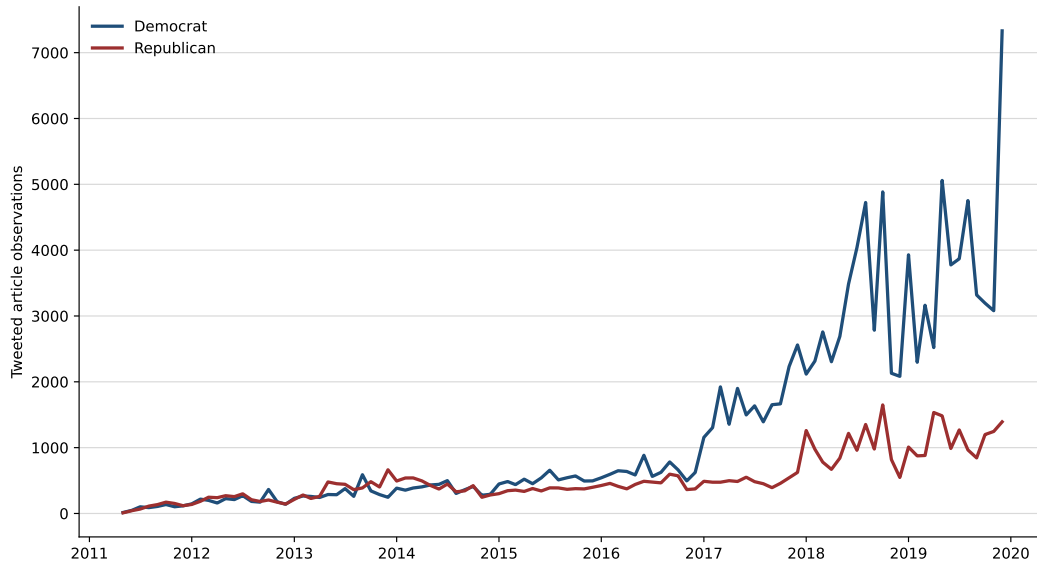
Appendix Figure C11: Republican Share by Topic in Politician-Tweeted Articles



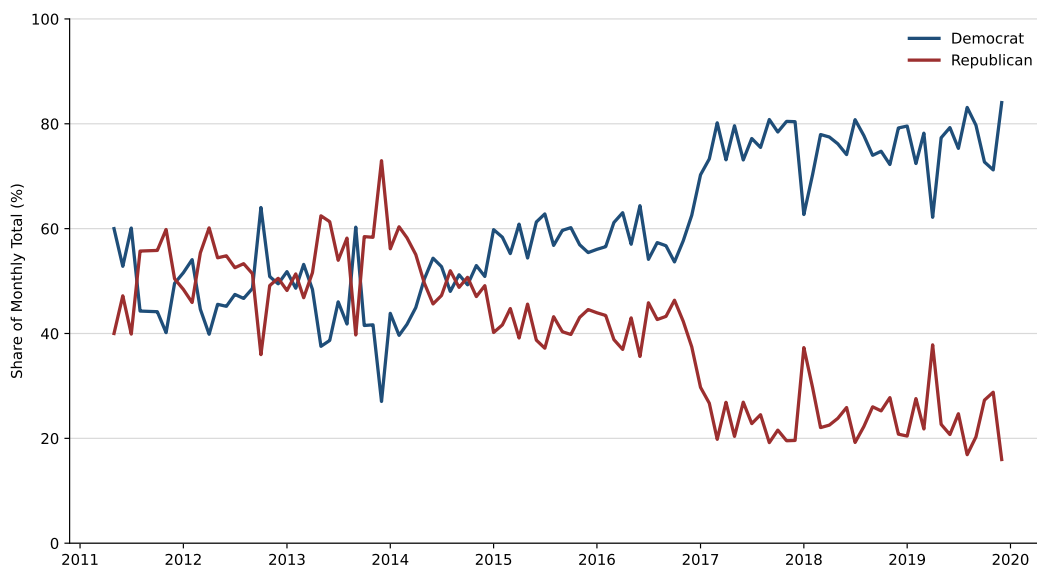
Notes: Figure reports the weighted share of politician-tweeted articles within each IPTC topic that were tweeted by Republican politicians, using article-tweet weights that balance the overall Democratic and Republican shares.

Appendix Figure C12: Politician-Tweeted Article Time Series

Panel A: Article-Tweet Counts by Party

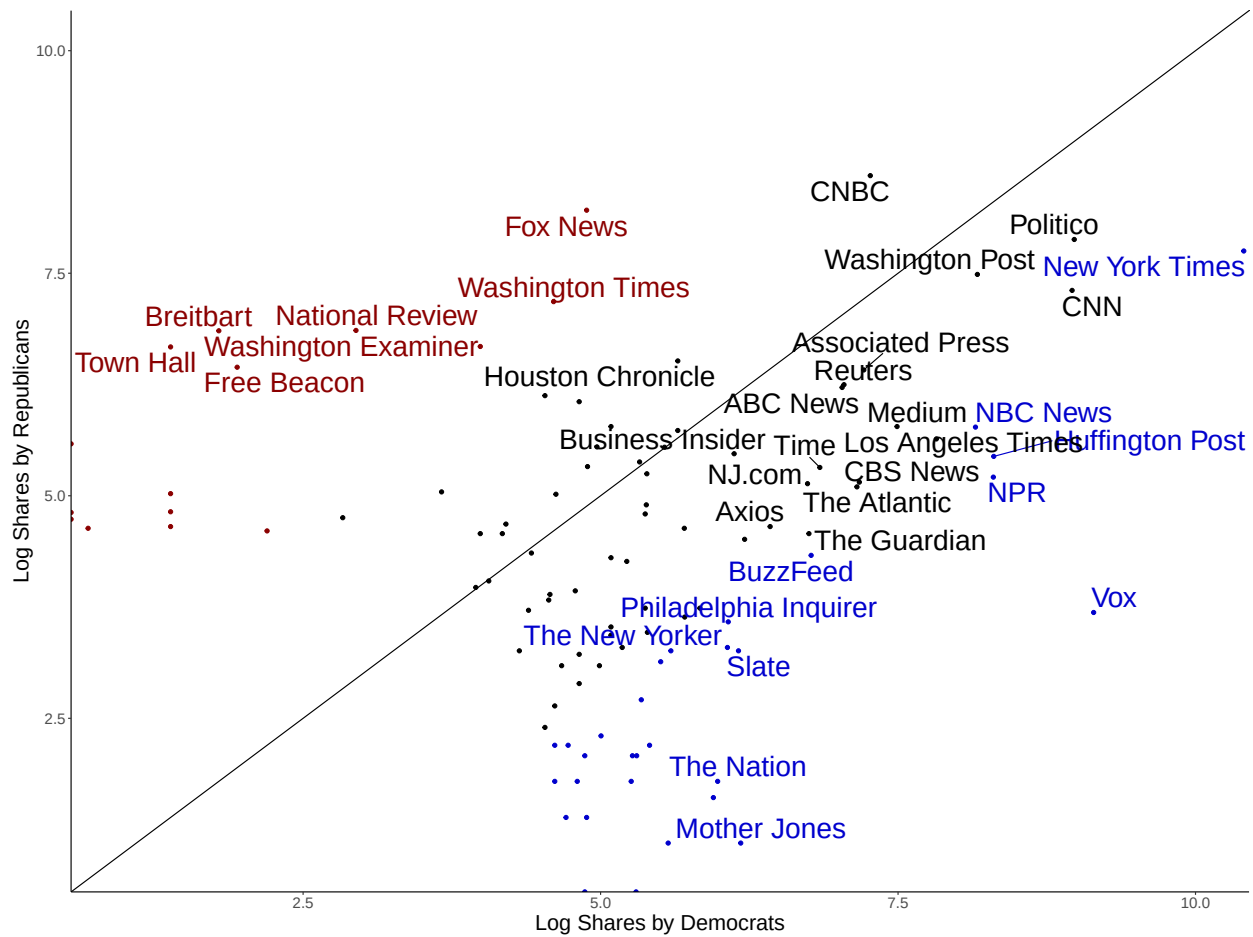


Panel B: Article-Tweet Party Shares



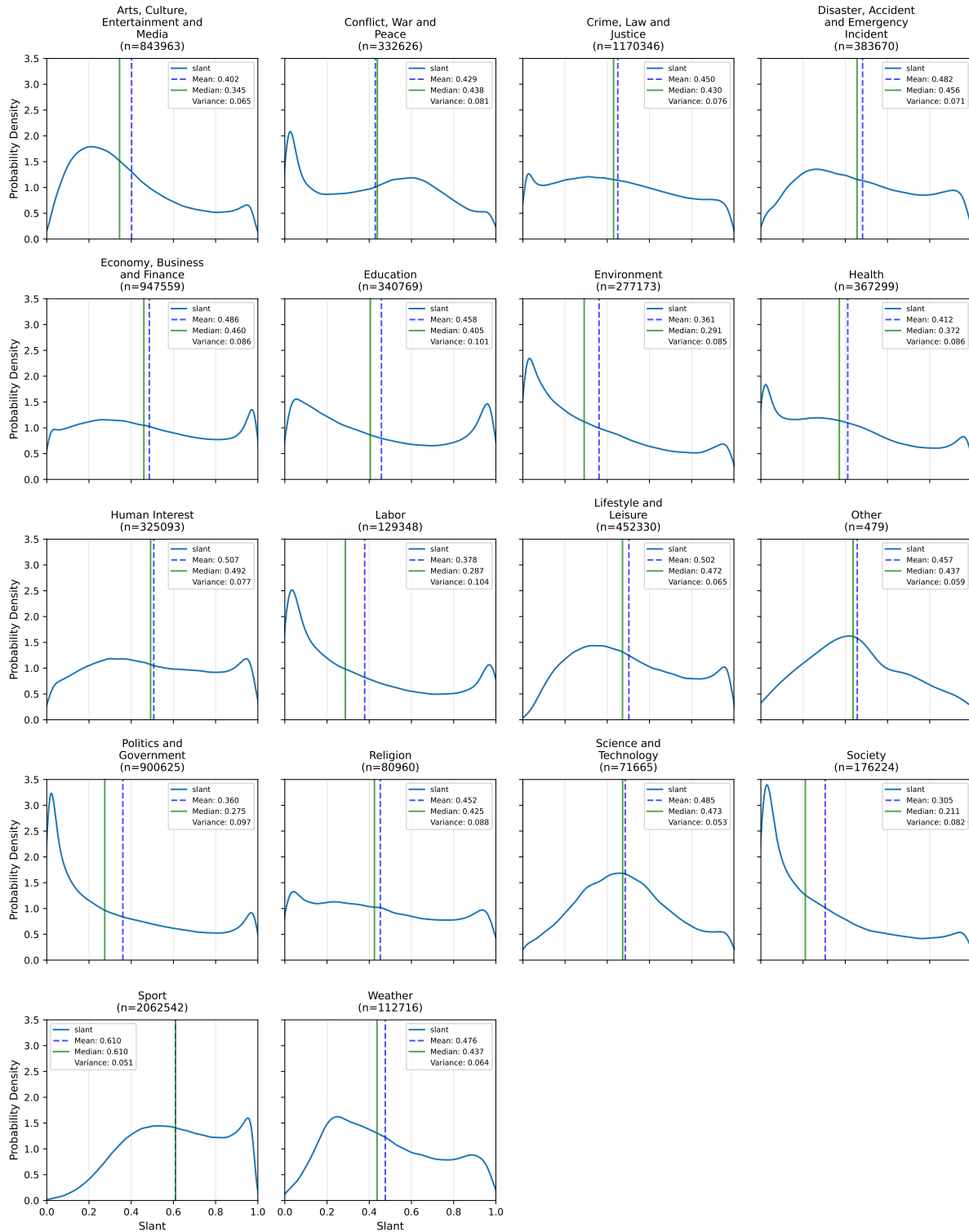
Notes: Figure summarizes the cleaned politician-tweeted article sample. Panel A reports article-tweet counts separately by month $\times$ the party of the politician who tweeted the article. Panel B reports the corresponding party shares among article-tweet observations. For visualization purposes, both panels restrict to the longest contiguous monthly span containing at least 25 tweeted-article observations per month.

Appendix Figure C13: Outlet-Level Sharing by Politician Party in the Tweeted-Article Sample



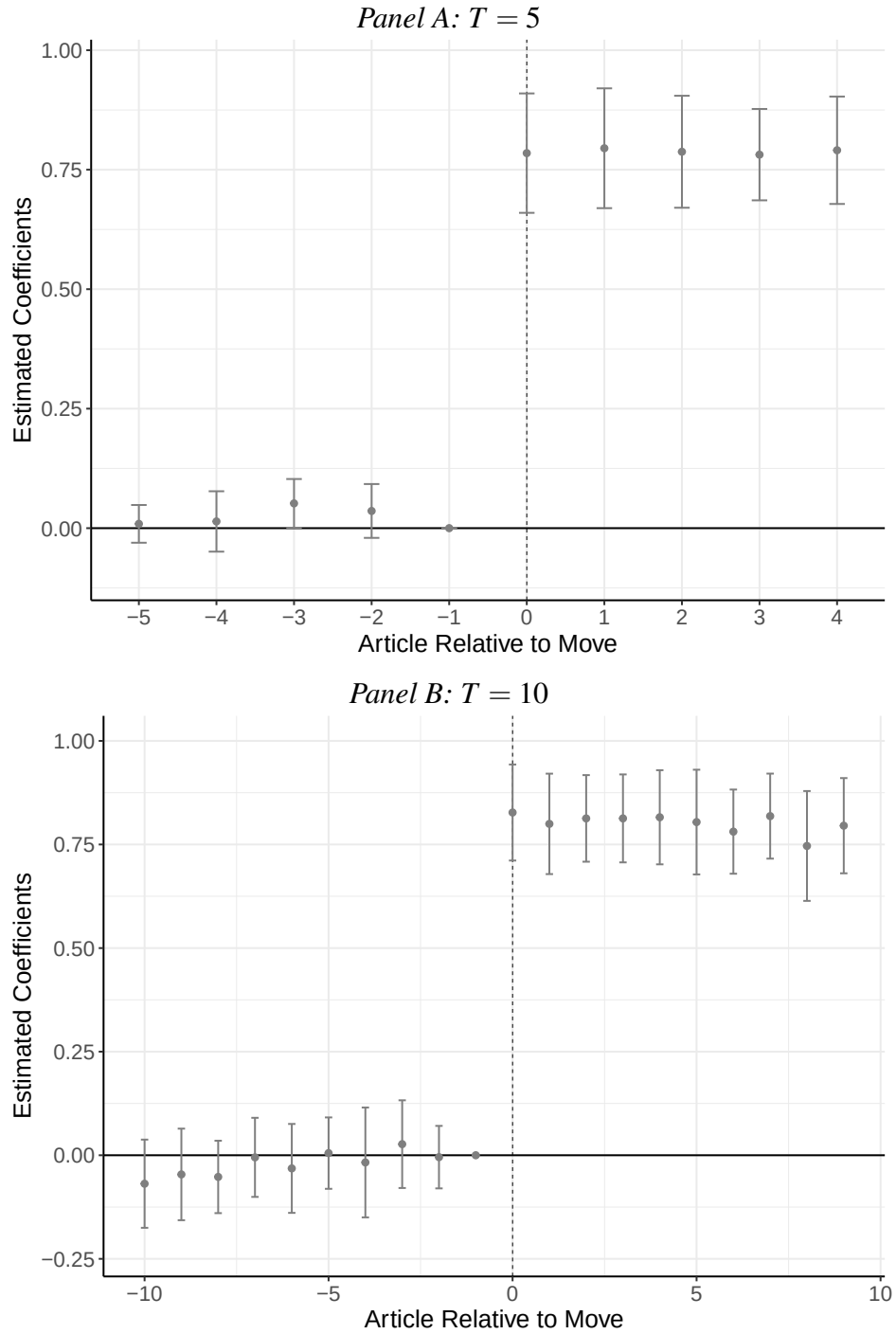
Notes: Figure plots, by outlet, the log number of article tweets shared by Democratic politicians on the x-axis and Republican politicians on the y-axis. Points above the 45-degree line are shared more often by Republican politicians, while points below the line are shared more often by Democratic politicians. Labeled outlets are selected high-frequency outlets or outlets with highly asymmetric sharing by politician party.

Appendix Figure C14: Slant Distributions by Topic in ProQuest



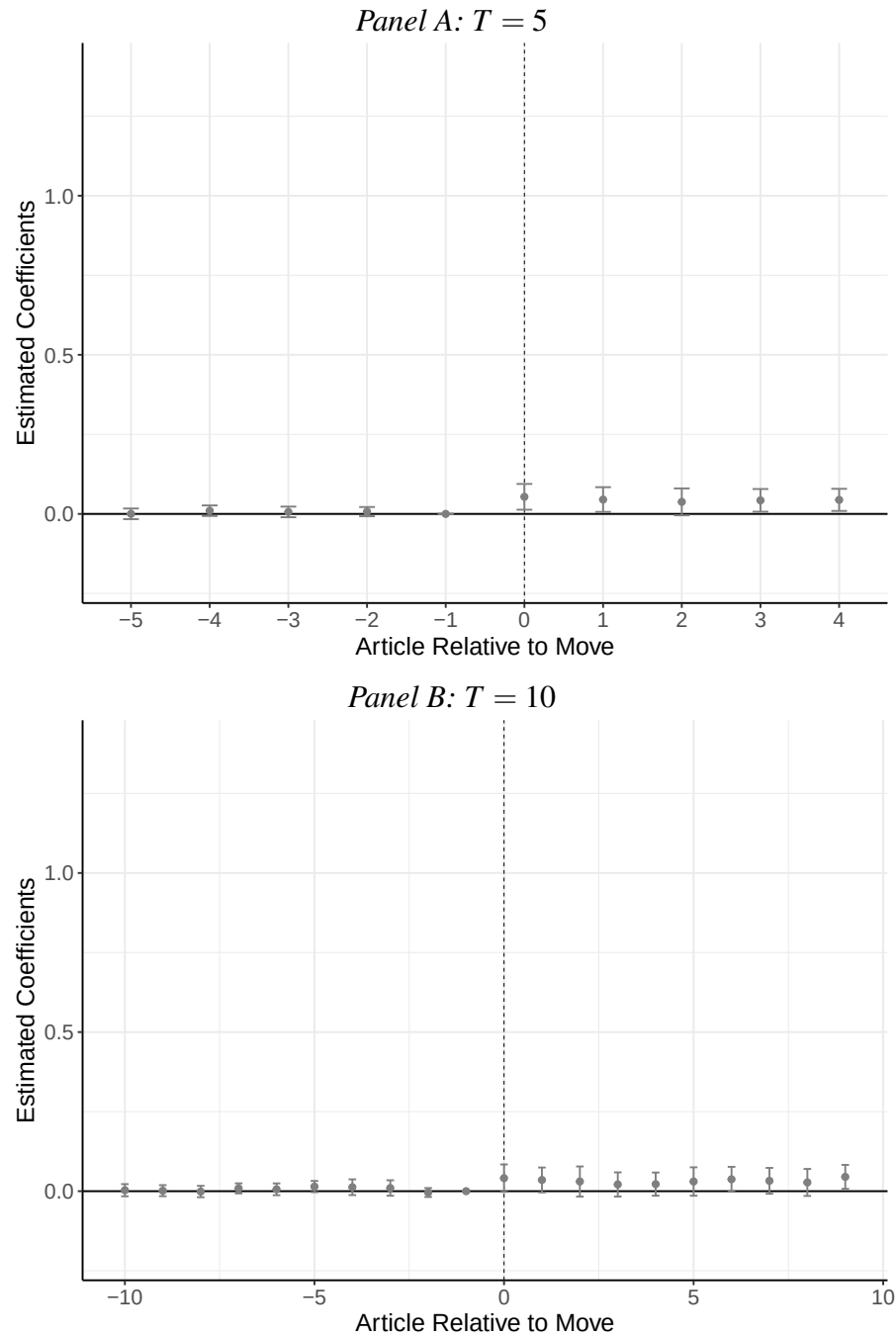
Notes: Each facet plots the distribution of article slant within a cleaned topic in the ProQuest sample. The dashed blue line marks the mean slant within the topic, and the solid green line marks the median.

Appendix Figure C15: Single-Author Event Study Changes in Slant Around Author Moves



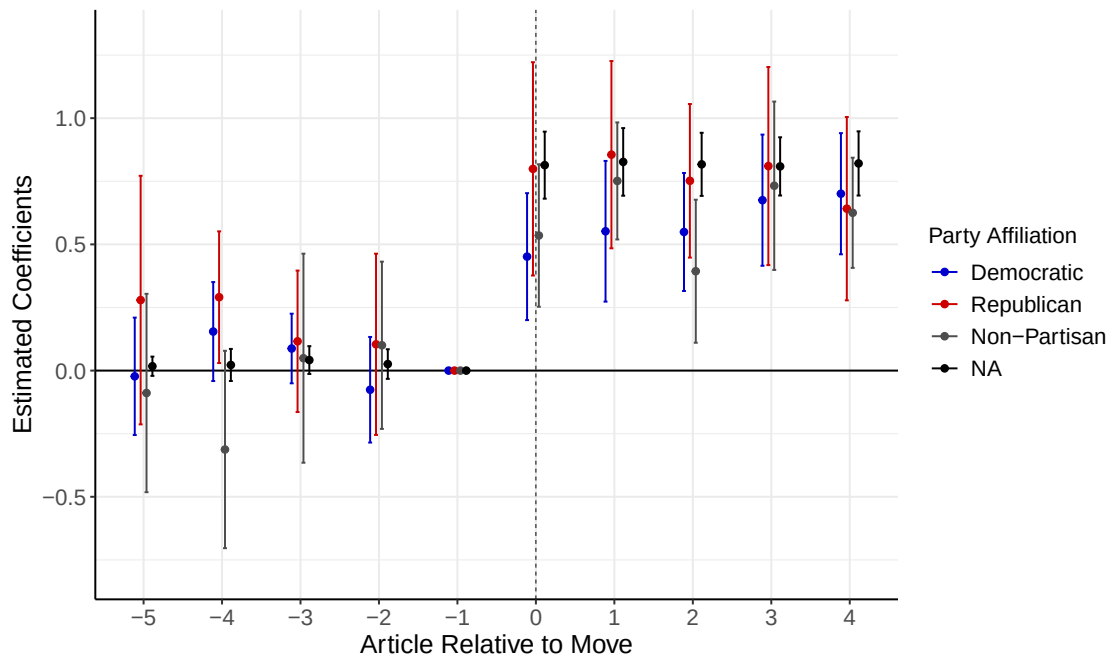
Notes: Figure follows Figure 7 in plotting the estimated mover event-study coefficients, but restricts the event-study sample to articles with a single author. Panel A uses the baseline $T = 5$ window and Panel B uses $T = 10$ as a horizon robustness check. The coefficients are normalized and weighted using the leave-focal-author-out origin-destination outlet slant gap constructed after removing publication-month means. Standard errors are two-way clustered at the origin- and destination-outlet levels.

Appendix Figure C16: Topic-Level Event Study Changes in Slant Around Author Moves



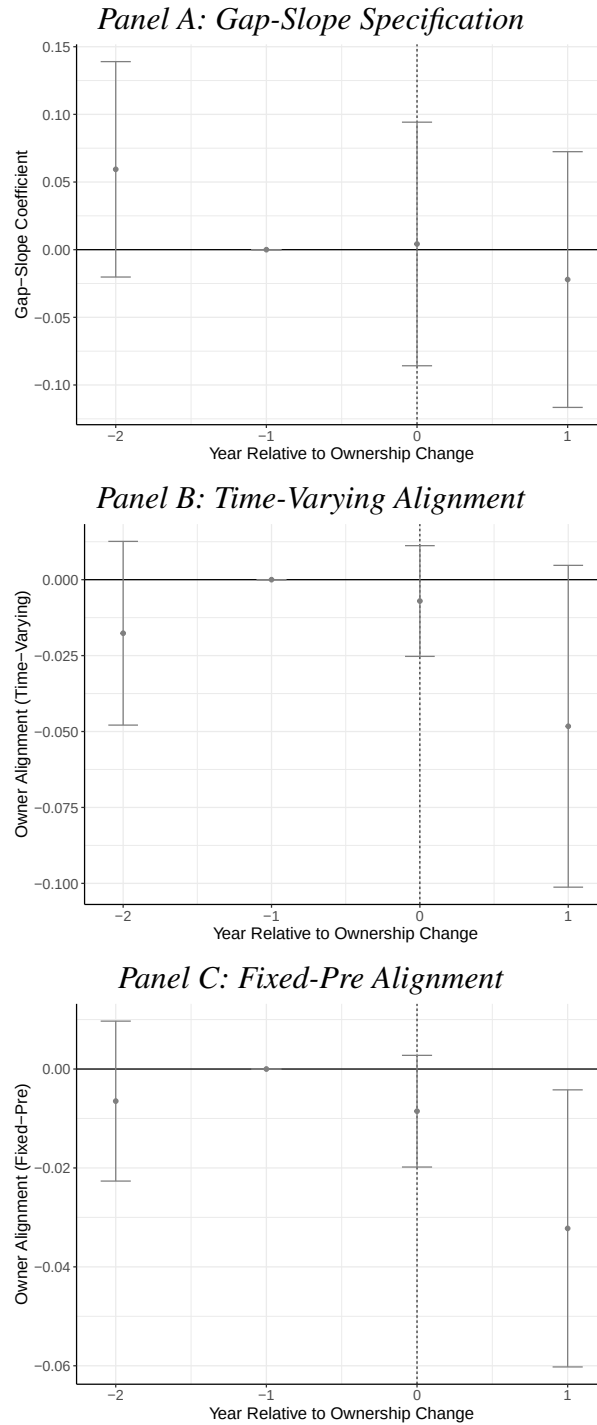
Notes: Figure follows Figure 7 in plotting the estimated mover event-study coefficients, but modifies the outcome variable to be the mean slant for the article's topic, rather than the slant of the individual article itself. Topic means are calculated across all eligible articles after removing publication-month means. Panel A uses the baseline $T = 5$ window and Panel B uses $T = 10$ as a horizon robustness check. The coefficients are normalized and weighted using the leave-focal-author-out origin-destination outlet slant gap, with move-pair fixed effects. Standard errors are two-way clustered at the origin- and destination-outlet levels.

Appendix Figure C17: Event Study Changes in Slant Around Author Moves by Party Affiliation



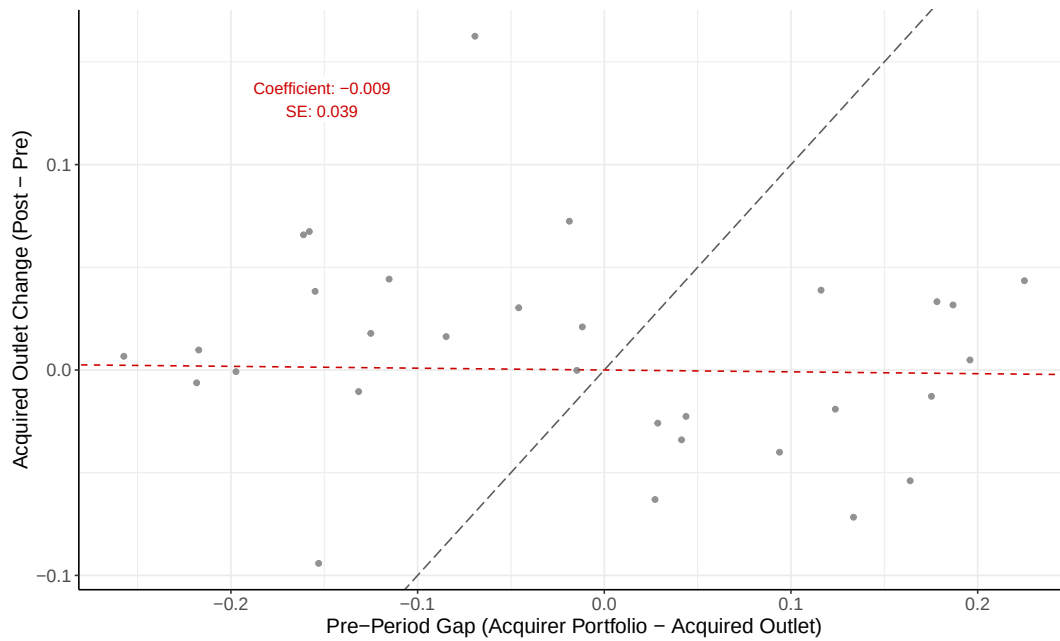
Notes: Figure follows Figure 7 in plotting estimated mover event-study coefficients for the base-line $T = 5$ estimation sample, but estimates the event study separately by journalist party-affiliation group. Party groups are Democratic, Republican, Non-Partisan, and NA. Coefficients are normalized and weighted as in the main event-study specification. Standard errors are two-way clustered at the origin- and destination-outlet levels.

Appendix Figure C18: Ownership Event Studies



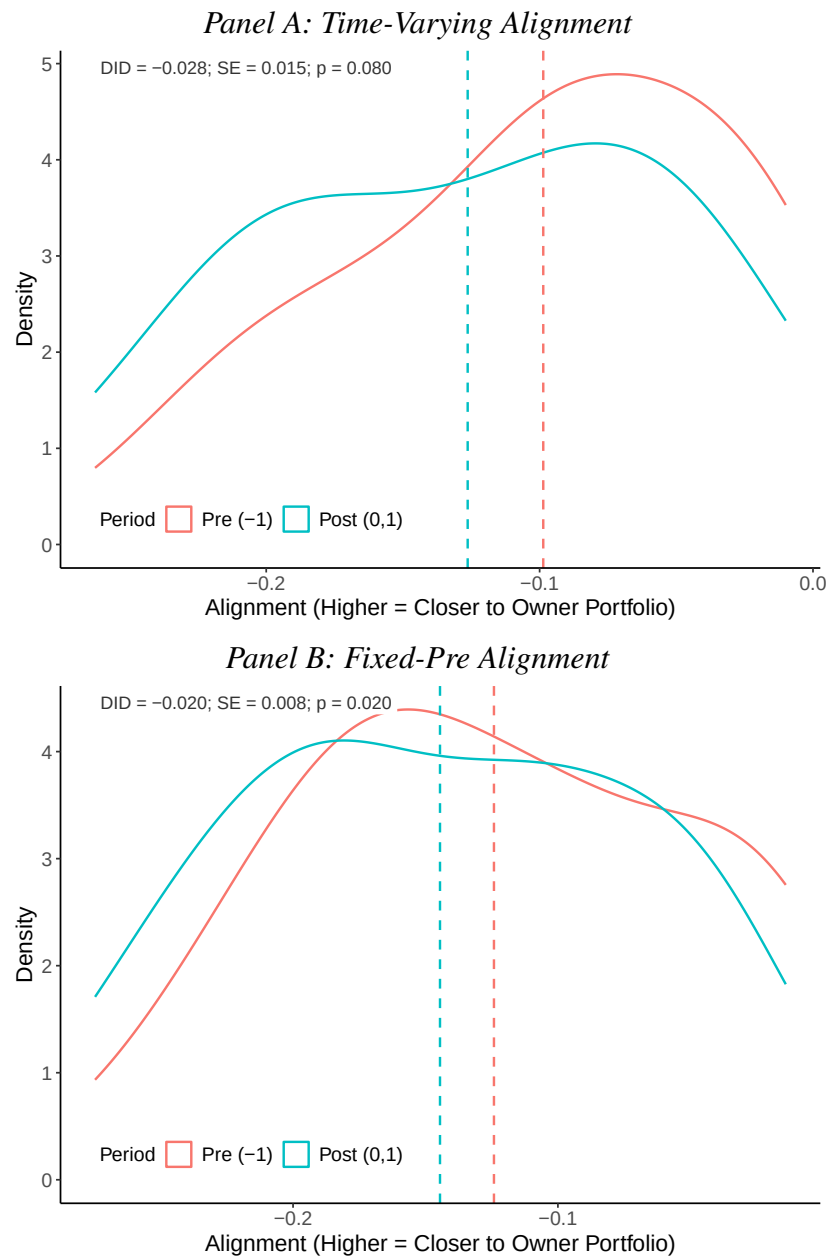
Notes: Figure reports dynamic event-study estimates around newspaper ownership changes using the balanced $[-2, +1]$ event window. All specifications omit the year before acquisition as the base period. Panel A uses the acquiring-owner pre-period slant gap as the treatment-intensity measure and weights by the squared gap in a mover-style analysis. Panels B and C report owner-alignment specifications with equal event-year weights. Standard errors are clustered at the event level. See Appendix Section B.5 for detail.

Appendix Figure C19: Ownership Change Movers-Style Variation



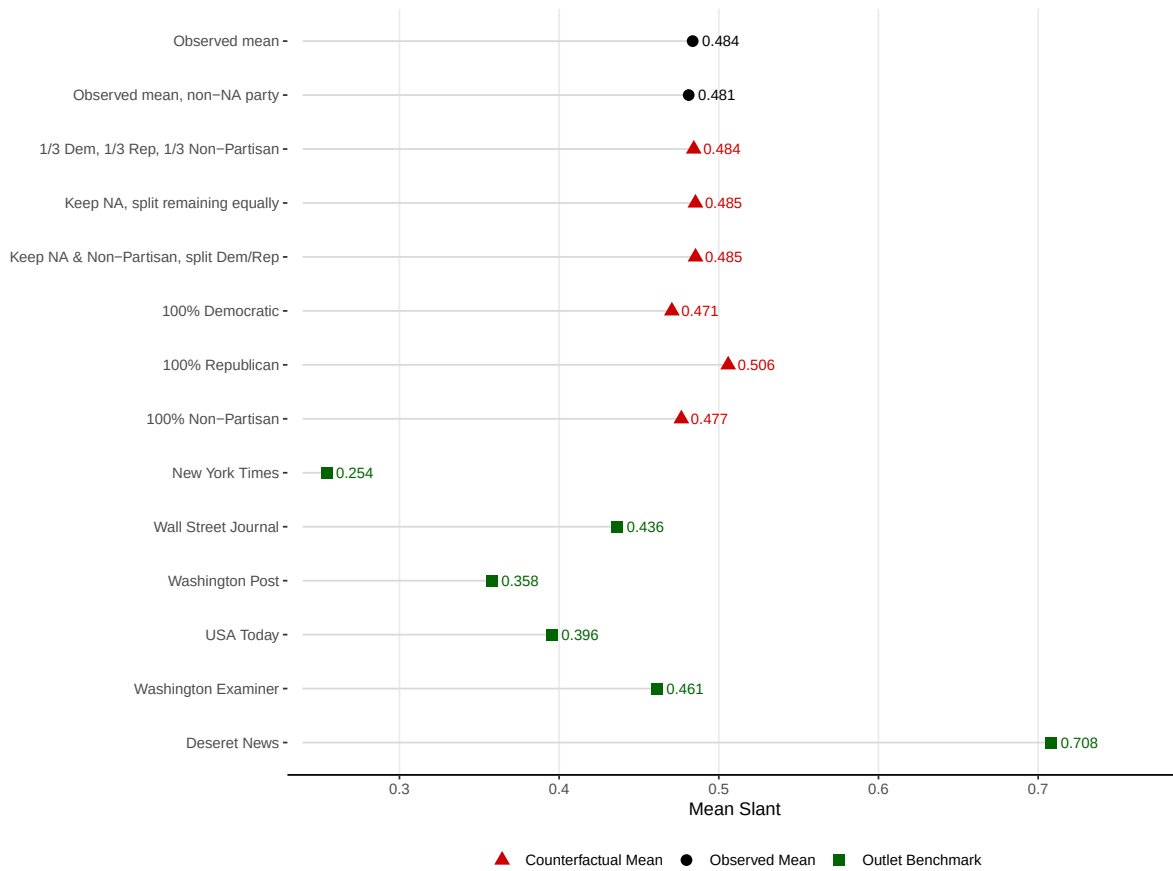
Notes: Figure reports the event-level variation underlying the pooled estimate for the gap-slope specification in Figure C18 Panel A. The figure plots the acquiring-owner pre-period slant gap against the acquired outlet's post-minus-pre change in calendar-year-residualized slant, where the pre period is event year -1 and the post period averages event years 0 and 1 . The red dashed line represents a best-fit line (without an intercept), with slope coefficient and standard error reported in the annotation. Standard errors are clustered at the event level. The dashed gray line represents a 45-degree line. See Appendix Section B.5 for detail.

Appendix Figure C20: Ownership Change Alignment Shift Variation



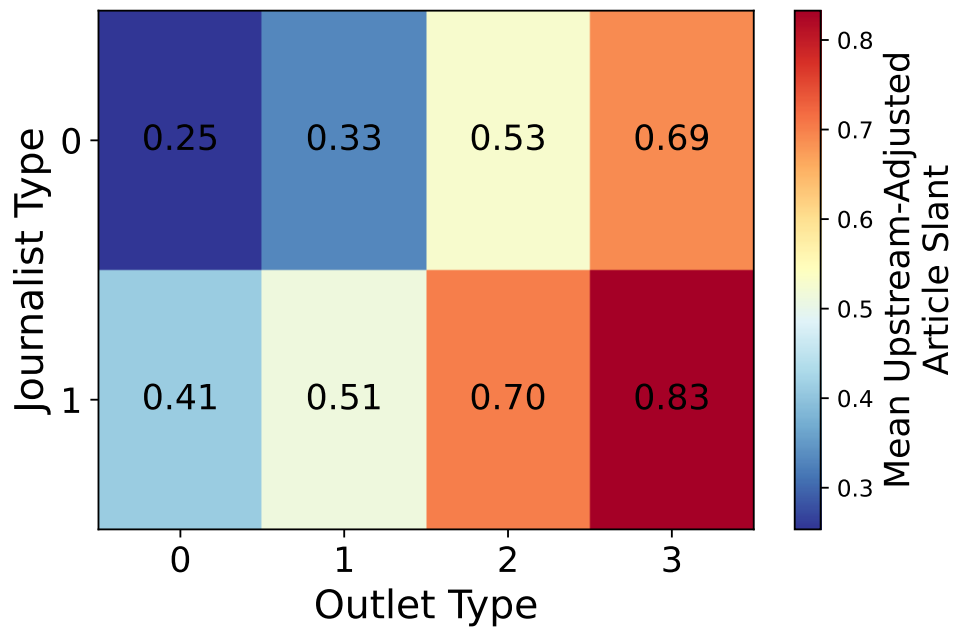
Notes: Figure reports the event-level alignment variation underlying the pooled estimates in Figure C18. Panels A and B plot pre- and post-period owner-alignment densities across events, with dashed lines indicating period means. The annotations report point estimates and standard errors for the difference in means. See Appendix Section B.5 for detail.

Appendix Figure C21: Counterfactual Mean Slant under Alternative Journalist Party Compositions



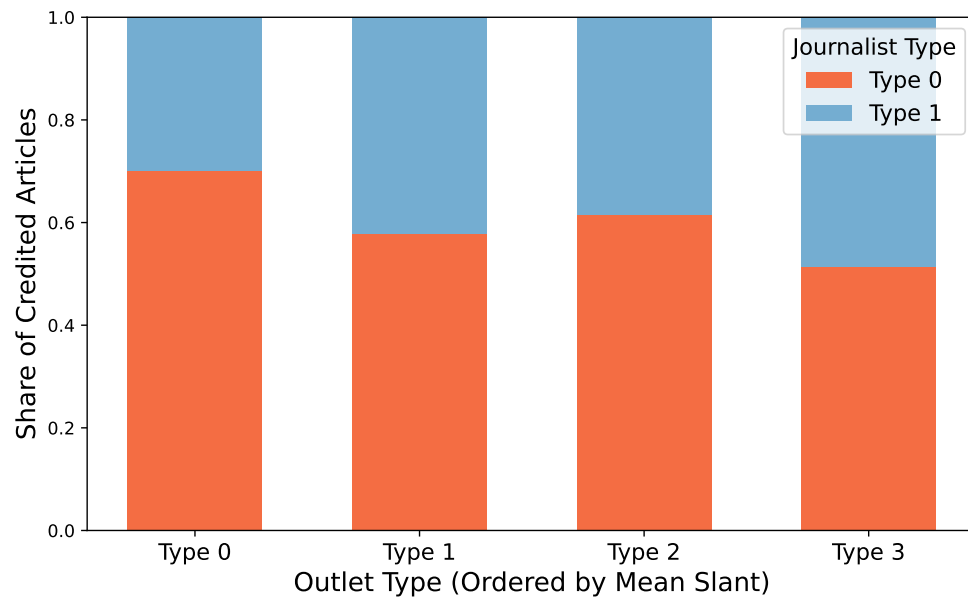
Notes: Figure plots observed and counterfactual mean slant estimates. Counterfactual means reweight the distribution of journalist party affiliations (and associated average AKM journalist fixed effects) while holding the average outlet and time effects fixed. See Section 5.3 for detail. The figure also reports benchmark observed mean slant for selected outlets, including *The New York Times*, *The Wall Street Journal*, *The Washington Post*, *USA Today*, the *Washington Examiner*, and the *Deseret News*.

Appendix Figure C22: Average Slant by Journalist and Outlet Types



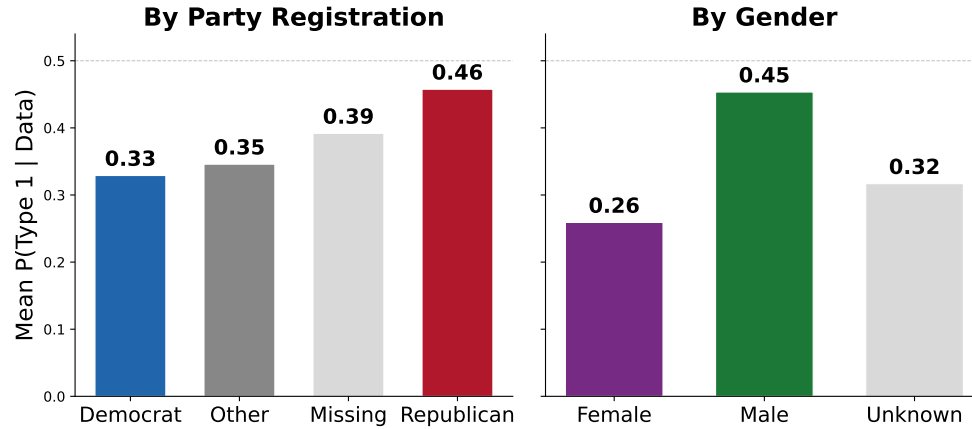
Notes: Figure shows the estimated mean slant for each combination of journalist type (rows) and outlet type (columns), from the BLM model with two journalist types and four outlet types. Outlet types are ordered by mean slant, from most liberal (type 0) to most conservative (type 3). The sample contains all credited articles in the two spells around each journalist's first move. The upstream calendar-adjusted article slant, with its common level restored, is used to estimate one shared type-by-class mean matrix across origin and destination roles. These means are displayed directly.

Appendix Figure C23: Distribution of Journalist Type by Outlet Type



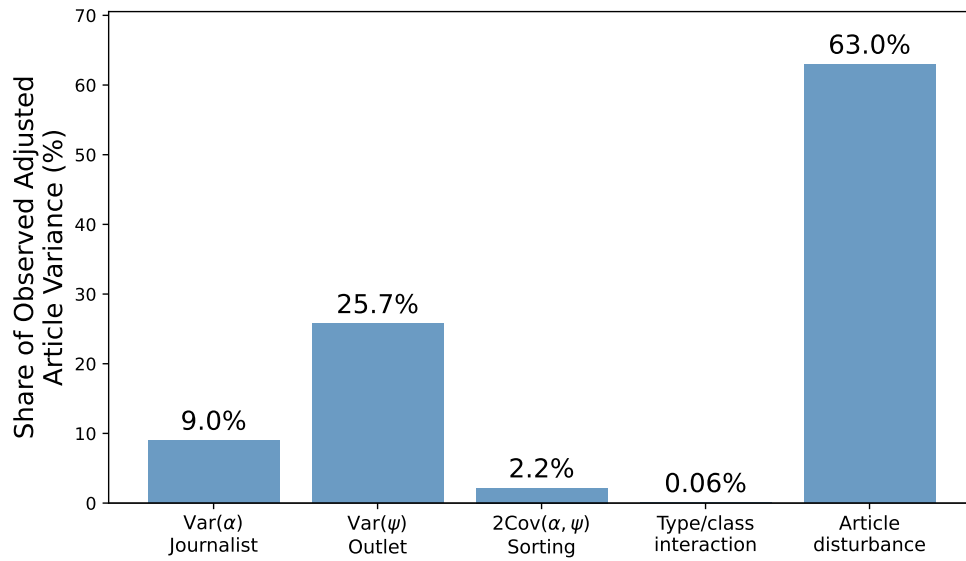
Notes: Figure shows the share of credited articles attributed to each journalist type within each outlet type, using soft posterior probabilities. The sample contains both complete spells around each journalist's first move. Outlet types are ordered by mean adjusted article slant, from most liberal (type 0) to most conservative (type 3).

Appendix Figure C24: Projection of Journalist Type onto Observable Characteristics



Notes: Figure shows the mean posterior probability of being classified as a type-1 (conservative) journalist by party registration (left panel) and gender (right panel). Posterior probabilities are computed from the BLM model with two journalist types and four outlet types, estimated on upstream calendar-adjusted articles in both spells around each journalist's first move. Each journalist receives equal weight. Party registration is obtained from voter file records; gender is inferred from first names.

Appendix Figure C25: BLM Variance Decomposition



Components sum to 99.94% of observed variance; model minus observed: -0.06 percentage points.

Notes: Figure reports model variance components as percentages of observed calendar-adjusted article variance in the two spells around each journalist's first move. The additive journalist and outlet effects project the shared type-by-class mean matrix onto additive type indicators using posterior article weights. The five components are journalist variance, outlet variance, twice their covariance (sorting), interaction residual variance, and article disturbance variance. Conditional disturbance variances remain role-specific. Publication-month means are removed upstream; no additional month effects are estimated. Components are not rescaled to sum to observed variance; the model-data gap is shown in the plot.

Appendix Table C1: Journalist Summary Statistics by L2 Linkage Status

	All journalists	Linked to L2
Journalists	33,571	10,670
Article-author observations	9,005,189	2,776,654
Outlets represented	279	276
Mean article slant	0.475	0.474
SD article slant	0.285	0.286
Mean outlet slant exposure	0.473	0.475
Mean articles per journalist	268.24	260.23
Mean active months per journalist	25.26	25.28
Mean outlets per journalist	5.36	4.07
Share eligible movers	0.093	0.084
Politics and government topic share	0.100	0.091

Notes: Table reports summary statistics for cleaned journalist identities in the ProQuest/TDM article sample by linkage status. Linked to L2 indicates that the journalist has a non-empty party affiliation in the author-level L2 output, including affiliations outside the Democratic, Republican, and Non-Partisan recode used in the party-summary table. Article-author observations count all cleaned articles by journalists in each column. The moving share counts journalists with at least one canonical valid move who satisfy the common career cap. Outlet slant exposure is the article-weighted mean outlet slant across each journalist's observed articles. Politics and government topic share is computed among articles with non-missing topic labels.

Appendix Table C2: Journalist Summary Statistics by Party Affiliation

	Democratic	Republican	Non-Partisan	NA
Journalists	5,438	2,243	2,891	22,999
Article-author observations	1,358,805	512,070	879,036	6,255,278
Outlets represented	270	261	264	279
Mean article slant	0.449	0.558	0.466	0.475
SD article slant	0.277	0.283	0.292	0.285
Mean outlet slant exposure	0.448	0.538	0.478	0.471
Mean articles per journalist	249.872	228.297	304.059	271.980
Mean active months per journalist	25.610	24.370	25.416	25.251
Mean outlets per journalist	4.080	3.544	4.480	5.946
Share eligible movers	0.081	0.071	0.101	0.097
Politics and government topic share	0.072	0.080	0.124	0.105

Notes: Table reports summary statistics for cleaned journalist identities in the ProQuest/TDM article sample by party affiliation. Party groups use the standard analysis recode of journalist party affiliation; NA indicates no observed party affiliation. The moving share counts journalists with at least one canonical valid move who satisfy the common career cap. Outlet slant exposure is the article-weighted mean outlet slant across each journalist's observed articles. Politics and government topic share is computed among articles with non-missing topic labels.

Appendix Table C3: Slant-measure anonymization comparison

	Baseline	Geography anonymized	Aggressively anonymized
Validation balanced accuracy	0.816	0.786	0.747
Validation ROC-AUC	0.901	0.876	0.833
Budak et al. correlation	0.307	0.279	0.217
$\hat{\rho}_0$ (event study, $T = 5$)	0.773	0.794	0.775
95% CI	[0.643, 0.903]	[0.668, 0.921]	[0.637, 0.914]
KSS journalist share	0.085	0.102	0.145
Outlet-decomp. journalist share	0.143	0.171	0.174
95% CI	[0.113, 0.189]	[0.128, 0.237]	[0.126, 0.252]
<i>Top 25 tokens by SHAP sum within direction</i>			
Democratic	trump, democrat, democratic, affordable, donald, climate, ibr, rights, gill, care, immigration, civil, violence, tammy, black, ud, repeal, brown, detroit, african, voting, shutdown, cummings, elijah, inequality	trump, democratic, democrat, climate, democrats, care, donald, people, change, women, brown, health, president, immigration, administration, black, affordable, ibr, gill, rights, act, public, violence, schumer, mr	democratic, democrat, democrats, climate, people, change, women, rights, health, president, civil, care, affordable, immigration, trump, environmental, public, group, black, immigrants, violence, racial, income, administration, families
Republican	said, state, republican, news, year, obama, percent, fox, years, house, texas, told, time, just, business, small, city, president, committee, sen, breitbart, gov, tax, government, national	state, republican, said, specific, news, fox, breitbart, obama, business, year, small, reported, committee, rob, jobs, told, abbott, percent, gov, local, paul, border, clinton, businesses, tuesday	republican, said, specific, state, individual, country, governor, conservative, business, year, agency, month, years, senator, small, region, time, news, economy, percent, military, jobs, economic, growth, businesses

Notes: The non-anonymized slant measure in the baseline column remains the canonical measure used in the paper. Validation statistics use a common hold-out sample of politician-tweeted articles, and the Budak et al. row reports Spearman rank correlations on a common sample of articles rated by blinded coders in Budak et al. (2016); the samples contain 26,499 and 5,673 articles, respectively. The event-study row reports $\hat{\rho}_0$, the event-time-zero gap-closure coefficient from the $T = 5$ specification corresponding to Figure 7, Panel A. Event-study article outcomes and leave-focal-journalist-outlet gaps are residualized by publication month separately for each slant measure; squared-gap weights are measure-specific, while structural event rows and estimator rules are held fixed. The KSS row reports the mover-only leave-match-out specification in Table 2, Panel B, column 2, and has no confidence interval. The outlet-decomposition row reports the calendar-residualized, article-weighted cross-fit specification corresponding to Table 3, column 1. Comparisons are across slant measures within a row, not across the three distinct event-study, KSS, and outlet-decomposition estimands. Brackets report 95 percent confidence intervals. SHAP uses Shapley values to attribute a model prediction to its input features (Lundberg and Lee, 2017). Token lists report the 25 largest absolute signed SHAP sums within each direction and separately trained model after restricting to meaningful tokens observed in at least ten sampled documents.

Appendix Table C4: Journalist Summary Statistics by Mover Status

	All journalists	Single-outlet journalists	$T = 5$ consistent movers	$T = 10$ consistent movers
Journalists	33,571	12,248	3,114	1,866
Article-author observations	9,005,189	1,726,099	1,999,915	1,302,504
Outlets represented	279	262	272	267
Mean article slant	0.475	0.488	0.470	0.467
SD article slant	0.285	0.283	0.286	0.288
Mean outlet slant exposure	0.473	0.478	0.472	0.472
Mean articles per journalist	268.243	140.929	642.233	698.019
Mean active months per journalist	25.264	19.706	40.368	43.189
Mean outlets per journalist	5.357	1.000	9.781	9.454
Share eligible movers	0.093	0.000	1.000	1.000
Politics and government topic share	0.100	0.099	0.107	0.112

Notes: Table reports summary statistics for cleaned journalist identities in the ProQuest/TDM article sample. Single-outlet journalists publish at one outlet in the broad cleaned sample. The moving share counts journalists with at least one canonical valid move who satisfy the common career cap. The $T = 10$ consistent mover column is a subset of the $T = 5$ consistent mover column. Article-author observations in mover columns count all cleaned articles by journalists with at least one qualifying consistent move, not only event-window observations. Outlet slant exposure is the article-weighted mean outlet slant across each journalist's observed articles. Politics and government topic share is computed among articles with non-missing topic labels.

Appendix Table C5: Move-Level Summary Statistics

	$T = 5$	$T = 10$
Move pairs	5,429	2,437
Unique journalists	3,114	1,866
Origin outlets	255	234
Destination outlets	243	228
Event-window article observations	54,290	48,740
Mean origin outlet slant	0.485	0.488
Mean destination outlet slant	0.474	0.466
Mean destination-origin outlet slant gap	-0.011	-0.022
Mean absolute outlet slant gap	0.117	0.124
Median absolute outlet slant gap	0.084	0.091
Share moving to more conservative outlet	0.46	0.43
Share large moves	0.44	0.47
Mean pre-move article slant	0.480	0.486
Mean post-move article slant	0.465	0.460
Mean post-minus-pre article slant	-0.015	-0.026
Event-study $t = 0$ gap-closure coefficient	0.773	0.809

Notes: Table reports move-pair summary statistics for the balanced event-study mover sample. Column T retains move pairs with exactly T articles before the move and T articles at or after the move, using rows with $|t| \leq T$ and excluding $t = T$. Outlet and article slant summary rows use raw slant; outlet rows use leave-focal-author-out means and gaps. More conservative moves and large moves (absolute gap greater than 0.1) are defined using these raw leave-out gaps. The event-study row reports the normalized gap-closure coefficient at $t = 0$ from the main weighted specification, which instead residualizes article slant and the leave-out outlet gap by publication month and includes move-pair fixed effects.

Appendix Table C6: Sample construction counts

Cleaning step	Unit	Rows	Unique articles	Author-articles
<i>Panel A. Tweet-linked articles</i>				
Tweeted URL occurrences	URL occurrences	1,196,782	1,107,105	–
Sampled for scraping	scrape rows	600,000	569,777	359,588
Join scraped text to tweets	article–tweets	1,365,407	569,777	359,588
Complete article/tweet metadata	article–tweets	844,437	328,966	355,021
Source text > 1,000 chars	article–tweets	332,500	214,816	306,004
Detected article authors	article–tweets	195,211	116,092	306,004
Exclude .gov URLs	article–tweets	192,933	114,450	303,503
Democratic or Republican shares	article–tweets	191,616	113,719	301,668
Final sentiment / cleaning input	article–tweets	191,584	113,704	301,631
Drop invalid articles	article–tweets	189,243	112,004	297,682
Drop criticizing tweets	article–tweets	184,384	108,986	290,317
Post-cleaning text > 1,000 chars	article–tweets	182,697	107,878	287,584
Keep English articles	article–tweets	182,538	107,720	287,255
Training split	article–tweets	127,777	78,213	209,403
Validation candidate	article–tweets	54,761	36,629	97,945
Validation after overlap removal	article–tweets	26,522	26,034	68,154
Training on/after 2011-06-01	article–tweets	127,707	78,143	209,215
Validation on/after 2011-06-01	article–tweets	26,499	26,011	68,084
<i>Panel B. Budak articles</i>				
Raw rows with URL and text	article rows	14,476	7,694	21,161
Drop invalid / clean non-text	article rows	14,029	7,490	20,023
Text > 1,000 chars	article rows	10,735	5,724	17,960
Deduplicate URLs / construct slant	article rows	5,724	5,724	17,960
<i>Panel C. ProQuest/TDM analysis data</i>				
Raw merged sample	article rows	11,006,646	11,006,646	11,637,445
News articles with authors	article rows	10,992,582	10,992,582	11,622,809
Keep clean parsed author names	author–articles	13,173,727	10,992,582	11,622,809
Drop duplicate author–articles	author–articles	11,622,800	10,992,582	11,622,800
Drop duplicate-mapped articles	author–articles	9,288,608	8,769,535	9,288,608
Serious authors with mapped group ID	author–articles	9,005,189	8,521,826	9,005,189

Notes: Panels A–C show how the tweet-linked, Budak, and ProQuest/TDM samples are built. Counts are measured after each listed step; when a row combines steps, they are measured after the final step. Unit says what the row counts. Unique articles are distinct articles, while author–articles are distinct author–article pairs. Training and validation are separate branches, not successive filters. The common-sample model inputs are the training split and the validation sample left after removing overlap with training. Date-filtered rows include observations on or after June 1, 2011.

Appendix Table C7: Prediction of Human Perceived Slant from Budak et al. (2016), Article Level

Dependent Variable:	Human Perceived Slant (SD)			
	(1)	(2)	(3)	(4)
RoBERTa	0.291 (0.013)	.	.	0.261 (0.015)
GST (re-estimated)	.	0.116 (0.013)	.	0.004 (0.014)
FastText	.	.	0.188 (0.013)	0.055 (0.015)
Observations	5,673	5,673	5,673	5,673
R^2	0.084	0.013	0.035	0.087

Notes: Table shows the results of regressing blinded-coder human perceived slant from Budak et al. (2016) on various text-based models of slant. Sample is composed of a random sample of articles from Budak et al. (2016). For some selected articles, we were unable to scrape the underlying text and these articles are removed. In Budak et al. (2016), each article is scored on a 5-point favorability score indicating the extent to which the article is favorable to Republicans and a separate score for how favorable the article is to Democrats. We take the difference between these two scores as our measure of perceived slant. Each article is rated twice—once by a coder who does not observe the outlet’s name, and once by a coder who does observe the outlet’s name. Following Budak et al. (2016), our human perceived slant measure uses ratings from coders who do not observe the outlet’s name. The R^2 from regressing unblinded coder slant on blinded coder slant is 0.151. The RoBERTa measure is our main measure of slant. The FastText measure is a measure of slant using Facebook’s FastText model and trained on a dataset similar to the RoBERTa model. The GST measure is the Republican posterior probability from a phrase-based classifier following Gentzkow et al. (2019), re-estimated on the same article training sample as RoBERTa, with equal party priors and the training-weighted average month effect. Human perceived slant is standardized to mean zero and standard deviation one in the full validation sample. All text-based model slant estimates are normalized to have a standard deviation of one in that sample. Regressions include an intercept; parentheses report conventional OLS standard errors.

Appendix Table C8: AKM Decompositions

	(1)	(2)	(3)	(4)
Sample:	Women	Men	White	Non-White
<i>Panel A: Plug-in Estimator</i>				
Share of Variance, x_{ijt}				
Journalist, $\text{var}(\gamma_i)$	0.180	0.214	0.191	0.272
Outlet, $\text{var}(\alpha_j)$	0.234	0.217	0.232	0.236
Sorting, $2 \times \text{cov}(\gamma_i, \alpha_j)$	-0.007	0.031	0.029	-0.072
Correlation				
Sorting, $\text{cor}(\gamma_i, \alpha_j)$	-0.018	0.072	0.068	-0.143
<i>Panel B: Leave-Out Estimator (Kline et al. 2020)</i>				
Share of Variance, x_{ijt}				
Journalist, $\text{var}(\gamma_i)$	0.116	0.166	0.152	0.206
Outlet, $\text{var}(\alpha_j)$	0.190	0.182	0.203	0.174
Sorting, $2 \times \text{cov}(\gamma_i, \alpha_j)$	0.074	0.098	0.083	0.041
Correlation				
Sorting, $\text{cor}(\gamma_i, \alpha_j)$	0.251	0.282	0.237	0.107

Notes: Table reports AKM variance decomposition exercises with publication-month fixed effects across various samples. Both panels use the leave-out connected set: Panel A reports the plug-in estimator and Panel B reports the Kline et al. (2020) estimator. Columns (1)–(4) restrict to various groups of journalists according to name-based predictions of gender and race.

Appendix Table C9: AKM Decompositions, Full Connected Sample

Sample:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	All	Dropping Stayers (Full)	Dropping Stayers (Large)	Max. 2 Outlets	Drop 1 Mo. Moves	Top 20 Outlets	$T = 5$ Full	$T = 5$ Large	$T = 10$ Full	Parent Company	Women	Men	White	Non-White
	Share of Variance, x_{ijt}													
Journalist, $\text{var}(\gamma_i)$	0.194	0.167	0.185	0.201	0.193	0.266	0.235	0.272	0.215	0.352	0.179	0.218	0.191	0.305
Outlet, $\text{var}(\alpha_j)$	0.217	0.224	0.264	0.216	0.227	0.094	0.188	0.248	0.210	0.082	0.246	0.223	0.237	0.275
Time, $\text{var}(\tau_t)$	0.025	0.027	0.022	0.025	0.025	0.045	0.029	0.024	0.029	0.022	0.027	0.024	0.024	0.035
Sorting, $2 \times \text{cov}(\gamma_i, \alpha_j)$	0.038	0.040	0.001	0.030	0.029	-0.001	0.056	-0.022	0.028	0.033	-0.012	0.022	0.025	-0.143
Other Components	0.526	0.542	0.529	0.527	0.526	0.596	0.492	0.477	0.518	0.510	0.561	0.513	0.524	0.528
	Correlation													
Sorting, $\text{cor}(\gamma_i, \alpha_j)$	0.091	0.103	0.001	0.073	0.069	-0.004	0.133	-0.042	0.066	0.096	-0.029	0.050	0.058	-0.246
	Sample Size													
Observations	7065499	1482338	489191	6790652	7021584	2437070	54290	21560	48700	4127869	2355089	4512701	5684989	583770
Journalists (All)	30504	4091	1399	29854	29991	11138	3114	1399	1864	20866	12135	17253	23984	2635
Journalists (2+ outlets)	4091	4091	1399	3460	3420	594	3114	1399	1864	1272	1443	2556	3261	411
Outlets	266	266	237	265	266	17	260	237	246	33	247	260	262	172
Months	72	72	72	72	72	72	72	72	72	72	72	72	72	72

Notes: Table reports AKM variance decomposition exercises using the plug-in estimator on the ordinary connected set with various samples. All columns use raw article slant, the raw outcome-variance denominator, and publication-month fixed effects. Column (1) uses the full sample. Column (2) drops journalists observed at only one retained outlet. Column (3) retains only origin and destination spells around moves whose absolute calendar-residualized leave-focal-author-out outlet gap exceeds 0.1, counting each author–article observation once. Column (4) drops journalists that appear in more than two outlets. Column (5) restricts to tenures of longer than one month in duration. Column (6) restricts to the top 20 outlets according to weekend circulation. Column (7) uses the $T = 5$ event study sample. Column (8) is the same as Column (7) except that it restricts to moves whose absolute calendar-residualized leave-focal-author-out outlet gap exceeds 0.1. Column (9) uses the $T = 10$ event study sample. Column (10) treats all outlets owned by a single “Parent Company” as the same outlet and restricts attention to outlets for which we observe the parent company. Columns (11)–(14) restrict to various groups of journalists according to name-based predictions of gender and race.

Appendix Table C10: Full Parameter Estimates

	Estimate	SE		Estimate	SE
β	0.674	0.053	δ ideal, Dem	0.537	0.027
$\beta/(\beta + 1)$	0.403	0.019	δ ideal, Rep	0.636	0.028
τ_ζ	1.162	0.057	δ ideal, NP	0.571	0.028
$\tau_{\delta, \text{Dem}}$	0.679	0.090	δ ideal, NA	0.562	0.017
$\tau_{\delta, \text{Rep}}$	0.793	0.081	class position 0	0.132	—
$\tau_{\delta, \text{NP}}$	0.937	0.184	class position 1	0.253	0.005
$\tau_{\delta, \text{NA}}$	0.876	0.067	class position 2	0.391	0.015
λ	0.046	0.008	class position 3	0.629	0.024
γ	0.019	0.004	class position 4	0.842	0.017
$\tilde{\kappa}_d$	0.350	0.020	$\tilde{\alpha}_1$	-0.276	0.049
κ_n	0.677	0.009	$\tilde{\alpha}_2$	-1.033	0.064
s	3.048	0.031	$\tilde{\alpha}_3$	-0.881	0.071
s_c	4.526	0.037	$\tilde{\alpha}_4$	-1.234	0.089
σ_e	0.204	first step			

Notes: The table reports the full parameter vector estimated by simulated maximum likelihood. Standard errors are clustered by journalist and are conditional on the first-step $\hat{\sigma}_e$. Ideal points and median class preferences $\delta_k = \Lambda(b_k)$ are in slant units; τ_ζ is in log-odds units. The stay, tenure, and same-class parameters are in utility units; the destination intercepts $\tilde{\alpha}_k$ (with $\tilde{\alpha}_0 = 0$) and the distance cost $\tilde{\kappa}_d$ are in within-nest units (utility divided by λ).

Appendix Table C11: Model-Implied Variance of Article Slant

Component	Variance (slant ²)	Share (%)
Journalist: $\text{Var}(J)$	0.005396	6.5
Outlet class: $\text{Var}(O)$	0.020103	24.3
Accepted offer deviation: $\text{Var}(X)$	0.013901	16.8
$2\text{Cov}(J, O)$	0.007707	9.3
$2\text{Cov}(J, X)$	0.002611	3.2
$2\text{Cov}(O, X)$	-0.008780	-10.6
Independent article noise: σ_e^2	0.041768	50.5
Systematic variance: $\text{Var}(J + O + X)$	0.040938	49.5
Total article variance	0.082706	100.0

Notes: The table reports fitted-model variance accounting for stationary published article credits at observed journalist party composition and fixed outlet-class shares. For an accepted contract from class k , define $J = \frac{\beta}{1+\beta} \delta_i$, $O = \frac{1}{1+\beta} \delta_k$, and $X = \frac{1}{1+\beta} \xi_{ikt}$, so $x_{ikt}^* = J + O + X$. The class baseline $\delta_k = \Lambda(b_k)$ is the median preference before selection; the accepted deviation X need not have mean zero. Weights reflect the stationary distribution of published credits, including entry selection and spell duration. All three doubled covariances retain their signs. The first seven rows sum to total model-implied article variance, including independent article noise σ_e^2 ; this total is the denominator for every percentage. The outlet term describes class baselines, and the components are accounting terms rather than causal effects. Entries are point estimates conditional on the fit and numerical integration grid.